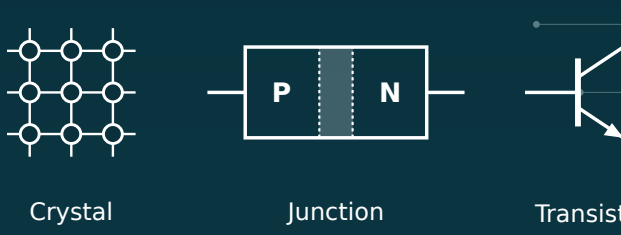
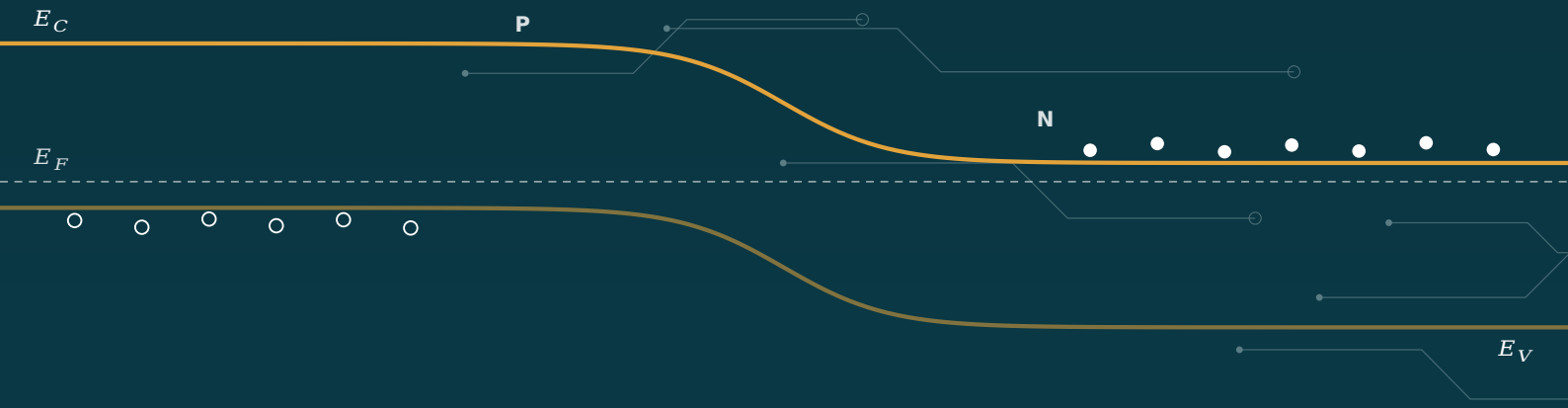
NATIONAL SCHOOL OF ENGINEERS OF SOUSSE (ENISO)
Department of Industrial Electronics





WORKED SOLUTIONS

Semiconductor Devices Physics

Complete solutions of the problems for Chapters 0 to 8

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ACADEMIC YEAR
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VERSION
Version 2

Chapters 0 to 8: why semiconductor devices? · transport phenomena · semiconductors at equilibrium · energy bands and the Fermi level · carriers out of equilibrium · the PN junction at and out of equilibrium · the capacitances of the junction · the bipolar junction transistor

How to use this collection

One set per chapter. Each chapter of the course has a set of problems, numbered by chapter: Problem 3.4 is the fourth problem of the Chapter 3 set. A set uses only what has been taught up to the end of its chapter; the box at the start of each set says which ideas it calls on and which data it uses.

Three levels. Within a set the problems are arranged in order of difficulty, and each one carries a label:

- **A · Understanding**: reasoning with the physical picture, orders of magnitude, short arguments. No long calculation.
- **B · Guided calculation**: a calculation in several steps, set out in the order in which the course builds the result.
- **C · Synthesis**: problems that combine several ideas, ask for a derivation, or apply the course to a situation it did not treat explicitly.

Hints. Many problems have a hint. It tells you where to start or which idea to use; it does not give the result. Try the problem first without it.

Figures. Figures placed with a statement show the device or the set-up; they appear in both documents. Figures inside a solution show the result, a profile, a curve or a band diagram, and appear only in the worked solutions. Curves are computed with the numbers of the solution; pictures labelled 'generic' show shapes, not values. A figure marked 'After Neamen, Fig. …' or 'After Sze & Lee, Fig. …' is redrawn from that textbook figure.

Sources. Under the title of each problem, a line gives its origin. *Original problem* means that the problem was written for this course. *Adapted from Neamen, Problem n, p. p* or *Adapted from Sze & Lee, Ch. c, Problem n, p. p* means that the problem starts from that textbook problem; it has been rewritten for the course, usually with the data of the course, more explicit assumptions and additional questions, so that its numbers differ from those of the book. The two textbooks are

- D. A. Neamen, *Semiconductor Physics and Devices: Basic Principles*, 4th ed., McGraw-Hill, 2012;
- S. M. Sze and M. K. Lee, *Semiconductor Devices: Physics and Technology*, 3rd ed., Wiley, 2012.

Page numbers are the printed page numbers of these editions.

Numbers. Give results with two or three significant figures, and always with their units. When a result looks surprising, check its order of magnitude before going on: many problems ask you to compare a computed value with another one, and that comparison is usually the point of the problem.

Data used throughout

Unless a problem states otherwise, the material is silicon at $T = 300$ K and the values below are used. They are the values of the course.

Quantity	Value	Remark
Elementary charge	$q = 1.602 \times 10^{-19}$ C	$q > 0$; electron charge $-q$
Electron mass	$m_0 = 9.11 \times 10^{-31}$ kg	
Boltzmann constant	$k = 1.381 \times 10^{-23}$ J/K $= 8.617 \times 10^{-5}$ eV/K	
Thermal energy, thermal voltage	$kT = 25.9$ meV, $V_T = kT/q = 25.9$ mV	at 300 K
Planck constant	$h = 6.626 \times 10^{-34}$ J·s, $hc = 1240$ eV·nm	
Vacuum permittivity	$\varepsilon_0 = 8.854 \times 10^{-14}$ F/cm	
Silicon permittivity	$\varepsilon = 11.7\,\varepsilon_0 = 1.036 \times 10^{-12}$ F/cm	
Atoms of silicon	5×10^{22} cm^{-3}	
Gap of silicon	$E_g = 1.12$ eV	
Intrinsic density of silicon	$n_i = 10^{10}$ cm^{-3}	reference value of the course
Effective densities of states	$N_C = 2.8 \times 10^{19}$, $N_V = 1.04 \times 10^{19}$ cm^{-3}	
Mobilities (low doping)	$\mu_n = 1350$, $\mu_p = 460$ cm²/(V·s)	
Diffusion coefficients	$D_n = 35$, $D_p = 12$ cm²/s	Einstein relation

Reference devices. The *reference junction* of Chapters 5 to 7 is an abrupt silicon P$^+$N junction with $N_A = 10^{18}$ cm^{-3} and $N_D = 10^{16}$ cm^{-3}, built-in voltage $V_d = 0.835$ V and equilibrium SCR width $W = 0.330$ µm. As a diode it has short neutral regions, $d_p = 2$ µm and $d_n = 10$ µm, minority lifetimes $\tau_n = 0.1$ µs and $\tau_p = 10$ µs, an area $S = 10^{-4}$ cm², and $J_0 = 2.2 \times 10^{-11}$ A/cm². The *reference transistor* of Chapter 8 is an NPN transistor with $N_E = 10^{18}$, $N_B = 10^{16}$ and $N_C = 10^{15}$ cm^{-3}, neutral widths $d_E = 0.5$ µm, $W_B = 1.0$ µm (at $V_{CB} = 5$ V) and $d_C = 10$ µm, and area $S = 10^{-4}$ cm²; its main parameters are recalled at the start of the Chapter 8 set.

Mobilities in doped material. The mobilities fall when the doping rises. The course uses the low-doping values above throughout, as a simplification; when a problem needs another value, the problem gives it.

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SEMICONDUCTOR DEVICES PHYSICS · SOLUTIONS

CHAPTER 0

Why Semiconductor Devices?

WHAT THIS SET USES

The building blocks of devices, the history from the vacuum tube to the integrated circuit, the two device laws quoted in the chapter (Child–Langmuir and Shockley), Moore's law and scaling, lithography. Neither textbook has problems on these topics: this set is original, and is a set of estimates rather than of physics.

DATA

$hc = 1240$ eV·nm, $\varepsilon_0 = 8.854\times10^{-12}$ F/m, $q = 1.602\times10^{-19}$ C, $m_0 = 9.11\times10^{-31}$ kg; silicon: 5×10^{22} atoms/cm³, density 2.33 g/cm³; $V_T = 25.9$ mV.

Problem 0.1 Four building blocks

A · Understanding

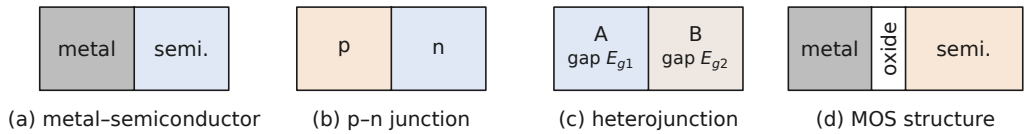
Original problem

The chapter claims that almost every device is one of four building blocks, metal–semiconductor contact, p–n junction, heterojunction, metal–oxide–semiconductor (MOS) structure, or a combination of them.

(a) Classify the following devices: silicon solar cell; Schottky diode; bipolar junction transistor; MOS transistor; GaN blue LED (whose active layer is a thin InGaN layer between GaN layers); heterojunction bipolar transistor.

(b) For the MOS transistor and the bipolar transistor, say which block appears more than once and how many times.

(c) The course studies the p–n junction and the bipolar transistor in detail. Using your answer to (a), explain why this choice already covers part of the physics of the MOS transistor.



(a) metal–semiconductor (b) p–n junction (c) heterojunction (d) MOS structure

The four building blocks of semiconductor devices.

HINT

A device is a combination of blocks when it contains several interfaces of different kinds; count the interfaces a carrier crosses.

(Answer: (a) p–n; metal–semiconductor; two p–n; MOS + two p–n; heterojunctions (+ p–n); heterojunction + p–n. (b) BJT: p–n twice; MOS transistor: p–n twice (source, drain) around one MOS structure.)

SOLUTION

(a) The **silicon solar cell** is a p–n junction under light. The **Schottky diode** is a rectifying metal–semiconductor contact. The **bipolar junction transistor** is two p–n junctions sharing a thin base. The **MOS transistor** is a MOS structure (the gate) placed between two p–n junctions (source and drain). The **GaN/InGaN LED** uses heterojunctions to confine the carriers in the thin InGaN layer, inside a p–n structure that injects them. The **heterojunction bipolar transistor** is a bipolar transistor in which one junction, usually the emitter–base junction, is a heterojunction.
(b) In both transistors the p–n junction appears twice. In the bipolar transistor the two junctions interact through the base; in the MOS transistor they are the source and drain, and the MOS structure controls the region between them.
(c) The MOS transistor contains two p–n junctions, whose electrostatics, currents and capacitances are those of Chapters 5 to 7. What remains to be learned is the MOS structure itself; the carriers, their densities, their currents and the junctions are already understood. This is why the chapter says that the MOS transistor becomes accessible once the framework is mastered.

Problem 0.2 Moore's law in numbers

B · Guided calculation

Original problem

The first microprocessor (1971) contained about 2300 transistors. A large processor chip of the early 2020s contains of the order of 5×10^{10} transistors.

(a) Assume that the number of transistors per chip doubles every two years from 1971. How many transistors would a chip of 2021 contain?

(b) Which doubling time is implied by the two data points (1971 and 2021)?

(c) By how much does the number of transistors grow each year, in percent, for a doubling time of 2 years?

(d) The chapter says that Moore's law is 'an industrial roadmap, not a law of physics'. Give one argument from your results that supports this statement.

HINT

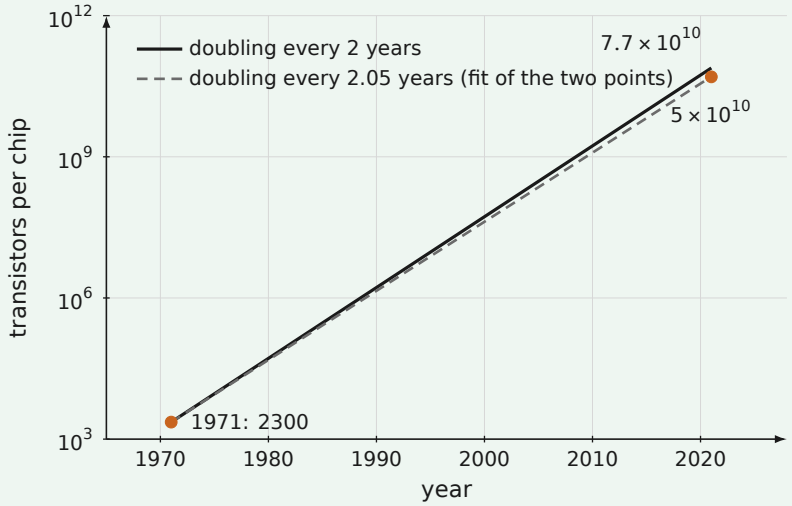
After t years with a doubling time T_2, the number is multiplied by $2^{t/T_2}$. To find T_2 from two values, take the base-2 logarithm of their ratio.

(Answer: (a) 7.7×10^{10}; (b) 2.05 years; (c) 41 % per year.)

SOLUTION

(a) From 1971 to 2021 there are 50 years, that is 25 doublings: $N = 2300 \times 2^{25} = 2300 \times 3.36 \times 10^7 = 7.7 \times 10^{10}$ transistors.
(b) The ratio of the two values is $5 \times 10^{10}/2300 = 2.2 \times 10^7$, and $\log_2(2.2 \times 10^7) = 24.4$ doublings. In 50 years this gives $T_2 = 50/24.4 = 2.05$ years.
(c) A doubling every 2 years is a factor $2^{1/2} = 1.414$ per year: a growth of 41 % per year.
(d) The pace of a physical law does not depend on decisions; here the doubling time was first one year (1965) and then two years (1975), and the industry has held the

second value for fifty years, within a few percent, by planning investments and technology generations to match it. A pace that is revised, then maintained on purpose, is a roadmap. The result of (b) also contains several contributions (smaller transistors, larger chips, better layouts), not a single physical mechanism; Problem 0.3 separates the first one.



Transistors per chip on a logarithmic axis: a constant doubling time is a straight line.

Problem 0.3 How much of Moore's law is shrinking? B · Guided calculation

Original problem

The minimum feature size went from about 10 µm in 1971 to about 20 nm around 2015 (Chapter 0, figure of scaling).

(a) By what factor did the feature size L decrease, and by what factor per year, on average?

(b) The area of a transistor scales as L^2. By what factor did the number of transistors per unit area increase? What doubling time of the density does this correspond to?

(c) Compare with the doubling time of about 2 years of Problem 0.2. What else must have grown to make up the difference?

HINT

Treat the decrease as geometric: a constant factor per year.

(Answer: (a) factor 500, i.e. ×0.868 per year (−13 %/year); (b) factor 2.5×10^5, doubling every 2.45 years; (c) the area of the chip and the efficiency of the layout.)

SOLUTION

(a) 10 μm/20 nm = 500 in 44 years, i.e. a factor $500^{-1/44} = 0.868$ per year: the feature size has decreased by about 13 % per year.
(b) The density of transistors scales as $1/L^2$: $500^2 = 2.5{\times}10^5$. Since $\log_2(2.5{\times}10^5) =$

17.9, the density doubled 17.9 times in 44 years, once every 44/17.9 = 2.45 years.
(c) Shrinking alone gives a doubling every 2.45 years, slower than the 2.05 years of Problem 0.2. The difference comes from the other factors in the number of transistors per chip: chips became larger (more area per chip), and transistors were packed more efficiently (better layouts, three-dimensional transistors such as FinFETs, which take less area for the same current). Shrinking is the main contribution, but not the only one.

Problem 0.4 Light for lithography

B · Guided calculation

Original problem

The chapter states that the smallest pattern that light can print is of the order of λ/NA, where NA is the numerical aperture of the optics.

(a) Compute the photon energy, in eV, of the mercury line at 365 nm, of the ArF excimer laser at 193 nm and of EUV light at 13.5 nm. Compare each with the gap of silicon, 1.12 eV.

(b) Estimate λ/NA for 193 nm light through water with NA = 1.35, and for EUV light with NA = 0.33.

(c) The tightest metal pitch of a '3 nm' process is about 20–25 nm. Which of the two techniques of (b) can print it in a single exposure?

(d) The chapter says that EUV light is absorbed by air and by glass. Use your result of (a) to explain why such light is absorbed by nearly every material.

HINT

$E = hc/\lambda$ with $hc = 1240$ eV·nm.

(Answer: (a) 3.40, 6.42 and 91.9 eV, i.e. 3, 6 and 82 times E_g; (b) 143 nm and 41 nm; (c) lines of 10–12 nm: immersion is more than 10 times too coarse, EUV within a factor of 2 to 4; (d) 92 eV exceeds the binding energies of the outer electrons of all atoms.)

SOLUTION

(a) $E = 1240/\lambda$: 365 nm gives 3.40 eV, 193 nm gives 6.42 eV, and 13.5 nm gives 91.9 eV. Compared with $E_g = 1.12$ eV these are factors of 3.0, 5.7 and 82. All three photons can break a bond of silicon; the EUV photon carries almost a hundred times the gap.
(b) 193/1.35 = 143 nm for immersion lithography; 13.5/0.33 = 41 nm for EUV.
(c) A pitch of 20–25 nm means lines and spaces of 10–12 nm each. Both values of (b) are larger. Immersion lithography is more than ten times too coarse and needs several successive exposures and etches (multiple patterning) to reach such pitches. EUV is within a factor of about 2 to 4; with the process tricks that lower the proportionality constant below 1, and sometimes with double patterning, it reaches the tightest layers. The order-of-magnitude rule explains why the industry needed EUV.
(d) Photons of a few eV are absorbed only by materials whose electrons can take that energy: glass and air are transparent to visible light because their gaps are larger than the photon energy. A 92 eV photon is larger than the binding energy of the outer

electrons of every atom, so every material, including the gases of air and the oxides of glass, absorbs it within a short distance. EUV optics must therefore be mirrors, used in vacuum, as the chapter says.

Problem 0.5 Two device laws

B · Guided calculation

Original problem

The chapter quotes two laws: the Child–Langmuir law of the vacuum diode, $J = \frac{4\varepsilon_0}{9}\sqrt{\frac{2q}{m}}\frac{V^{3/2}}{d^2}$, and the Shockley law of the junction diode, $I = I_s[\exp(V/\eta V_T) - 1]$.

(a) Compute J for a vacuum diode with $d = 1$ mm and $V = 100$ V. What current flows through a plate of 1 cm²?

(b) By what factor does the current of the vacuum diode increase when V is doubled?

(c) For the junction diode ($\eta = 1$, well above V_T), by how much must V increase to multiply the current by 10? By 2.83?

(d) Compare the voltages involved in the two devices, and comment on the advantage of the junction diode for low-voltage electronics.

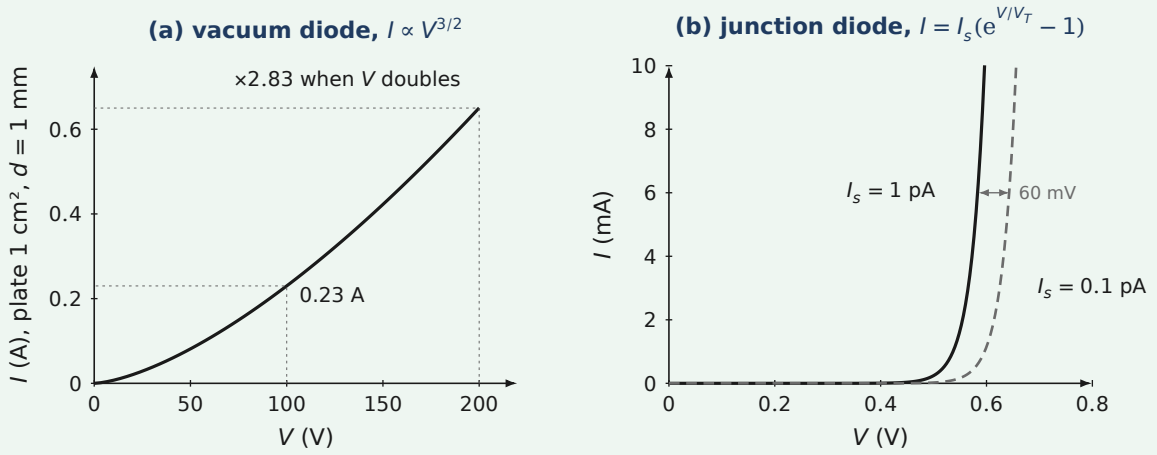
HINT

Work in SI units in (a). In (c), for $V \gg V_T$, $I \simeq I_s e^{V/V_T}$, so a factor r on the current needs $\Delta V = V_T \ln r$.

(Answer: (a) $J = 2.3 \times 10^3$ A/m² = 0.23 A/cm², 0.23 A; (b) $2^{3/2} = 2.83$; (c) 59.6 mV, 26.9 mV.)

SOLUTION

(a) $\sqrt{2q/m} = \sqrt{2 \times 1.602 \times 10^{-19}/9.11 \times 10^{-31}} = 5.93 \times 10^5$ (SI), $4\varepsilon_0/9 = 3.94 \times 10^{-12}$ F/m, $V^{3/2} = 1000$ V$^{3/2}$ and $d^2 = 10^{-6}$ m². Hence $J = 3.94 \times 10^{-12} \times 5.93 \times 10^5 \times 10^3/10^{-6} = 2.3 \times 10^3$ A/m², that is 0.23 A/cm²: a plate of 1 cm² collects 0.23 A.
(b) $J \propto V^{3/2}$: doubling V multiplies J by $2^{3/2} = 2.83$.
(c) $\Delta V = V_T \ln 10 = 25.9 \times 2.303 = 59.6$ mV for a factor 10, and $V_T \ln 2.83 = 26.9$ mV for the factor 2.83 that the vacuum diode needed 100 V to obtain.
(d) The vacuum diode works at tens to hundreds of volts, and its current rises slowly, as a power law. The junction diode switches from blocking to conducting within a few tenths of a volt, and its current changes by a decade every 60 mV. This exponential sensitivity, together with the absence of a heated filament, is what makes semiconductor devices suitable for electronics running at a volt or less. The law of the junction diode is derived in Chapter 6.



A power law against an exponential: 100 V for a factor 2.83 in the vacuum diode, 27 mV in the junction diode.

Challenge: a new situation

These problems apply the method of the chapter to a device, a material or a situation that the chapter did not treat. The reasoning does not change: state the physical picture, choose the model, compute, check.

Problem 0.6 From a wafer to a chip, and why not tubes

C · Synthesis

Original problem

A 300 mm silicon wafer is 0.775 mm thick. It carries square chips of 1 cm², each holding 5×10^{10} transistors.

(a) How many silicon atoms does the wafer contain, and what is its mass?

(b) The number of whole chips on a wafer of diameter D is estimated by $N \simeq \pi D^2/(4A) - \pi D/\sqrt{2A}$, where A is the chip area; the second term counts the chips lost at the edge. Evaluate N.

(c) The transistors are built in the top micrometre of the silicon. How many silicon atoms of this layer are there per transistor on one chip?

(d) Suppose that each of the switches of one chip were a vacuum tube whose filament needs only 1 W. What power would the chip need? A modern processor dissipates of the order of 100 W: what power does that leave per transistor?

HINT

In (c), the volume of the layer under one chip is 1 cm² × 1 µm.

(Answer: (a) 2.7×10^{24} atoms, 128 g; (b) about 640 chips; (c) 10^8 atoms per transistor; (d) 5×10^{10} W = 50 GW, against 2 nW per transistor.)

SOLUTION

(a) The volume is $\pi \times (15\ \text{cm})^2 \times 0.0775\ \text{cm} = 54.8\ \text{cm}^3$: $54.8 \times 5 \times 10^{22} = 2.7 \times 10^{24}$ atoms, and $54.8 \times 2.33 = 128$ g.
(b) With $D = 30$ cm and $A = 1\ \text{cm}^2$: $\pi \times 900/4 = 707$, and $\pi \times 30/\sqrt{2} = 67$, so $N \simeq 640$ chips.
(c) The layer under one chip has $1\ \text{cm}^2 \times 10^{-4}\ \text{cm} = 10^{-4}\ \text{cm}^3$ and 5×10^{18} atoms; divided among 5×10^{10} transistors, 10^8 atoms each. The transistor is small, but it is still a crystal of a hundred million atoms; this is why the physics of the bulk crystal, built in this course, applies to it.
(d) 5×10^{10} filaments of 1 W would need 50 GW, the output of several dozen large power stations, for one chip. The real chip dissipates about 100 W, $100/5 \times 10^{10} = 2$ nW per transistor. The absence of a heated filament, together with the size, is the decisive advantage of the transistor listed in the chapter.

SEMICONDUCTOR DEVICES PHYSICS · SOLUTIONS

CHAPTER 1

Transport Phenomena: Drift, Diffusion and Continuity

WHAT THIS SET USES

The current density $J = q\,n\,v$; the Drude model, $v_d = q\tau_c\mathcal{E}/m^*$ and $\mu = q\tau_c/m^*$; the thermal velocity; the diffusion flux $-D\,\mathrm{d}n/\mathrm{d}x$; the Einstein relation $D = \mu\,kT/q$; the total currents $J_n = q\mu_n n\mathcal{E} + qD_n\,\mathrm{d}n/\mathrm{d}x$ and $J_p = q\mu_p p\mathcal{E} - qD_p\,\mathrm{d}p/\mathrm{d}x$; the continuity equation without generation or recombination.

DATA

Silicon electrons: conductivity effective mass $m^* = 0.26\,m_0$. $kT/q = 25.9$ mV at 300 K. Other data from the data sheet at the front.

Problem 1.1 From a drift velocity to a collision time

B · Guided calculation

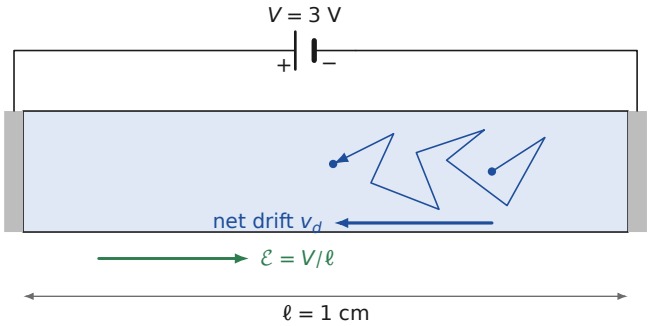
Adapted from Neamen, Problem 5.10, p. 185

A voltage of 3 V is applied across a semiconductor bar 1 cm long. The electrons acquire an average drift velocity of 10^4 cm/s.

(a) What is the field in the bar, assumed uniform? Deduce the electron mobility.

(b) With the Drude relation $\mu = q\tau_c/m^*$, find the mean time between collisions, taking $m^* = 0.26\,m_0$.

(c) In another material the electron mobility is 800 cm²/(V·s). What drift velocity does the same voltage produce? By what factor is τ_c shorter, for the same m^*?



Random flights between collisions, with a small net drift of the electrons against the field.

HINT

Convert the mobility to m²/(V·s) before using $\tau_c = \mu m^*/q$ in SI units.

(Answer: (a) 3 V/cm, $\mu = 3.3\times10^3$ cm²/(V·s); (b) $\tau_c = 0.49$ ps; (c) 2.4×10^3 cm/s, 4.2 times shorter.)

SOLUTION

(a) In a uniform bar the field is the voltage divided by the length, $\mathcal{E} = 3/1 = 3$ V/cm. The mobility is the drift velocity per unit field: $\mu = v_d/\mathcal{E} = 10^4/3 = 3.3\times10^3$ cm²/(V·s).
(b) $\mu = 3.33\times10^3$ cm²/(V·s) = 0.333 m²/(V·s). Then $\tau_c = \mu m^*/q = 0.333\times0.26\times9.11\times10^{-31}/1.602\times10^{-19} = 4.9\times10^{-13}$ s ≈ 0.5 ps. Between two collisions an electron accelerates for about half a picosecond.
(c) $v_d = \mu\mathcal{E} = 800\times3 = 2.4\times10^3$ cm/s. Since $\tau_c \propto \mu$ at fixed m^*, it is 3333/800 = 4.2 times shorter: the carriers collide more often and gain less velocity between collisions.

Problem 1.2 How much energy does drift give an electron? B · Guided calculation

Adapted from Neamen, Problem 5.20, p. 186

Electrons in silicon at 300 K have $\mu_n = 1350$ cm²/(V·s) and $m^* = 0.26\,m_0$. The Drude model assumes that the field only adds a small drift to a fast random thermal motion.

(a) Compute the drift velocity and the kinetic energy of the drift motion, $\frac{1}{2}m^*v_d^2$, for $\mathcal{E} = 10$ V/cm and for $\mathcal{E} = 1$ kV/cm.

(b) Compute the thermal velocity $v_{th} = \sqrt{3kT/m^*}$ and the mean thermal kinetic energy $\frac{3}{2}kT$.

(c) Compare. Up to what order of magnitude of field can the assumption of the Drude model, $v_d \ll v_{th}$, be considered safe?

HINT

Work in SI units and convert the energies to eV by dividing by q.

(Answer: (a) 1.35×10^4 cm/s and 1.3×10^{-8} eV; 1.35×10^6 cm/s and 1.3×10^{-4} eV; (b) 2.3×10^7 cm/s, 38.8 meV; (c) $v_d/v_{th} = 0.06$ at 1 kV/cm: safe up to a few kV/cm.)

SOLUTION

(a) $v_d = \mu_n\mathcal{E}$. At 10 V/cm, $v_d = 1.35\times10^4$ cm/s = 135 m/s and $\frac{1}{2}m^*v_d^2 = \frac{1}{2}\times0.26\times9.11\times10^{-31}\times135^2 = 2.16\times10^{-27}$ J = 1.35×10^{-8} eV. At 1 kV/cm, $v_d = 1.35\times10^6$ cm/s, a hundred times larger, and the energy is 10^4 times larger: 1.35×10^{-4} eV.
(b) $v_{th} = \sqrt{3\times1.381\times10^{-23}\times300/(0.26\times9.11\times10^{-31})} = 2.3\times10^5$ m/s = 2.3×10^7 cm/s, and $\frac{3}{2}kT = 1.5\times25.9 = 38.8$ meV.
(c) At 10 V/cm the drift is 6×10^{-4} of the thermal velocity and its energy 3×10^{-7} of the thermal energy: the field barely perturbs the random motion, exactly as the Drude model assumes. At 1 kV/cm, $v_d/v_{th} = 0.06$ and the drift energy is still 3×10^{-3} of $\frac{3}{2}kT$: the model holds. Since v_d grows as $\mathcal{E}$, it reaches a sizeable fraction of v_{th} at some 10 kV/cm; the energy gained between collisions then heats the electrons, the collision time changes, and the drift velocity no longer grows in proportion to the field (velocity saturation, which the course meets again in the collector of Chapter 8). The linear law $v_d = \mu\mathcal{E}$ is safe up to a few kV/cm.

Problem 1.3 Several kinds of collisions

B · Guided calculation

Adapted from Neamen, Problem 5.24, p. 187

In a semiconductor several scattering mechanisms act at the same time (vibrations of the lattice, ionised impurities, …). Each one, alone, would give a collision time τ_i and a mobility $\mu_i = q\tau_i/m^*$.

(a) The mechanisms are independent: in a short time $\mathrm{d}t$, the probability of a collision of type i is $\mathrm{d}t/\tau_i$. Show that the total collision rate is $1/\tau_c = \sum_i 1/\tau_i$, and deduce that $1/\mu = \sum_i 1/\mu_i$ (Matthiessen's rule).

(b) Three mechanisms alone would give 2000, 1500 and 500 cm²/(V·s). What is the mobility? Which mechanism controls it?

(c) Two mechanisms alone would give 250 and 500 cm²/(V·s) (Sze & Lee, Ch. 2, Problem 3, p. 79). What is the mobility?

> **HINT**
> Probabilities of independent, rare events in the same short interval add.

(Answer: (b) 316 cm²/(V·s), controlled by the 500 cm²/(V·s) mechanism; (c) 167 cm²/(V·s).)

> **SOLUTION**
> (a) In $\mathrm{d}t$, a collision of type i occurs with probability $\mathrm{d}t/\tau_i$. For a short interval these events are rare and independent, so the probability that some collision occurs is the sum, $\mathrm{d}t\sum_i 1/\tau_i$. By definition this is $\mathrm{d}t/\tau_c$: $1/\tau_c = \sum_i 1/\tau_i$. With $\mu = q\tau_c/m^*$ and $\mu_i = q\tau_i/m^*$, dividing by q/m^* gives $1/\mu = \sum_i 1/\mu_i$: **the collision rates add, so the inverse mobilities add**.
> (b) $1/\mu = 1/2000 + 1/1500 + 1/500 = (0.5 + 0.667 + 2.0)\times 10^{-3} = 3.17\times 10^{-3}$, so $\mu = 316$ cm²/(V·s). The result is below the smallest of the three: the mechanism that, alone, gives 500 cm²/(V·s) is the most frequent collision and controls the mobility.
> (c) $1/\mu = 1/250 + 1/500 = 6\times 10^{-3}$: $\mu = 167$ cm²/(V·s).

Problem 1.4 Mobility, diffusion and temperature

B · Guided calculation

Adapted from Neamen, Problem 5.25, p. 187

The electron mobility of a silicon sample is 1300 cm²/(V·s) at 300 K. It is limited by the vibrations of the lattice, and then varies as $T^{-3/2}$.

(a) Explain in one sentence, with the Drude picture, why the mobility falls when the temperature rises.

(b) Compute the mobility at 200 K and at 400 K.

(c) Compute the diffusion coefficient at 200, 300 and 400 K with the Einstein relation. How does D vary with T?

> **HINT**
> $kT/q = 25.9\times(T/300)$ mV.

(Answer: (b) 2390 and 844 cm²/(V·s); (c) 41.2, 33.7, 29.2 cm²/s: $D \propto T^{-1/2}$.)

SOLUTION

(a) The lattice vibrates more strongly at higher temperature, so an electron meets a vibration sooner: τ_c is shorter and so is $\mu = q\tau_c/m^*$.
(b) $\mu(200) = 1300 \times (200/300)^{-3/2} = 1300 \times 1.837 = 2390$ cm²/(V·s); $\mu(400) = 1300 \times (400/300)^{-3/2} = 1300 \times 0.650 = 844$ cm²/(V·s).
(c) $D = \mu kT/q$: at 200 K, $2390 \times 0.01727 = 41.2$ cm²/s; at 300 K, $1300 \times 0.0259 = 33.7$ cm²/s; at 400 K, $844 \times 0.03453 = 29.2$ cm²/s. Since $\mu \propto T^{-3/2}$ and $kT/q \propto T$, $D \propto T^{-1/2}$: it falls, but more slowly than the mobility, because faster thermal motion partly makes up for the shorter flights.

Problem 1.5 Two diffusion currents

B · Guided calculation

Adapted from Neamen, Problems 5.31 and 5.34, p. 188

Use $D_p = 12$ cm²/s and $D_n = 35$ cm²/s.

(a) The hole density in a silicon sample is $p(x) = 5 \times 10^{15}\, e^{-x/L_p}$ cm^{-3} for $x \geq 0$, with $L_p = 50$ µm. Compute the hole flux and the hole diffusion current density at $x = 0$ and at $x = L_p$. In which direction do the holes move, and in which direction does the current flow?

(b) The electron diffusion current density in another sample is uniform, $J_n = 2$ A/cm², and $n(0) = 10^{15}$ cm^{-3}. What is the shape of $n(x)$? Compute n at $x = 20$ µm. Do the electrons move towards $+x$ or $-x$?

(c) In (a) the current varies with x, in (b) it does not. Using the continuity equation, say what would happen to the hole density in case (a) if nothing else acted on the holes. (Chapter 4 adds the missing process.)

HINT

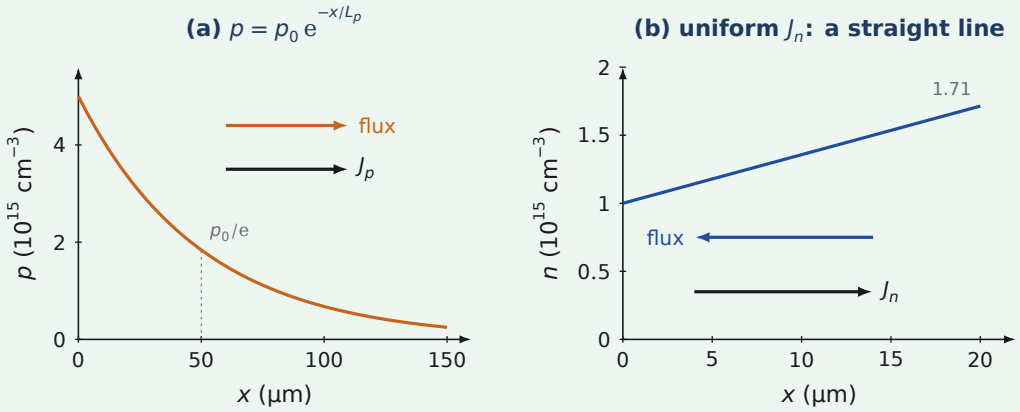
Flux $\Phi = -D\,\mathrm{d}n/\mathrm{d}x$; current of holes $q\Phi_p$, of electrons $-q\Phi_n$.

(Answer: (a) $J_p(0) = 1.92$ A/cm², $J_p(L_p) = 0.71$ A/cm², both towards $+x$; (b) linear, $n(20\ \mu\mathrm{m}) = 1.71 \times 10^{15}$ cm^{-3}, electrons move towards $-x$.)

SOLUTION

(a) $\mathrm{d}p/\mathrm{d}x = -(p_0/L_p)\, e^{-x/L_p}$, so the flux $\Phi_p = -D_p\,\mathrm{d}p/\mathrm{d}x = (D_p p_0/L_p)\, e^{-x/L_p}$ is positive: holes move towards $+x$, down the gradient. At $x = 0$, $\Phi_p = 12 \times 5 \times 10^{15}/5 \times 10^{-3} = 1.2 \times 10^{19}$ cm^{-2}s^{-1}, and $J_p = q\Phi_p = 1.92$ A/cm². At $x = L_p$ both are divided by e: $J_p = 0.71$ A/cm². For holes, current and flux have the same direction, $+x$.
(b) $J_n = -q\Phi_n = qD_n\,\mathrm{d}n/\mathrm{d}x$. A uniform J_n means a uniform slope: $n(x)$ is a straight line, $n(x) = n(0) + J_n x/(qD_n)$. The slope is $2/(1.602 \times 10^{-19} \times 35) = 3.57 \times 10^{17}$ cm^{-4}, and $n(20\ \mu\mathrm{m}) = 10^{15} + 3.57 \times 10^{17} \times 2 \times 10^{-3} = 1.71 \times 10^{15}$ cm^{-3}. The density increases with x, so the electrons diffuse towards $-x$; the current flows towards $+x$, opposite to the electron flow.
(c) The continuity equation without generation or recombination reads $\partial p/\partial t = -\partial \Phi_p/\partial x$. In (a) the flux decreases with x: each slice receives more holes on its

left than it loses on its right, so p would increase everywhere and the profile could not be stationary. A stationary exponential profile requires a process that removes holes in each slice at the rate $-\partial\Phi_p/\partial x$; this is recombination, introduced in Chapter 4, and L_p is the diffusion length defined there. In (b) the flux is uniform and the linear profile is stationary without any such process.



(a) Holes diffuse down the gradient, and their current has the same direction. (b) A uniform current needs a uniform slope; electrons flow towards $-x$, the current towards $+x$.

Problem 1.6 Drift and diffusion share one current

B · Guided calculation

Adapted from Sze & Lee, Ch. 2, Problem 16, p. 80

In a silicon sample the total current density is constant, $J = 4.8$ A/cm², and is the sum of an electron drift current and a hole diffusion current. The electron density is uniform, $n = 10^{16}\,\text{cm}^{-3}$, and the holes have $p(x) = 10^{15}\,e^{-x/L}\,\text{cm}^{-3}$ for $x \geq 0$, with $L = 12$ µm. Take $\mu_n = 1350$ cm²/(V·s) and $D_p = 12$ cm²/s.

(a) Compute the hole diffusion current density $J_p(x)$.

(b) Deduce the electron current density $J_n(x)$.

(c) Deduce the field $\mathcal{E}(x)$ and its values at $x = 0$ and far from the origin. Why can the electron diffusion current and the hole drift current be neglected?

HINT

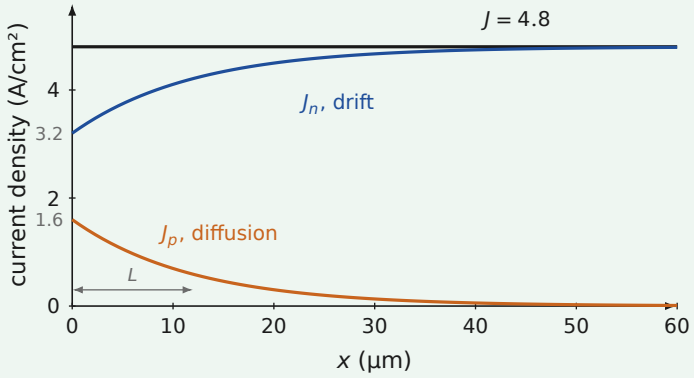
The total current is the same at every x; what one carrier loses, the other gains.

(Answer: (a) $J_p = 1.60\,e^{-x/L}$ A/cm²; (b) $J_n = 4.8 - 1.60\,e^{-x/L}$ A/cm²; (c) $\mathcal{E}(0) = 1.48$ V/cm, $\mathcal{E}(\infty) = 2.22$ V/cm.)

SOLUTION

(a) $J_p = -qD_p\,dp/dx = qD_p p(0)\,e^{-x/L}/L = 1.602 \times 10^{-19} \times 12 \times 10^{15}/1.2 \times 10^{-3}\,e^{-x/L} = 1.60\,e^{-x/L}$ A/cm², towards $+x$.
(b) $J_n(x) = J - J_p(x) = 4.8 - 1.60\,e^{-x/L}$ A/cm²: 3.2 A/cm² at $x = 0$, rising to 4.8 A/cm² far away, where the holes have disappeared.

(c) The electron current is a drift current, $J_n = q\mu_n n\mathcal{E}$, with $q\mu_n n = 1.602\times10^{-19}\times 1350\times10^{16} = 2.16$ S/cm. Hence $\mathcal{E}(x) = (4.8 - 1.60\,\mathrm{e}^{-x/L})/2.16$ V/cm: 1.48 V/cm at $x = 0$ and 2.22 V/cm far away. The electron density is uniform, so there is no electron diffusion current. The hole drift current, $q\mu_p p\mathcal{E} \le 1.602\times10^{-19}\times460\times10^{15}\times2.2 = 0.16$ A/cm², is ten times smaller than the hole diffusion current; neglecting it is the approximation the problem makes.



The total current is constant: the electron drift current takes over from the hole diffusion current over a few L.

Problem 1.7 A field without a current

B · Guided calculation

Adapted from Neamen, Problem 5.40, p. 189

An n-type sample is in thermal equilibrium, with no current. Its electron density follows the donors, $n(x) = N_0\,\mathrm{e}^{-x/L}$ for $0 \le x \le L$, with $N_0 = 10^{16}\,\mathrm{cm}^{-3}$ and $L = 10$ µm.

(a) In equilibrium $J_n = 0$. Write J_n as the sum of its drift and diffusion terms and show, with the Einstein relation, that $\mathcal{E}(x) = -\dfrac{kT}{q}\dfrac{1}{n}\dfrac{\mathrm{d}n}{\mathrm{d}x}$.

(b) Compute $\mathcal{E}$ for the exponential profile. In which direction does it point, and why must it point that way?

(c) Compute the potential difference between $x = 0$ and $x = L$. Which end is at the higher potential?

(d) Does a voltmeter connected between the two ends read this voltage? Answer with an argument about equilibrium.

HINT

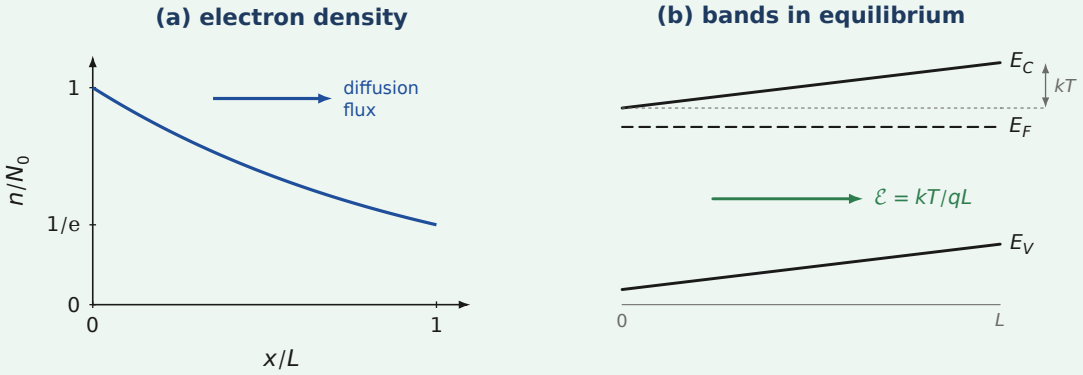
The field must push the electrons back up the gradient, against diffusion.

(Answer: (b) $\mathcal{E} = kT/(qL) = 25.9$ V/cm along $+x$; (c) $V(0) - V(L) = 25.9$ mV, $x = 0$ positive; (d) no.)

SOLUTION

(a) $J_n = q\mu_n n\mathcal{E} + qD_n\,\mathrm{d}n/\mathrm{d}x = 0$, so $\mathcal{E} = -(D_n/\mu_n)(1/n)\,\mathrm{d}n/\mathrm{d}x = -(kT/q)(1/n)\,\mathrm{d}n/\mathrm{d}x$ by the Einstein relation.

(b) $(1/n)\,\mathrm{d}n/\mathrm{d}x = -1/L$, so $\mathcal{E} = kT/(qL) = 0.0259/10^{-3} = 25.9$ V/cm, uniform and pointing towards $+x$. The electrons diffuse towards $+x$, down the gradient; to cancel this flux the field must push them towards $-x$, and since electrons move against the field, the field points towards $+x$.
(c) $V(0) - V(L) = \int_0^L \mathcal{E}\,\mathrm{d}x = \mathcal{E}L = kT/q = 25.9$ mV, with $x = 0$ at the higher potential. More generally $V(0) - V(L) = (kT/q)\ln[n(0)/n(L)]$, here $\ln \mathrm{e} = 1$.
(d) No. The sample is in equilibrium: if a voltmeter could draw a current from this voltage, the sample would deliver energy indefinitely without any source. The internal potential difference is compensated by the contact potentials at the two ends, where the leads meet the semiconductor, and the sum around the closed circuit is zero. The same argument returns in Chapter 5 for the built-in voltage of the junction.



In equilibrium the Fermi level is flat; the bands tilt by kT over L, and the field holds the electrons back against diffusion.

Problem 1.8 A spreading packet of carriers

C · Synthesis

Adapted from Neamen, Problem 6.28, p. 237

A short pulse of light creates, at $t = 0$, a thin sheet of N_s holes per cm² at $x = 0$ in a silicon bar. Ignore recombination (the time scale is short) and, at first, any field.

(a) Combine the continuity equation without generation or recombination and the diffusion flux to obtain the diffusion equation $\partial p/\partial t = D_p\,\partial^2 p/\partial x^2$.

(b) Show that $p(x,t) = \dfrac{N_s}{\sqrt{4\pi D_p t}}\exp\!\left(-\dfrac{x^2}{4D_p t}\right)$ is a solution, and that $\int_{-\infty}^{+\infty} p\,\mathrm{d}x = N_s$ at all times. What does the second result express?

(c) The width of the packet is measured by its standard deviation $\sqrt{2D_p t}$. Compute it after 1 µs.

(d) A field of 10 V/cm is now applied along $+x$. Where is the centre of the packet after 1 µs? Compare the drift distance with the width. Show that their ratio is $\dfrac{\mathcal{E}}{kT/q}\sqrt{\dfrac{D_p t}{2}}$, and explain why measuring the arrival time and the width of the packet at a second contact gives both μ_p and D_p (the principle of the Haynes–Shockley experiment, Neamen, Problem 6.27, p. 237).

HINT

In (b), write $p = At^{-1/2}\mathrm{e}^{-x^2/4D_p t}$ and differentiate. For the integral, use $\int_{-\infty}^{+\infty}\mathrm{e}^{-ax^2}\,\mathrm{d}x = \sqrt{\pi/a}$.

(Answer: (c) 49 µm; (d) 46 µm along $+x$, comparable to the width.)

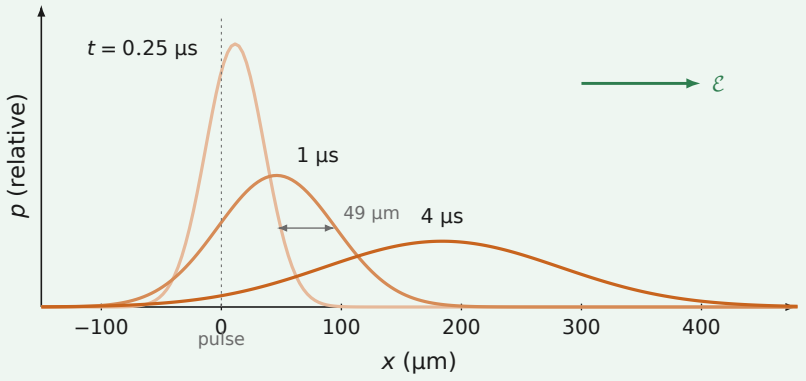
SOLUTION

(a) Continuity: $\partial p/\partial t = -\partial \Phi_p/\partial x$; Fick: $\Phi_p = -D_p\,\partial p/\partial x$. With D_p uniform, $\partial p/\partial t = D_p\,\partial^2 p/\partial x^2$.

(b) Let $A = N_s(4\pi D_p)^{-1/2}$, so $p = At^{-1/2}\mathrm{e}^{-x^2/4D_p t}$. Then $\partial p/\partial t = p\,[-1/(2t) + x^2/(4D_p t^2)]$; $\partial p/\partial x = -p\,x/(2D_p t)$; $\partial^2 p/\partial x^2 = p\,[x^2/(4D_p^2 t^2) - 1/(2D_p t)]$. Multiplying the last one by D_p gives the first one: the equation is satisfied. With $a = 1/(4D_p t)$, $\int \mathrm{e}^{-x^2/4D_p t}\,\mathrm{d}x = \sqrt{4\pi D_p t}$, so $\int p\,\mathrm{d}x = N_s$. The number of holes is conserved: without recombination diffusion only spreads them out.

(c) $\sqrt{2D_p t} = \sqrt{2\times 12\times 10^{-6}} = 4.9\times 10^{-3}$ cm = 49 µm.

(d) With a field, the whole packet drifts at $v_d = \mu_p \mathcal{E}$ while it spreads (the solution is the same function of $x - v_d t$). After 1 µs the centre is at $\mu_p \mathcal{E} t = 460\times 10\times 10^{-6} = 4.6\times 10^{-3}$ cm = 46 µm, about the width of the packet. The ratio is $\mu_p \mathcal{E} t/\sqrt{2D_p t}$; with $\mu_p = D_p/(kT/q)$ it becomes $\frac{\mathcal{E}}{kT/q}\frac{D_p t}{\sqrt{2D_p t}} = \frac{\mathcal{E}}{kT/q}\sqrt{\frac{D_p t}{2}}$. At a second contact at distance d, the arrival time t_d of the peak gives $\mu_p = d/(\mathcal{E} t_d)$, and the duration of the pulse seen there gives its width $\sqrt{2D_p t_d}$, hence D_p. Measuring both checks the Einstein relation directly.



The packet drifts at $\mu_p \mathcal{E}$ and spreads as $\sqrt{2D_p t}$; its area stays N_s.

Practice exercises from the course notes

These exercises closed Chapter 1 of the course notes; they now live here, with their answers. Tables, equations and sections cited by number are those of the course notes. Use the values of the notation caption unless stated otherwise.

Problem 1.9 Drift in a household cable

B · Guided calculation

From the course notes, Chapter 1

A copper conductor of 2.5 mm² carries 10 A. Find the drift velocity and the time an electron takes to travel 1 m. Why does a lamp light up as soon as the switch is closed?

(Answer: $J = 10/2.5\times10^{-6} = 4\times10^{6}$ A/m²; $v_d = J/(en) = 2.9\times10^{-4}$ m/s; 1 m takes 3.4×10^{3} s, about an hour. The lamp lights at once because the field is established along the whole circuit almost at the speed of light, and all the electrons start drifting together; no electron has to travel from the switch to the lamp.)

Problem 1.10 Aluminium

B · Guided calculation

From the course notes, Chapter 1

Aluminium has $\sigma = 3.77\times10^{7}$ S/m and three free electrons per atom, $n = 1.81\times10^{29}$ m^{-3}. Find τ_c and μ, and compare with copper.

(Answer: $\tau_c = \sigma m/(ne^2) = 3.77\times10^{7}\times9.11\times10^{-31}/(1.81\times10^{29}\times2.57\times10^{-38}) = 7.4\times10^{-15}$ s; $\mu = e\tau_c/m = 1.3\times10^{-3}$ m²/(V·s) = 13 cm²/(V·s). More electrons than copper, but three times shorter flights: a slightly lower conductivity.)

Problem 1.11 A real wire

B · Guided calculation

From the course notes, Chapter 1

Find the resistance of 100 m of copper wire of 1.5 mm², and the field inside the copper when it carries 10 A.

(Answer: $R = L/(\sigma S) = 100/(5.96\times10^{7}\times1.5\times10^{-6}) = 1.12\ \Omega$. $\mathcal{E} = J/\sigma = (10/1.5\times10^{-6})/5.96\times10^{7} = 0.11$ V/m.)

Problem 1.12 Thermal velocities

B · Guided calculation

From the course notes, Chapter 1

Compute v_{th} for an electron at 77 K and 300 K, and for a sodium ion Na^+ ($m = 3.82\times10^{-26}$ kg) at 300 K.

(Answer: Electron: $v_{th} = \sqrt{3kT/m} = 5.9\times10^{4}$ m/s at 77 K, 1.2×10^{5} m/s at 300 K. Na^+: $\sqrt{3\times1.381\times10^{-23}\times300/3.82\times10^{-26}} = 570$ m/s: the heavier the carrier, the slower its thermal motion.)

Problem 1.13 A diffusion current

B · Guided calculation

From the course notes, Chapter 1

In silicon, the electron concentration falls linearly from 10^{16} cm^{-3} to zero over 10 µm; $D_n = 35$ cm²/s. Find the electron flux and the current density, with their directions.

(Answer: $dn/dx = -10^{16}/10^{-3} = -10^{19}$ cm^{-4}. Flux $\Phi = -D\,dn/dx = 3.5\times10^{20}$ cm^{-2} s^{-1}, towards $+x$ (down the gradient). Current $J = q\Phi = -e\Phi = -56$ A/cm²: towards $-x$, opposite to the electron flow.)

Problem 1.14 Einstein

B · Guided calculation

From the course notes, Chapter 1

Holes in silicon have $\mu_p = 460$ cm²/(V·s). Find D_p at 300 K. Na^+ in water has $\mu = 5.19 \times 10^{-8}$ m²/(V·s); find its D (measured: 1.33×10^{-9} m²/s).

(Answer: $D_p = \mu_p kT/e = 460 \times 0.0259 = 11.9$ cm²/s. Na^+: $D = 5.19 \times 10^{-8} \times 0.0259 = 1.34 \times 10^{-9}$ m²/s, in agreement with the measurement: the Einstein relation also holds for ions in a liquid.)

Problem 1.15 Copper

B · Guided calculation

From the course notes, Chapter 1

Compute the plasma frequency ω_p of copper (with $\varepsilon = \varepsilon_0$), its period, $\omega_p \tau_c$ and $\tau_d = \varepsilon_0/\sigma$. Check the identity $\omega_p^2 \tau_c \tau_d = 1$. Which regime?

(Answer: $\omega_p = \sqrt{8.5 \times 10^{28} \times (1.602 \times 10^{-19})^2/(8.854 \times 10^{-12} \times 9.11 \times 10^{-31})} = 1.64 \times 10^{16}$ rad/s, period $2\pi/\omega_p = 0.38$ fs. $\omega_p \tau_c = 1.64 \times 10^{16} \times 2.5 \times 10^{-14} \approx 410$. $\tau_d = 8.854 \times 10^{-12}/5.96 \times 10^7 = 1.5 \times 10^{-19}$ s. $\omega_p^2 \tau_c \tau_d = 2.7 \times 10^{32} \times 2.5 \times 10^{-14} \times 1.5 \times 10^{-19} \approx 1.0$. Strongly underdamped: a plasma oscillation, gone in a few τ_c (~100 fs).)

Problem 1.16 Silicon at 10^{16} cm^{-3}

B · Guided calculation

From the course notes, Chapter 1

Electrons with $m^* = 0.26\, m_0$, $\mu = 1350$ cm²/(V·s), $\varepsilon_r = 11.7$. Find τ_c, σ, τ_d and $\omega_p \tau_c$. Which regime?

(Answer: $\tau_c = \mu m^*/e = 0.135 \times 2.37 \times 10^{-31}/1.602 \times 10^{-19} = 2.0 \times 10^{-13}$ s; $\sigma = ne\mu = 10^{22} \times 1.602 \times 10^{-19} \times 0.135 = 216$ S/m; $\tau_d = 11.7 \times 8.854 \times 10^{-12}/216 = 4.8 \times 10^{-13}$ s; $\omega_p \tau_c = \sqrt{\tau_c/\tau_d} = 0.65$: just above the critical value 1/2, a weakly damped oscillation.)

Problem 1.17 The slow root

B · Guided calculation

From the course notes, Chapter 1

For $\omega_p \tau_c \ll 1/2$, expand the roots of $r^2 + r/\tau_c + \omega_p^2 = 0$ to show that $r_1 \simeq -1/\tau_c$ and $r_2 \simeq -1/\tau_d$.

(Answer: $r = -\frac{1}{2\tau_c}\left[1 \mp \sqrt{1 - 4\omega_p^2 \tau_c^2}\right]$. With $\sqrt{1-u} \simeq 1 - u/2$ for $u = 4\omega_p^2 \tau_c^2 \ll 1$: the minus sign gives $r_2 \simeq -\frac{1}{2\tau_c} \cdot 2\omega_p^2 \tau_c^2 = -\omega_p^2 \tau_c = -1/\tau_d$ by the identity $\omega_p^2 \tau_c \tau_d = 1$; the plus sign gives $r_1 \simeq -1/\tau_c$.)

Problem 1.18 Dielectric relaxation

B · Guided calculation

From the course notes, Chapter 1

Compute τ_d for sea water ($\sigma = 5$ S/m, $\varepsilon_r = 80$) and for pure water ($\sigma = 5.5 \times 10^{-6}$ S/m). Is the local Ohm's law justified in these cases (ionic collision times are below 10^{-13} s)?

(Answer: Sea water: $\tau_d = 80 \times 8.85 \times 10^{-12}/5 = 1.4 \times 10^{-10}$ s. Pure water: $7.1 \times 10^{-10}/5.5 \times 10^{-6} = 1.3 \times 10^{-4}$ s. Both are much longer than the ionic collision times ($\omega_p \tau_c \ll 1/2$), so Ohm's law and the RC law (1.23) are valid there, unlike in a metal.)

Challenge: a new situation
These problems apply the method of the chapter to a device, a material or a situation that the chapter did not treat. The reasoning does not change: state the physical picture, choose the model, compute, check.

Problem 1.19 Drift and diffusion in gallium arsenide

C · Synthesis

Adapted from Neamen, Problem 5.37, p. 189

A uniform field $\mathcal{E} = 12$ V/cm along $+x$ exists in n-type GaAs between $x = 0$ and $x = 50$ µm, at 300 K. The total electron current density is constant, $J_n = 100$ A/cm², and at $x = 0$ the drift and diffusion currents are equal. Take $\mu_n = 8000$ cm²/(V·s).

(a) Find D_n, and write the equation $J_n = q\mu_n n\mathcal{E} + qD_n\,\mathrm{d}n/\mathrm{d}x$ as a first-order differential equation for $n(x)$.

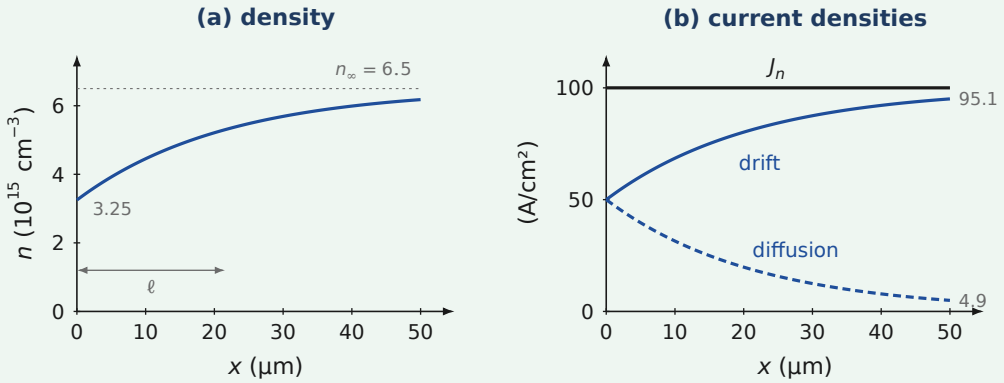
(b) Solve it with the condition at $x = 0$. What length characterises the variation of n, and what is its physical meaning?

(c) Compute $n(0)$ and $n(50\ \mu\mathrm{m})$, and the drift and diffusion currents at $x = 50$ µm.

HINT
The equation has a constant particular solution, for which the whole current is drift.

(Answer: (a) $D_n = 207$ cm²/s; (b) $n(x) = n_\infty(1 - \frac{1}{2}\mathrm{e}^{-x/\ell})$ with $n_\infty = 6.50 \times 10^{15}\,\mathrm{cm}^{-3}$ and $\ell = kT/(q\mathcal{E}) = 21.6$ µm; (c) 3.25×10^{15} and $6.18 \times 10^{15}\,\mathrm{cm}^{-3}$; 95.1 and 4.9 A/cm².)

SOLUTION
(a) $D_n = \mu_n kT/q = 8000 \times 0.0259 = 207$ cm²/s. Dividing by qD_n: $\mathrm{d}n/\mathrm{d}x + n/\ell = J_n/(qD_n)$, with $\ell = D_n/(\mu_n\mathcal{E}) = (kT/q)/\mathcal{E} = 0.0259/12 = 2.16 \times 10^{-3}$ cm $= 21.6$ µm.
(b) The constant solution $n_\infty = J_n/(q\mu_n\mathcal{E}) = 100/(1.602 \times 10^{-19} \times 8000 \times 12) = 6.50 \times 10^{15}\,\mathrm{cm}^{-3}$ carries the whole current by drift. The general solution is $n = n_\infty + A\,\mathrm{e}^{-x/\ell}$. At $x = 0$, drift = diffusion $= J_n/2$, so $q\mu_n n(0)\mathcal{E} = J_n/2$ and $n(0) = n_\infty/2$: $A = -n_\infty/2$, and $n(x) = n_\infty(1 - \frac{1}{2}\mathrm{e}^{-x/\ell})$. The length ℓ is the distance over which the potential energy $q\mathcal{E}\ell$ equals kT: over shorter distances diffusion competes with drift, over longer ones drift wins.
(c) $n(0) = 3.25 \times 10^{15}\,\mathrm{cm}^{-3}$; at 50 µm, $\mathrm{e}^{-50/21.6} = 0.099$ and $n = 6.50 \times 10^{15} \times (1 - 0.049) = 6.18 \times 10^{15}\,\mathrm{cm}^{-3}$. Drift: $q\mu_n n\mathcal{E} = 1.602 \times 10^{-19} \times 8000 \times 6.18 \times 10^{15} \times 12 = 95.1$ A/cm²; diffusion: $100 - 95.1 = 4.9$ A/cm². Check: $qD_n\,\mathrm{d}n/\mathrm{d}x = qD_n n_\infty \mathrm{e}^{-x/\ell}/(2\ell) = 4.9$ A/cm².



The density approaches n_∞ over the length $\ell = kT/q\mathcal{E}$; diffusion hands the current over to drift.

SEMICONDUCTOR DEVICES PHYSICS · SOLUTIONS

CHAPTER 2

Pure and Doped Semiconductors at Equilibrium: A Qualitative View

WHAT THIS SET USES

Bonds, electron–hole pairs and holes; generation and recombination, $R = \alpha\, np$ and $G_i = \alpha n_i^2$; equilibrium and the mass-action law $n_0 p_0 = n_i^2$; donors and acceptors; the three laws for the doped sample: neutrality $n_0 + N_A = p_0 + N_D$, mass action, complete ionisation; majority and minority carriers; the conductivity $\sigma = q(n\mu_n + p\mu_p)$ of Chapter 1. The Fermi level is not used in this set.

DATA

Silicon at 300 K: $n_i = 10^{10}\,\mathrm{cm}^{-3}$, $\mu_n = 1350$, $\mu_p = 460$ cm²/(V·s) unless a problem gives other values. Germanium: $n_i = 2.4 \times 10^{13}\,\mathrm{cm}^{-3}$; GaAs: $n_i = 1.8 \times 10^{6}\,\mathrm{cm}^{-3}$.

Problem 2.1 Which atom gives which carrier? A · Understanding

Adapted from Sze & Lee, Ch. 1, Problem 6, p. 41

A dopant is a donor or an acceptor according to the number of valence electrons it brings compared with the atom it replaces.

(a) Silicon (column IV) is doped with phosphorus (column V), then with boron (column III). Draw the bonds around each impurity at 0 K and say which one is a donor.

(b) Gallium arsenide is a compound of gallium (column III) and arsenic (column V). Tin (column IV) is added and replaces gallium atoms. Is tin a donor or an acceptor? Is the material n- or p-type?

(c) Silicon atoms can also be added to GaAs. Explain why a silicon atom on a gallium site is a donor, and a silicon atom on an arsenic site an acceptor. (Problem 2.16 uses this.)

HINT

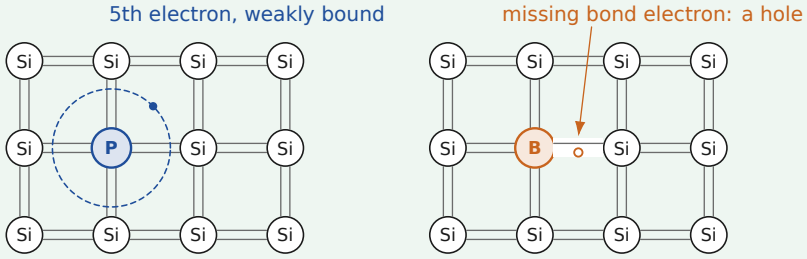
Count the electrons available for the four bonds of the site, and compare with the eight shared electrons of a complete set of bonds.

(Answer: (a) P donor, B acceptor; (b) donor, n-type; (c) one electron in excess on a Ga site, one missing on an As site.)

SOLUTION

(a) Phosphorus has five valence electrons. Four of them complete the four bonds with the neighbouring silicon atoms; the fifth is bound to the impurity only by a weak attraction (the hydrogen-like orbit of the chapter) and is freed at 300 K: phosphorus is a **donor**. Boron has three: one bond lacks an electron, and an electron from a neighbouring bond can fill it at 300 K, which leaves a hole that moves through the crystal: boron is an **acceptor**.

(b) On a gallium site, the atom must provide three electrons to the bonds with its four arsenic neighbours, just as gallium does. Tin has four: one is in excess, and is freed as in the case of phosphorus in silicon. Tin on a gallium site is a donor, and the material is n-type.
(c) The same counting: silicon (four electrons) on a gallium site (three needed) brings one electron too many, a donor; on an arsenic site (five needed) it brings one too few, an acceptor. The same atom can be either, depending on its site: such an impurity is called amphoteric.



A flattened picture of the silicon bonds at 0 K around a donor and an acceptor.

Problem 2.2 Compensated samples

B · Guided calculation

Adapted from Neamen, Problems 4.34 and 4.39, pp. 151–152

Silicon at 300 K, complete ionisation.

(a) Write the three laws for a sample containing N_D donors and N_A acceptors, and show that, when $N_D > N_A$, $n_0 = \frac{N_D - N_A}{2} + \sqrt{\left(\frac{N_D - N_A}{2}\right)^2 + n_i^2}$.

(b) Find n_0 and p_0 for: (i) $N_D = 10^{15}$, $N_A = 4 \times 10^{15}\,\mathrm{cm^{-3}}$; (ii) $N_D = 3 \times 10^{16}\,\mathrm{cm^{-3}}$, $N_A = 0$; (iii) $N_D = N_A = 2 \times 10^{15}\,\mathrm{cm^{-3}}$. Say in each case whether the sample is n-type, p-type or behaves as pure silicon.

(c) A sample contains 2×10^{15} arsenic atoms and 1.2×10^{15} boron atoms per cm³. Find n_0 and p_0.

(d) Boron is added to the sample of (c) until $p_0 = 4 \times 10^{15}\,\mathrm{cm^{-3}}$. How many boron atoms per cm³ must be added? What is the new n_0?

HINT

When $|N_D - N_A| \gg n_i$ the square root reduces to $|N_D - N_A|/2$: the majority density is the net doping, and the minority density follows from mass action.

(Answer: (b) (i) p-type, $p_0 = 3 \times 10^{15}$, $n_0 = 3.3 \times 10^4\,\mathrm{cm^{-3}}$; (ii) n-type, $n_0 = 3 \times 10^{16}$, $p_0 = 3.3 \times 10^3\,\mathrm{cm^{-3}}$; (iii) $n_0 = p_0 = 10^{10}\,\mathrm{cm^{-3}}$; (c) $n_0 = 8 \times 10^{14}$, $p_0 = 1.25 \times 10^5\,\mathrm{cm^{-3}}$; (d) $4.8 \times 10^{15}\,\mathrm{cm^{-3}}$, $n_0 = 2.5 \times 10^4\,\mathrm{cm^{-3}}$.)

SOLUTION

(a) Neutrality: $n_0 + N_A = p_0 + N_D$ (all dopants ionised). Mass action: $p_0 = n_i^2/n_0$. Substituting, $n_0 - n_i^2/n_0 = N_D - N_A$, that is $n_0^2 - (N_D - N_A)n_0 - n_i^2 = 0$, whose positive root is the formula given. For $N_A > N_D$ the same steps give p_0 with $N_A - N_D$ in place of $N_D - N_A$.
(b) (i) $N_A - N_D = 3\times10^{15} \gg n_i$: p-type, $p_0 = 3\times10^{15}\,\mathrm{cm^{-3}}$ and $n_0 = 10^{20}/3\times10^{15} = 3.3\times10^{4}\,\mathrm{cm^{-3}}$. (ii) n-type, $n_0 = 3\times10^{16}\,\mathrm{cm^{-3}}$, $p_0 = 3.3\times10^{3}\,\mathrm{cm^{-3}}$. (iii) $N_D - N_A = 0$ and the formula gives $n_0 = p_0 = n_i = 10^{10}\,\mathrm{cm^{-3}}$: the donors give their electrons to the acceptors, and the carriers are those of pure silicon. A perfectly compensated sample is not pure, however: its ionised impurities still scatter the carriers and lower the mobility.
(c) $N_D - N_A = 8\times10^{14}\,\mathrm{cm^{-3}}$: $n_0 = 8\times10^{14}\,\mathrm{cm^{-3}}$ and $p_0 = 10^{20}/8\times10^{14} = 1.25\times10^{5}\,\mathrm{cm^{-3}}$; n-type.
(d) Now $p_0 = 4\times10^{15} \gg n_i$, so $N_A - N_D = 4\times10^{15}$ and $N_A = 6\times10^{15}\,\mathrm{cm^{-3}}$. The sample already has 1.2×10^{15}: 4.8×10^{15} boron atoms per cm³ must be added. Then $n_0 = 10^{20}/4\times10^{15} = 2.5\times10^{4}\,\mathrm{cm^{-3}}$. The type of a sample is set by the net doping, and each added acceptor first neutralises a donor.

Problem 2.3 When n_i is not negligible — B · Guided calculation

Adapted from Sze & Lee, Ch. 1, Problem 18, p. 42, and Neamen, Problem 4.45, p. 152

(a) A germanium sample is doped with $N_D = 2.5\times10^{13}\,\mathrm{cm^{-3}}$ donors, fully ionised, at 300 K, where n_i of germanium is taken as $2.5\times10^{13}\,\mathrm{cm^{-3}}$. Why can the majority-carrier approximation $n_0 \simeq N_D$ not be used? Compute n_0 and p_0, and the error that the approximation would make on n_0.

(b) In another material, doped with $N_D = 2\times10^{14}$ and $N_A = 1.2\times10^{14}\,\mathrm{cm^{-3}}$, the electron density is measured: $n_0 = 1.1\times10^{14}\,\mathrm{cm^{-3}}$. Using the neutrality and mass-action laws, find n_i and p_0.

(c) In (b), check your result by verifying neutrality with the computed values.

HINT

In (b) eliminate p_0 between the two laws: $n_i^2 = n_0^2 - n_0(N_D - N_A)$.

(Answer: (a) $n_0 = 4.05\times10^{13}$, $p_0 = 1.55\times10^{13}\,\mathrm{cm^{-3}}$; the approximation underestimates n_0 by 38 %; (b) $n_i = 5.7\times10^{13}\,\mathrm{cm^{-3}}$, $p_0 = 3.0\times10^{13}\,\mathrm{cm^{-3}}$.)

SOLUTION

(a) The approximation requires $N_D \gg n_i$; here they are equal, so the thermally created pairs are as numerous as the electrons given by the donors. With $(N_D/2) = 1.25\times10^{13}$: $n_0 = 1.25\times10^{13} + \sqrt{(1.25\times10^{13})^2 + (2.5\times10^{13})^2} = 1.25\times10^{13} + 2.80\times10^{13} = 4.05\times10^{13}\,\mathrm{cm^{-3}}$, and $p_0 = n_i^2/n_0 = 6.25\times10^{26}/4.05\times10^{13} = 1.55\times10^{13}\,\mathrm{cm^{-3}}$. The approximation $n_0 = 2.5\times10^{13}$ would be 38 % too low, and would give $p_0 = n_i$, which ignores the reduction of the minority carriers. Check: $n_0 - p_0 = 2.5\times10^{13} = N_D$.

(b) With $p_0 = n_i^2/n_0$, neutrality $n_0 = p_0 + N_D - N_A$ becomes $n_i^2 = n_0^2 - n_0(N_D - N_A) = (1.1\times10^{14})^2 - 1.1\times10^{14}\times0.8\times10^{14} = 1.21\times10^{28} - 0.88\times10^{28} = 3.3\times10^{27}$, so $n_i = 5.7\times10^{13}\,\text{cm}^{-3}$ and $p_0 = 3.3\times10^{27}/1.1\times10^{14} = 3.0\times10^{13}\,\text{cm}^{-3}$.
(c) $n_0 + N_A = 1.1\times10^{14} + 1.2\times10^{14} = 2.3\times10^{14}$ and $p_0 + N_D = 0.3\times10^{14} + 2\times10^{14} = 2.3\times10^{14}$: neutrality holds.

Problem 2.4 Designing a resistor

B · Guided calculation

Adapted from Neamen, Problem 5.51, p. 190

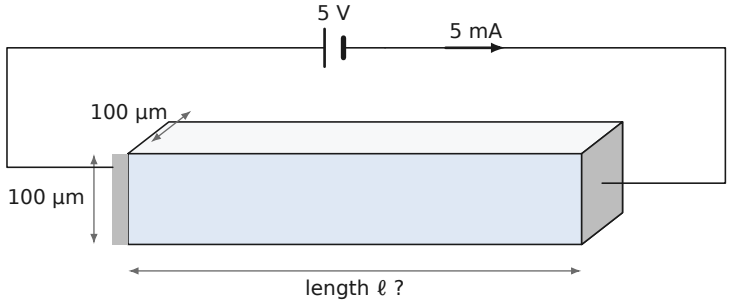
A silicon resistor must carry 5 mA under 5 V. Its cross-section is a square of 100 µm × 100 µm.

(a) What resistance is required?

(b) The silicon is n-type with $N_D = 3\times10^{14}\,\text{cm}^{-3}$ and $N_A = 0$ ($\mu_n = 1350$ cm²/(V·s)). Compute the conductivity, neglecting the holes, and the length of the resistor. Check that neglecting the holes is justified.

(c) The resistor is now made of compensated silicon, $N_D = 3\times10^{16}$ and $N_A = 2.5\times10^{16}\,\text{cm}^{-3}$, in which the electron mobility, lowered by the many ionised impurities, is measured as 1000 cm²/(V·s). Compute n_0, the conductivity and the new length.

(d) The electron density is multiplied by 16.7 between (b) and (c). Is the length multiplied by the same factor? Explain.



The bar resistor (not to scale).

HINT

$R = L/(\sigma A)$, so $L = R\sigma A$.

(Answer: (a) 1 kΩ; (b) $\sigma = 0.0649$ S/cm, $L = 64.9$ µm; (c) $n_0 = 5\times10^{15}\,\text{cm}^{-3}$, $\sigma = 0.801$ S/cm, $L = 801$ µm; (d) no, 12.3, because of the lower mobility.)

SOLUTION

(a) $R = 5/5\times10^{-3} = 1$ kΩ.
(b) $n_0 = N_D = 3\times10^{14}\,\text{cm}^{-3}$ and $\sigma = qn_0\mu_n = 1.602\times10^{-19}\times3\times10^{14}\times1350 = 0.0649$ S/cm. With $A = 10^{-4}$ cm², $L = R\sigma A = 10^3\times0.0649\times10^{-4} = 6.49\times10^{-3}$ cm = 64.9 µm. The holes: $p_0 = 10^{20}/3\times10^{14} = 3.3\times10^5\,\text{cm}^{-3}$; their conductivity is 10^9 times smaller.

(c) $n_0 = N_D - N_A = 5 \times 10^{15}\,\text{cm}^{-3}$ (minority holes $2 \times 10^4\,\text{cm}^{-3}$, negligible). $\sigma = 1.602 \times 10^{-19} \times 5 \times 10^{15} \times 1000 = 0.801$ S/cm and $L = 10^3 \times 0.801 \times 10^{-4} = 8.01 \times 10^{-2}$ cm = 801 µm.
(d) The length is multiplied by 801/64.9 = 12.3, not 16.7: $L \propto \sigma = qn_0\mu_n$, and the mobility fell from 1350 to 1000 cm²/(V·s). The compensating impurities do not contribute carriers but they do scatter them. The relation between doping and resistor length is therefore not linear.

Problem 2.5 Is pure silicon the most resistive? B · Guided calculation

Adapted from Neamen, Problem 5.22, p. 187

The conductivity of silicon at 300 K is $\sigma = q(n_0\mu_n + p_0\mu_p)$, with $n_0 p_0 = n_i^2$. Consider it as a function of p_0.

(a) Show that σ is smallest for $p_0 = n_i\sqrt{\mu_n/\mu_p}$, and that the minimum is $\sigma_{min} = 2qn_i\sqrt{\mu_n\mu_p}$.

(b) Show that $\sigma_{min}/\sigma_i = 2\sqrt{\mu_n\mu_p}/(\mu_n + \mu_p)$, where σ_i is the conductivity of pure silicon, and evaluate the ratio. Compute ρ_i and ρ_{max}.

(c) Is the material at the minimum n- or p-type? What net doping $N_A - N_D$ gives it?

(d) Why does a small number of acceptors *increase* the resistivity of pure silicon?

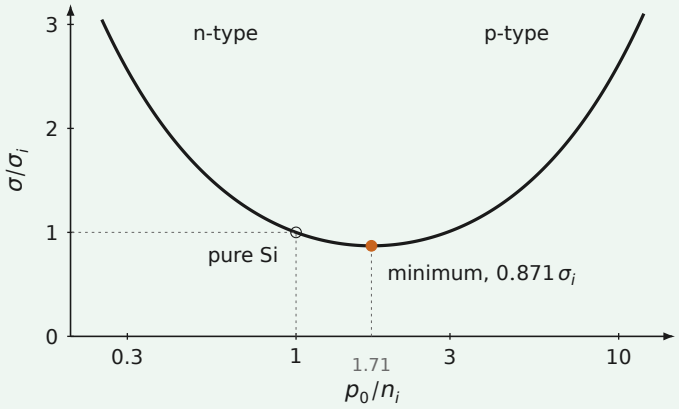
HINT

Write $\sigma(p_0) = q(\mu_n n_i^2/p_0 + \mu_p p_0)$ and set $\mathrm{d}\sigma/\mathrm{d}p_0 = 0$.

(Answer: (a) $p_0 = 1.71 \times 10^{10}\,\text{cm}^{-3}$; (b) 0.871; $\rho_i = 3.45 \times 10^5$ Ω·cm, $\rho_{max} = 3.96 \times 10^5$ Ω·cm; (c) p-type, $N_A - N_D = 1.13 \times 10^{10}\,\text{cm}^{-3}$.)

SOLUTION

(a) $\mathrm{d}\sigma/\mathrm{d}p_0 = q(-\mu_n n_i^2/p_0^2 + \mu_p) = 0$ gives $p_0^2 = n_i^2\mu_n/\mu_p$, so $p_0 = n_i\sqrt{\mu_n/\mu_p}$ and $n_0 = n_i^2/p_0 = n_i\sqrt{\mu_p/\mu_n}$. The second derivative is positive: it is a minimum. Then $\sigma_{min} = q(\mu_n n_i\sqrt{\mu_p/\mu_n} + \mu_p n_i\sqrt{\mu_n/\mu_p}) = 2qn_i\sqrt{\mu_n\mu_p}$. With $\sqrt{1350/460} = 1.713$, $p_0 = 1.71 \times 10^{10}\,\text{cm}^{-3}$.
(b) $\sigma_i = qn_i(\mu_n + \mu_p)$, hence the ratio $2\sqrt{\mu_n\mu_p}/(\mu_n + \mu_p) = 2 \times 788/1810 = 0.871$. $\sigma_i = 1.602 \times 10^{-19} \times 10^{10} \times 1810 = 2.90 \times 10^{-6}$ S/cm, $\rho_i = 3.45 \times 10^5$ Ω·cm, and $\rho_{max} = \rho_i/0.871 = 3.96 \times 10^5$ Ω·cm.
(c) $p_0 > n_0$: slightly p-type. Neutrality gives $N_A - N_D = p_0 - n_0 = n_i(1.713 - 0.584) = 1.13 \times 10^{10}\,\text{cm}^{-3}$.
(d) Adding acceptors replaces fast electrons by slow holes: mass action removes electrons ($\mu_n = 1350$) while adding holes ($\mu_p = 460$). As long as the electrons still carry a large share of the current, the loss outweighs the gain. Beyond the minimum, the holes become numerous enough for the conductivity to rise again.



The conductivity is smallest in slightly p-type material, because holes are less mobile than electrons.

Problem 2.6 Halving a resistance by counter-doping

B · Guided calculation

Adapted from Sze & Lee, Ch. 2, Problem 9, p. 79

A silicon bar doped with $N_D \gg n_i$ donors has a resistance R_1. Acceptors are then added to the same bar, $N_A > N_D$, and its resistance becomes $R_1/2$. Take the mobilities constant, $\mu_n/\mu_p = 1350/460$.

(a) Write the conductivity before and after the addition of the acceptors.

(b) Deduce N_A in terms of N_D.

(c) Why must N_A be several times larger than N_D to halve the resistance, and not simply about $2N_D$?

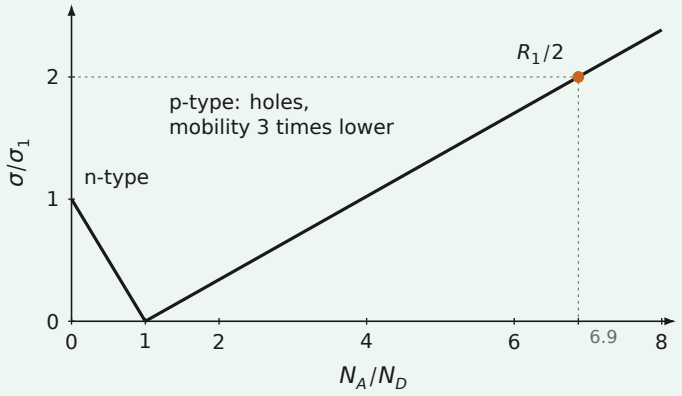
HINT

After the addition the majority carriers are holes, and their density is the net doping $N_A - N_D$.

(Answer: (b) $N_A = (1 + 2\mu_n/\mu_p)N_D = 6.9\,N_D$.)

SOLUTION

(a) Before: $\sigma_1 = qN_D\mu_n$. After: holes are the majority carriers, $p_0 = N_A - N_D$, and $\sigma_2 = q(N_A - N_D)\mu_p$ (minority electrons negligible).
(b) $R_2 = R_1/2$ means $\sigma_2 = 2\sigma_1$: $(N_A - N_D)\mu_p = 2N_D\mu_n$, so $N_A = N_D(1 + 2\mu_n/\mu_p) = N_D(1 + 5.87) = 6.9\,N_D$.
(c) Two effects. First, N_D acceptors only cancel the donors: the net doping is $N_A - N_D$. Second, the new carriers are holes, 2.9 times less mobile than the electrons they replace; twice the conductivity therefore needs $2 \times 2.9 = 5.9$ times more holes than there were electrons. In practice the mobility would also fall with the total number of ionised impurities, $N_A + N_D$, and even more acceptors would be needed.



Counter-doping: the conductivity first falls to almost zero at compensation, then rises slowly, carried by the less mobile holes.

Problem 2.7 The generic recombination model in numbers B · Guided calculation

Adapted from Neamen, Problems 6.1 and 6.3, p. 233

In the generic model of the chapter, the recombination rate is $R = \alpha np$ and thermal generation is $G_i = \alpha n_i^2$. In an n-type sample, an individual hole disappears at the rate αn_0, so its mean lifetime is $\tau_p = 1/(\alpha n_0)$.

(a) A silicon sample has $N_D = 10^{16}\,\text{cm}^{-3}$ and a measured hole lifetime $\tau_p = 20$ µs. Find α, the thermal generation rate G_i and the equilibrium recombination rate R_0.

(b) What is the mean time that an electron of this sample waits before recombining? Comment on the difference between the two lifetimes.

(c) Another sample has $N_D = 5\times10^{15}\,\text{cm}^{-3}$ and $\tau_p = 0.2$ µs. Find α and R_0. Why is R_0 the same as G_i in each sample, whatever τ_p?

HINT

At equilibrium $R_0 = \alpha n_0 p_0 = \alpha n_i^2$.

(Answer: (a) $\alpha = 5\times10^{-12}$ cm³/s, $G_i = R_0 = 5\times10^{8}$ cm^{-3}s^{-1}; (b) 2×10^{7} s; (c) $\alpha = 10^{-9}$ cm³/s, $R_0 = 10^{11}$ cm^{-3}s^{-1}.)

SOLUTION

(a) $\alpha = 1/(\tau_p n_0) = 1/(2\times10^{-5}\times10^{16}) = 5\times10^{-12}$ cm³/s. $G_i = \alpha n_i^2 = 5\times10^{-12}\times10^{20} = 5\times10^{8}$ pairs/(cm³·s). At equilibrium $R_0 = \alpha n_0 p_0 = \alpha n_i^2 = G_i$.
(b) An electron recombines with a hole; it meets one at the rate $\alpha p_0 = 5\times10^{-12}\times10^{4} = 5\times10^{-8}$ s^{-1}, so it waits $1/(\alpha p_0) = 2\times10^{7}$ s, about eight months. The two lifetimes differ by $n_0/p_0 = 10^{12}$: a hole is surrounded by electrons and is soon captured; an electron finds a hole only rarely. Yet the numbers of recombinations per second are the same for both carriers, because each recombination removes one of each: $n_0/\tau_n = p_0/\tau_p = R_0$.
(c) $\alpha = 1/(2\times10^{-7}\times5\times10^{15}) = 10^{-9}$ cm³/s and $R_0 = G_i = \alpha n_i^2 = 10^{11}$ cm^{-3}s^{-1}. In equilibrium, generation and recombination balance by definition; the value of α,

which depends on the impurities and defects of the sample, fixes how fast pairs are created and destroyed, but not their balance. A shorter lifetime means a faster exchange, not a different equilibrium.

Practice exercises from the course notes
These exercises closed Chapter 2 of the course notes; they now live here, with their answers. Tables, equations and sections cited by number are those of the course notes. Use the data at the start of the chapter, the reference value $n_i = 10^{10}\,\text{cm}^{-3}$ for silicon, and $q = 1.6\times10^{-19}$ C, unless stated otherwise. All situations are at 300 K and at equilibrium.

Problem 2.8 Broken bonds — B · Guided calculation

From the course notes, Chapter 2

Germanium has 4.4×10^{22} atoms per cm³. How many bonds per cm³? What fraction of them is broken at 300 K ($n_i = 2.4\times10^{13}\,\text{cm}^{-3}$)? Compare with silicon.

(Answer: Bonds: $4.4\times10^{22}\times4/2 = 8.8\times10^{22}\,\text{cm}^{-3}$. Fraction broken: $2.4\times10^{13}/8.8\times10^{22} = 2.7\times10^{-10}$, which is 2700 times larger than in silicon ($10^{10}/10^{23} = 10^{-13}$): the gap is smaller.)

Problem 2.9 Intrinsic resistivities — B · Guided calculation

From the course notes, Chapter 2

Compute the resistivity of pure germanium, silicon and gallium arsenide with Tables 2.3 (reference value for Si) and 2.4. In each case, what fraction of the current do the electrons carry?

(Answer: Ge: $\rho = 1/(1.6\times10^{-19}\times2.4\times10^{13}\times5800) \approx 45$ Ω·cm, electrons 3900/5800 = 67%. Si: $\rho \approx 3.4\times10^{5}$ Ω·cm, electrons 75%. GaAs: $\rho = 1/(1.6\times10^{-19}\times1.8\times10^{6}\times8900) \approx 3.9\times10^{8}$ Ω·cm, electrons 8500/8900 = 96%.)

Problem 2.10 The generic balance — B · Guided calculation

From the course notes, Chapter 2

In silicon take $\alpha = 10^{-11}$ cm³/s (an effective value; Chapter 4 discusses it). (i) Find the thermal generation rate G_i. (ii) A sample doped with $N_D = 10^{16}\,\text{cm}^{-3}$ has $n_0 = 10^{16}\,\text{cm}^{-3}$: under the assumptions of Section 2.11, find p_0 and the recombination rate R_0. (iii) Compute the lifetime $1/(\alpha p_0)$ of an individual electron and $1/(\alpha n_0)$ of an individual hole in this sample, and comment.

(Answer: (i) $G_i = \alpha n_i^2 = 10^{-11}\times10^{20} = 10^{9}\,\text{cm}^{-3}\,\text{s}^{-1}$. (ii) $p_0 = n_i^2/n_0 = 10^{4}\,\text{cm}^{-3}$ and $R_0 = \alpha n_0 p_0 = 10^{9}\,\text{cm}^{-3}\,\text{s}^{-1} = G_i$: the same rate as in the pure material. (iii) Electron: $1/(10^{-11}\times10^{4}) = 10^{7}$ s, about four months; hole: $1/(10^{-11}\times10^{16}) = 10^{-5}$ s = 10 µs. A majority electron waits a very long time for a rare hole; a minority hole is surrounded by electrons and disappears quickly. Chapter 4 shows that the second time governs the return to equilibrium.)

Problem 2.11 The hydrogen-like model elsewhere

B · Guided calculation

From the course notes, Chapter 2

Estimate the ionisation energy and the orbit radius of a donor in germanium ($\varepsilon_r = 16$, $m^* = 0.12\,m_0$) and in gallium arsenide ($\varepsilon_r = 13.1$, $m^* = 0.067\,m_0$). Compare with kT at 77 K and at 300 K. Measurement gives about 12 meV for phosphorus in germanium: comment.

(Answer: Ge: $E_d \approx 13.6 \times 0.12/256 = 6.4\,\text{meV}$, $a_d \approx 0.053 \times 16/0.12 = 7.1$ nm. GaAs: $E_d \approx 13.6 \times 0.067/171.6 = 5.3\,\text{meV}$, $a_d \approx 10$ nm. Both are comparable to kT at 77 K (6.6 meV) and four to five times smaller than at 300 K: donors stay ionised down to very low temperatures. The 12 meV measured in Ge: the model still gives the order of magnitude, the core of the impurity does the rest.)

Problem 2.12 p-type silicon

B · Guided calculation

From the course notes, Chapter 2

Silicon is doped with $N_A = 2 \times 10^{16}\,\text{cm}^{-3}$ (boron). How many silicon atoms per boron atom? Find p_0, n_0 and the resistivity. Is the majority approximation justified?

(Answer: $5\times10^{22}/2\times10^{16} = 2.5\times10^6$ Si atoms per boron atom. $p_0 \simeq 2\times10^{16}\,\text{cm}^{-3}$, $n_0 = 10^{20}/2\times10^{16} = 5\times10^3\,\text{cm}^{-3}$, $\rho = 1/(1.6\times10^{-19}\times2\times10^{16}\times460) = 0.68\ \Omega\cdot\text{cm}$. $N_A = 2\times10^6\,n_i \gg 10\,n_i$: the approximation is amply justified.)

Problem 2.13 Lightly doped

B · Guided calculation

From the course notes, Chapter 2

An n-type sample has $N_D = 1.5 \times 10^{10}\,\text{cm}^{-3}$. Why can the majority approximation not be used? Find n_0 and p_0 with the exact formula, and the error the approximation would have made.

(Answer: $N_D = 1.5\,n_i$ is not $\geq 10\,n_i$. Exact: $n_0 = 0.75 \times 10^{10} + \sqrt{0.5625\times10^{20} + 10^{20}} = 2.0 \times 10^{10}\,\text{cm}^{-3}$, $p_0 = 10^{20}/2\times10^{10} = 5\times10^9\,\text{cm}^{-3}$. Check: $n_0 - p_0 = 1.5\times10^{10} = N_D$. The approximation would have given $n_0 = 1.5\times10^{10}$, an error of 25%.)

Problem 2.14 Compensation

B · Guided calculation

From the course notes, Chapter 2

An n-type silicon wafer ($N_D = 3\times10^{16}\,\text{cm}^{-3}$) receives $N_A = 2\times10^{16}\,\text{cm}^{-3}$ boron atoms. (i) Find n_0, p_0 and the resistivity, keeping the mobility $\mu_n = 1350$ cm²/(V·s). (ii) What happens to n_0 and p_0 if exactly $N_A = 3\times10^{16}\,\text{cm}^{-3}$ is added? (iii) Which assumption on the mobility is questionable?

(Answer: (i) $n_0 \simeq N_D - N_A = 10^{16}\,\text{cm}^{-3}$, $p_0 = 10^4\,\text{cm}^{-3}$, $\rho = 1/(1.6\times10^{-19}\times10^{16}\times1350) = 0.46\ \Omega\cdot\text{cm}$. (ii) $N_D = N_A$: neutrality gives $n_0 = p_0$, so $n_0 = p_0 = n_i$: the sample behaves as pure silicon (perfect compensation, impossible to achieve in practice). (iii) The 5×10^{16} dopant ions per cm³ scatter the carriers: the real mobility is lower than in the pure crystal, and the resistivity higher (outside the model).)

Problem 2.15 Purity requirement

B · Guided calculation

From the course notes, Chapter 2

We want 'pure' silicon to have, at 300 K, an electron concentration that exceeds n_i by at most 1%. What concentration of residual donors can be tolerated? To how many silicon atoms per foreign atom does this correspond? Comment.

(Answer: For $N_D \ll n_i$, $n_0 \approx n_i + N_D/2$; $n_0 \leq 1.01\, n_i$ requires $N_D \leq 0.02\, n_i = 2 \times 10^8\,\text{cm}^{-3}$, that is one foreign atom per 2.5×10^{14} silicon atoms. This is beyond what the best purification achieves (residual levels of a few 10^{11} to $10^{12}\,\text{cm}^{-3}$): 'intrinsic' silicon at 300 K is an ideal, and the resistivity of 3.4×10^5 Ω·cm is not measured in practice.)

Challenge: a new situation
These problems apply the method of the chapter to a device, a material or a situation that the chapter did not treat. The reasoning does not change: state the physical picture, choose the model, compute, check.

Problem 2.16 An impurity that is both donor and acceptor

C · Synthesis

Adapted from Neamen, Problem 4.62, p. 154

Silicon is added to gallium arsenide at a concentration of $7 \times 10^{15}\,\text{cm}^{-3}$. Five percent of the silicon atoms replace gallium atoms and 95 percent replace arsenic atoms (Problem 2.1). All are ionised. GaAs at 300 K: $n_i = 1.8 \times 10^6\,\text{cm}^{-3}$.

(a) Find N_D and N_A. Is the material n- or p-type?

(b) Find p_0 and n_0.

(c) The value of n_0 is far below one electron per cm³. What does such a density mean physically, and is the mass-action law still meaningful?

(d) Compare with silicon doped with the same net doping. Why is the minority density so much smaller in GaAs?

HINT
A density is an average over volume and time.

(Answer: (a) $N_D = 3.5 \times 10^{14}$, $N_A = 6.65 \times 10^{15}\,\text{cm}^{-3}$, p-type; (b) $p_0 = 6.3 \times 10^{15}\,\text{cm}^{-3}$, $n_0 = 5.1 \times 10^{-4}\,\text{cm}^{-3}$; (d) silicon: $n_0 = 1.6 \times 10^4\,\text{cm}^{-3}$.)

SOLUTION
(a) $N_D = 0.05 \times 7 \times 10^{15} = 3.5 \times 10^{14}\,\text{cm}^{-3}$ (Si on Ga sites), $N_A = 0.95 \times 7 \times 10^{15} = 6.65 \times 10^{15}\,\text{cm}^{-3}$ (Si on As sites). $N_A > N_D$: p-type.
(b) $p_0 = N_A - N_D = 6.3 \times 10^{15}\,\text{cm}^{-3}$, and $n_0 = n_i^2/p_0 = (1.8 \times 10^6)^2/6.3 \times 10^{15} = 5.1 \times 10^{-4}\,\text{cm}^{-3}$.
(c) It means that a cubic centimetre contains one electron in the conduction band only 5×10^{-4} of the time on average, or that one electron is present, on average, in a volume of about 2000 cm³. The mass-action law still holds as a statement about

averages: in a large sample observed over a long time, the rate of generation αn_i^2 is balanced by the recombination $\alpha n_0 p_0$. In a device it means that the minority electrons play no role in this region unless they are injected.
(d) For silicon, $n_0 = 10^{20}/6.3\times10^{15} = 1.6\times10^{4}\,\mathrm{cm}^{-3}$. The ratio is $(n_{i,\mathrm{Si}}/n_{i,\mathrm{GaAs}})^2 = (10^{10}/1.8\times10^{6})^2 = 3\times10^{7}$: the larger gap of GaAs makes thermal pairs much rarer, and minority densities scale as n_i^2.

SEMICONDUCTOR DEVICES PHYSICS · SOLUTIONS

CHAPTER 3

Pure and Doped Semiconductors at Equilibrium: Energy Bands and the Fermi Level

WHAT THIS SET USES

Bands and gap; the Fermi–Dirac function and its Boltzmann limit; effective mass and density of states $g_C(E) \propto \sqrt{E-E_C}$; $n_0 = N_C\,\mathrm{e}^{-(E_C-E_F)/kT}$, $p_0 = N_V\,\mathrm{e}^{-(E_F-E_V)/kT}$ with $N_{C,V} = 2(2\pi m^* kT/h^2)^{3/2}$; $n_i = \sqrt{N_C N_V}\,\mathrm{e}^{-E_g/2kT}$ and the intrinsic level E_i; $n_0 = n_i\,\mathrm{e}^{(E_F-E_i)/kT}$; ionised donors $N_D^+ = N_D/[1+2\,\mathrm{e}^{(E_F-E_D)/kT}]$; the position of E_F in doped material.

DATA

Silicon: $E_g = 1.12$ eV (held constant with T), $N_C = 2.8\times10^{19}$, $N_V = 1.04\times10^{19}\,\mathrm{cm}^{-3}$ at 300 K, both $\propto T^{3/2}$; $n_i = 10^{10}\,\mathrm{cm}^{-3}$ at 300 K and $n_i(T) = 10^{10}(T/300)^{3/2}\exp[-\frac{E_g}{2k}(\frac{1}{T}-\frac{1}{300})]\,\mathrm{cm}^{-3}$; $k = 8.617\times10^{-5}$ eV/K; donor and acceptor levels 45 meV from their band edge.

Problem 3.1 Where are the conduction electrons? B · Guided calculation

Adapted from Neamen, Problems 4.6 and 4.7, p. 150

In the Boltzmann limit, the number of electrons per unit energy in the conduction band is $n(E) = g_C(E)f(E) \propto \sqrt{E-E_C}\,\mathrm{e}^{-(E-E_F)/kT}$.

(a) Sketch $\sqrt{E-E_C}$, $\mathrm{e}^{-(E-E_F)/kT}$ and their product above E_C. Explain why the product vanishes at E_C and at high energy.

(b) Show that $n(E)$ is largest at $E = E_C + kT/2$. Give this energy at 300 K.

(c) Compute the ratio $n(E_C+4kT)/n(E_C+kT/2)$. What does it say about the energies of the conduction electrons?

(d) Without calculation, where is the maximum of the hole distribution $g_V(E)[1-f(E)]$?

HINT

Set $u = (E-E_C)/kT$ and maximise $\sqrt{u}\,\mathrm{e}^{-u}$.

(Answer: (b) 13 meV above E_C; (c) 0.085; (d) $kT/2$ below E_V.)

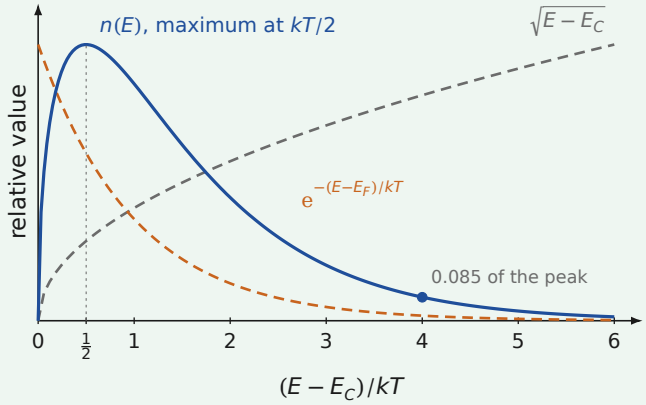
SOLUTION

(a) The density of states starts at zero at E_C and grows as $\sqrt{E-E_C}$; the occupation decreases exponentially. Near E_C there are few states; far above, the states are almost empty. The product is a skewed bell, zero at E_C, with a long exponential tail.

(b) With $u = (E - E_C)/kT$, $n \propto \sqrt{u}\,\mathrm{e}^{-u}$ (the factor $\mathrm{e}^{-(E_C - E_F)/kT}$ is a constant). Its derivative, $\mathrm{e}^{-u}(1/(2\sqrt{u}) - \sqrt{u})$, vanishes at $u = 1/2$: $E = E_C + kT/2 = E_C + 13$ meV at 300 K.

(c) $\dfrac{n(E_C + 4kT)}{n(E_C + kT/2)} = \sqrt{\dfrac{4}{0.5}}\,\mathrm{e}^{-(4-0.5)} = 2.83 \times 0.0302 = 0.085$.

(d) By symmetry of the argument, the hole distribution peaks at $kT/2$ below E_V. The conduction electrons sit within a few kT of the band edge: a few hundredths of an electronvolt, compared with a gap of 1.12 eV. This is why a band can be described by its edge and an effective mass, and why only the edges appear in the formulas for n_0 and p_0.



The number of electrons per unit energy is the product of a rising density of states and a falling occupation.

Problem 3.2 Effective masses from the densities of states B · Guided calculation

Adapted from Sze & Lee, Ch. 1, Problem 9, p. 41

The effective density of states is $N_V = 2\left(\dfrac{2\pi m_p^* kT}{h^2}\right)^{3/2}$, and similarly for N_C with m_n^*.

(a) Explain why N_V can be read as 'the number of states per cm³ that would be needed at the single energy E_V to hold all the holes'.

(b) From $N_V = 1.04 \times 10^{19}\,\mathrm{cm}^{-3}$ and $N_C = 2.8 \times 10^{19}\,\mathrm{cm}^{-3}$ at 300 K, compute m_p^* and m_n^* for silicon in units of m_0.

(c) From $N_V = 7.0 \times 10^{18}\,\mathrm{cm}^{-3}$ for GaAs, compute m_p^*. The table of the course gives $m_p^* = 0.48\,m_0$ for GaAs: what value of N_V does this mass give? Comment.

HINT

Work in SI units: N in m^{-3}, then $m^* = \dfrac{h^2}{2\pi kT}\left(\dfrac{N}{2}\right)^{2/3}$.

(Answer: (b) $m_p^* = 0.56\,m_0$, $m_n^* = 1.08\,m_0$; (c) $0.43\,m_0$; $0.48\,m_0$ gives $N_V = 8.4 \times 10^{18}\,\mathrm{cm}^{-3}$.)

SOLUTION

(a) In the Boltzmann limit $p_0 = N_V\,\mathrm{e}^{-(E_F-E_V)/kT}$: this is the number of holes that N_V states placed all at E_V would hold, each with the occupation by a hole $\mathrm{e}^{-(E_F-E_V)/kT}$. N_V is the real distribution of states, weighted by the occupation and integrated, gathered at the band edge.
(b) $h^2/(2\pi kT) = (6.626\times10^{-34})^2/(2\pi\times1.381\times10^{-23}\times300) = 1.69\times10^{-46}$ kg·m². For $N_V = 1.04\times10^{25}$ m^{-3}, $(N_V/2)^{2/3} = (5.2\times10^{24})^{2/3} = 3.00\times10^{16}$ m^{-2}, so $m_p^* = 5.07\times10^{-31}$ kg $= 0.56\,m_0$. The same steps with N_C give $m_n^* = 1.08\,m_0$. These are the density-of-states masses of the course table.
(c) $N_V = 7.0\times10^{24}$ m^{-3} gives $m_p^* = 0.43\,m_0$. Conversely $N_V \propto m^{*3/2}$, so $0.48\,m_0$ gives $7.0\times10^{18}\times(0.48/0.43)^{3/2} = 8.4\times10^{18}$ cm^{-3}. Tables of material constants gather values measured by different methods and rounded; they are not always exactly consistent with each other. The difference here (20 % on N_V) changes n_i by 10 % only, because $n_i \propto \sqrt{N_C N_V}$.

Problem 3.3 Placing the Fermi level

B · Guided calculation

Adapted from Neamen, Problems 4.17, 4.24 and 4.25, pp. 150–151

(a) Silicon at 300 K is doped with arsenic so that $n_0 = 7\times10^{15}$ cm^{-3}. Find $E_C - E_F$, $E_F - E_V$, p_0 and $E_F - E_i$. Which carrier is the minority carrier?

(b) Silicon at 300 K is doped with boron so that $p_0 = 5\times10^{15}$ cm^{-3}. Find $E_F - E_V$, $E_C - E_F$, n_0 and $E_i - E_F$.

(c) The sample of (b) is heated to 400 K; p_0 stays 5×10^{15} cm^{-3}. Compute N_V and n_i at 400 K, then $E_F - E_V$, n_0 and $E_i - E_F$. Does the Fermi level move towards or away from E_i? Why?

HINT

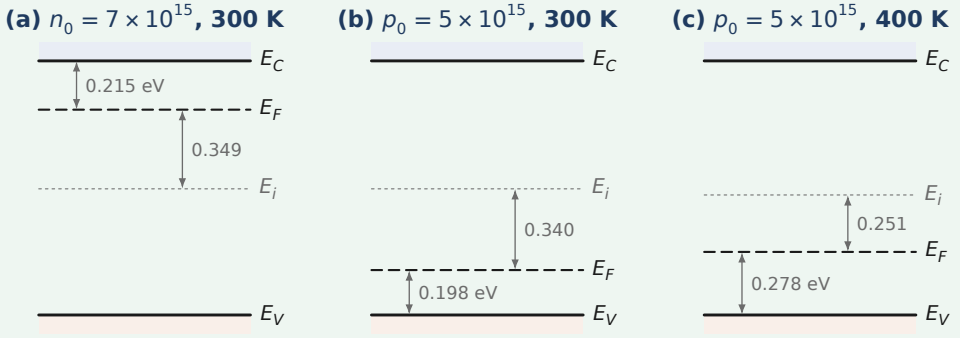
Use $E_C - E_F = kT\ln(N_C/n_0)$ and $E_F - E_i = kT\ln(n_0/n_i)$, and the same forms for holes; $kT = 34.5$ meV at 400 K.

(Answer: (a) 0.215, 0.905 eV, $p_0 = 1.43\times10^4$ cm^{-3}, $E_F - E_i = 0.349$ eV; (b) 0.198, 0.922 eV, $n_0 = 2\times10^4$ cm^{-3}, 0.340 eV; (c) $N_V = 1.60\times10^{19}$, $n_i = 3.5\times10^{12}$ cm^{-3}, 0.278 eV, $n_0 = 2.4\times10^9$ cm^{-3}, 0.251 eV: towards E_i.)

SOLUTION

(a) $E_C - E_F = 0.0259\ln(2.8\times10^{19}/7\times10^{15}) = 0.0259\times8.29 = 0.215$ eV, $E_F - E_V = 1.12 - 0.215 = 0.905$ eV. $p_0 = n_i^2/n_0 = 1.43\times10^4$ cm^{-3}: holes are the minority carriers. $E_F - E_i = 0.0259\ln(7\times10^5) = 0.349$ eV.
(b) $E_F - E_V = 0.0259\ln(1.04\times10^{19}/5\times10^{15}) = 0.198$ eV, $E_C - E_F = 0.922$ eV, $n_0 = 10^{20}/5\times10^{15} = 2\times10^4$ cm^{-3}, $E_i - E_F = 0.0259\ln(5\times10^5) = 0.340$ eV.
(c) $N_V(400) = 1.04\times10^{19}\times(4/3)^{3/2} = 1.60\times10^{19}$ cm^{-3}; $n_i(400) = 10^{10}\times1.540\times\exp[-\frac{1.12}{2\times8.617\times10^{-5}}(\frac{1}{400}-\frac{1}{300})] = 3.5\times10^{12}$ cm^{-3}. Then $E_F - E_V = 0.0345\ln(1.60\times10^{19}/5\times10^{15}) = 0.278$ eV, $n_0 = n_i^2/p_0 = 2.4\times10^9$ cm^{-3}, and $E_i - E_F = 0.0345\ln(5\times10^{15}/3.5\times10^{12}) = 0.251$ eV. The Fermi level moves **towards** E_i (0.251 instead of

0.340 eV), although p_0 is the same. At the higher temperature the band edges have more accessible states (N_V larger) and the thermal pairs are more numerous (n_i larger): the same number of holes is a smaller departure from the intrinsic material, so E_F sits closer to E_i. At still higher temperature, when n_i approaches N_A, it reaches E_i.



The Fermi level in the three samples, to scale. At 400 K the larger n_i brings E_F closer to E_i.

Problem 3.4 Doping for a given Fermi level

B · Guided calculation

Adapted from Sze & Lee, Ch. 1, Problem 14, p. 42

A silicon sample at 300 K contains $N_A = 10^{16}\,\text{cm}^{-3}$ acceptors. Donors are added so that the material becomes n-type with E_F 0.20 eV below E_C.

(a) What electron density does this position of E_F correspond to?

(b) Using neutrality with complete ionisation, find the required N_D.

(c) Check that the donors are indeed fully ionised at this Fermi level ($E_C - E_D = 45$ meV).

HINT

The added donors must first compensate the acceptors.

(Answer: (a) $n_0 = 1.24\times10^{16}\,\text{cm}^{-3}$; (b) $N_D = 2.24\times10^{16}\,\text{cm}^{-3}$; (c) $N_D^+/N_D = 0.995$.)

SOLUTION

(a) $n_0 = N_C\,\text{e}^{-0.20/0.0259} = 2.8\times10^{19}\times4.43\times10^{-4} = 1.24\times10^{16}\,\text{cm}^{-3}$.
(b) Neutrality: $n_0 + N_A = p_0 + N_D$, with $p_0 = n_i^2/n_0 \approx 8\times10^3\,\text{cm}^{-3}$ negligible: $N_D = n_0 + N_A = 2.24\times10^{16}\,\text{cm}^{-3}$.
(c) $E_F - E_D = (E_C - E_D) - (E_C - E_F) = 0.045 - 0.20 = -0.155$ eV, so $N_D^+/N_D = 1/(1 + 2\,\text{e}^{-0.155/0.0259}) = 1/(1 + 2\times2.5\times10^{-3}) = 0.995$. The Fermi level is six kT below the donor level: the donors are empty of their electron, i.e. ionised, as assumed.

Problem 3.5 A small difference of gap, a large difference of n_i calculation — B · Guided

Adapted from Neamen, Problem 4.5, p. 150

Two semiconductors A and B have identical properties except their gaps: 0.90 eV for A and 1.10 eV for B.

(a) Show that $n_{iB}/n_{iA} = \exp[-(E_{gB} - E_{gA})/2kT]$.

(b) Evaluate the ratio at 200, 300 and 400 K.

(c) Explain physically why the ratio approaches 1 when the temperature rises.

(Answer: (b) 3.0×10^{-3}, 2.1×10^{-2}, 5.5×10^{-2}.)

SOLUTION

(a) $n_i = \sqrt{N_C N_V}\,e^{-E_g/2kT}$; with the same N_C and N_V, the ratio is $e^{-(E_{gB}-E_{gA})/2kT} = e^{-0.1/kT}$.

(b) kT = 17.2, 25.9 and 34.5 meV: ratios $e^{-5.80} = 3.0\times10^{-3}$, $e^{-3.86} = 2.1\times10^{-2}$ and $e^{-2.90} = 5.5\times10^{-2}$.

(c) The ratio depends on $\Delta E_g/kT$. A difference of 0.2 eV in the gap is eight kT at 300 K, so it changes the probability of breaking a bond (of lifting an electron across the gap) by a large factor; as kT grows, the same 0.2 eV is a smaller number of kT and matters less. This exponential sensitivity is why silicon and GaAs, whose gaps differ by 0.3 eV, have intrinsic densities that differ by four orders of magnitude.

Problem 3.6 How hot can a device run? — B · Guided calculation

Adapted from Neamen, Problems 4.3 and 4.50, pp. 149 and 153

Use the expression of $n_i(T)$ given in the data, with E_g constant.

(a) A silicon device works properly only if $n_i \le 5\times10^{11}\,\text{cm}^{-3}$. Find the maximum temperature, by trial. Repeat for $5\times10^{12}\,\text{cm}^{-3}$.

(b) A device is doped with $N_D = 10^{15}\,\text{cm}^{-3}$. It requires that thermal pairs contribute no more than 5 % of the electron density, i.e. $n_0 \le 1.05\,N_D$. Show that this means $p_0 \le 0.05\,N_D$, and deduce the maximum value of n_i, then the maximum temperature.

(c) At that temperature, compute $E_C - E_F$ and $E_F - E_i$, and compare with 300 K. Is the Fermi level closer to or further from the intrinsic level?

HINT

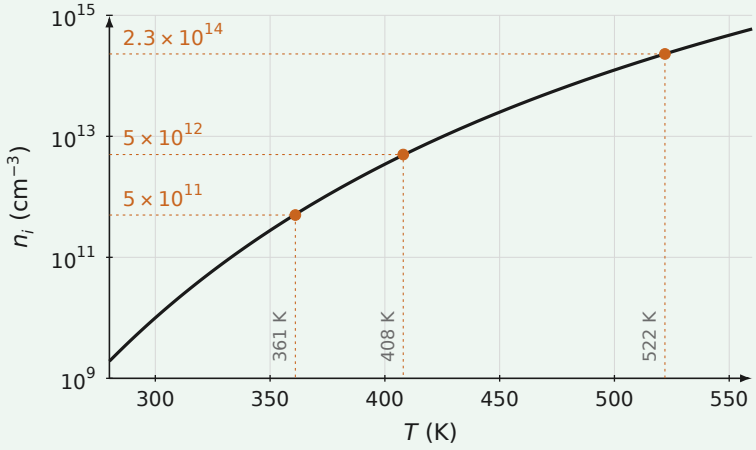
In (b), neutrality gives $n_0 - p_0 = N_D$, so $n_0 = 1.05N_D$ means $p_0 = 0.05N_D$, and $n_i^2 = n_0 p_0$.

(Answer: (a) 361 K and 408 K; (b) $n_i \le 2.3\times10^{14}\,\text{cm}^{-3}$, $T_{max} \approx 522$ K; (c) $E_C - E_F$: 0.265 eV at 300 K, 0.496 eV at 522 K; $E_F - E_i$: 0.298 and 0.068 eV, closer.)

SOLUTION

(a) $n_i(T)$ grows by about a factor 10 every 25 to 30 K near 350 K. Trials: $n_i(350)$ =

2.8×10^{11}, $n_i(360)=4.9\times10^{11}$, $n_i(362)=5.4\times10^{11}$: $T_{max}\approx361$ K (88 °C). For $5\times10^{12}\,\mathrm{cm}^{-3}$: $n_i(408)=5\times10^{12}$, $T_{max}\approx408$ K.
(b) Neutrality with all donors ionised: $n_0=N_D+p_0$. Then $n_0\le1.05N_D$ means $p_0\le0.05N_D=5\times10^{13}\,\mathrm{cm}^{-3}$, and $n_i^2=n_0p_0\le1.05\times10^{15}\times5\times10^{13}=5.25\times10^{28}$, so $n_i\le2.3\times10^{14}\,\mathrm{cm}^{-3}$. Trials give $T_{max}\approx522$ K (about 250 °C).
(c) At 300 K: $E_C-E_F=0.0259\ln(2.8\times10^{19}/10^{15})=0.265$ eV and $E_F-E_i=0.0259\ln(10^5)=0.298$ eV. At 522 K: $kT=0.0450$ eV, $N_C=2.8\times10^{19}\times(522/300)^{3/2}=6.4\times10^{19}\,\mathrm{cm}^{-3}$, $E_C-E_F=0.0450\ln(6.4\times10^{19}/1.05\times10^{15})=0.496$ eV, and $E_F-E_i=0.0450\ln(1.05\times10^{15}/2.3\times10^{14})=0.068$ eV. The Fermi level is much closer to E_i: the sample is near the end of its extrinsic range. The more lightly doped a device, the lower its maximum temperature; the gap sets the scale.



The intrinsic density of silicon versus temperature, and the three limits of the problem.

Problem 3.7 Is complete ionisation justified? B · Guided calculation

Adapted from Sze & Lee, Ch. 1, Problem 20, p. 42

Silicon at 300 K is doped with phosphorus ($E_C-E_D=45$ meV) at $N_D=10^{15}$, 10^{17} and $10^{19}\,\mathrm{cm}^{-3}$.

(a) Assuming complete ionisation, compute E_C-E_F for each doping.

(b) With this E_F, compute $N_D^+/N_D=1/[1+2\,\mathrm{e}^{(E_F-E_D)/kT}]$. For which dopings is the assumption justified?

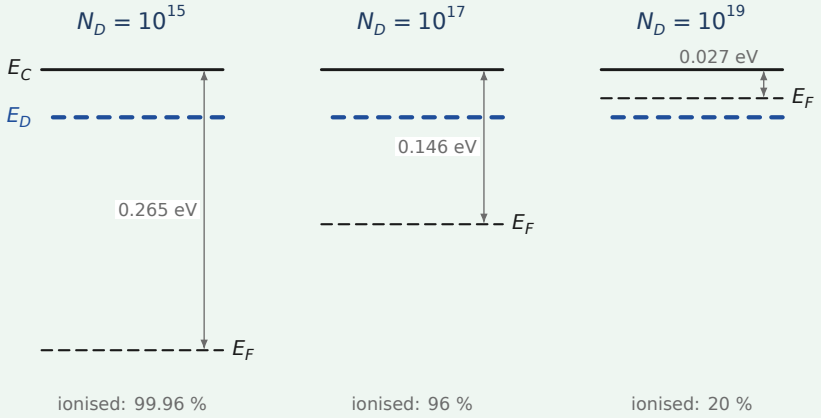
(c) For the highest doping, what goes wrong besides incomplete ionisation? Which assumption of the formula $n_0=N_C\,\mathrm{e}^{-(E_C-E_F)/kT}$ fails?

(Answer: (a) 0.265, 0.146, 0.027 eV; (b) 0.9996, 0.96, 0.20: justified for 10^{15}, acceptable for 10^{17}, not for 10^{19}; (c) E_F is within kT of E_C: the Boltzmann approximation fails.)

SOLUTION

(a) $E_C-E_F=kT\ln(N_C/N_D)$: $0.0259\ln(2.8\times10^4)=0.265$ eV; $0.0259\ln(280)=$

0.146 eV; $0.0259\ln(2.8) = 0.027$ eV.
(b) $E_F - E_D = 0.045 - (E_C - E_F)$: −0.220, −0.101 and +0.018 eV. Then $N_D^+/N_D = 1/(1+2\mathrm{e}^{-8.50}) = 0.9996$, $1/(1+2\mathrm{e}^{-3.90}) = 0.96$, and $1/(1+2\mathrm{e}^{+0.71}) = 0.20$. Complete ionisation is excellent at 10^{15}, acceptable (4 % error) at 10^{17}, and completely wrong at $10^{19}\,\mathrm{cm}^{-3}$: the result contradicts the assumption. A self-consistent calculation would solve neutrality $n_0 = N_D^+$ for E_F (see the chapter).
(c) At $10^{19}\,\mathrm{cm}^{-3}$ the Fermi level is about kT below E_C: the occupation of the conduction states is no longer small, and the Boltzmann form of Fermi–Dirac, on which $n_0 = N_C\,\mathrm{e}^{-(E_C-E_F)/kT}$ rests, is no longer valid. The material is close to degenerate. (At such dopings the donor levels also merge into a band, which the hydrogen-like model ignores.)



The top of the gap, to scale. When E_F comes close to or above E_D, the donors keep their electrons.

Practice exercises from the course notes

These exercises closed Chapter 3 of the course notes; they now live here, with their answers. Tables, equations and sections cited by number are those of the course notes. Use the values of Table 3.3, the reference value $n_i = 10^{10}\,\mathrm{cm}^{-3}$ for silicon and $kT = 25.9\,\mathrm{meV}$ at 300 K unless stated otherwise ($m_0 = 9.11\times10^{-31}$ kg, $\hbar = 1.055\times10^{-34}$ J·s, $h = 6.626\times10^{-34}$ J·s, $k = 1.381\times10^{-23}$ J/K).

Problem 3.8 A one-dimensional metal, and counting bands B · Guided calculation

From the course notes, Chapter 3

(i) A chain of atoms spaced by $a = 0.25$ nm, one electron per atom. Find k_F, E_F, and the fraction of the Brillouin zone that is filled. (ii) A crystal of germanium has N primitive cells of 2 atoms. How many valence electrons, how many states per band, how many full bands at 0 K?

(Answer: (i) $k_F = \pi/2a = 6.28\times10^{9}\ \mathrm{m}^{-1}$; $E_F = \hbar^2k_F^2/2m = 1.50\,\mathrm{eV}$; half of the zone. (ii) $4\times2\times N = 8N$ electrons; $2N$ states per band; 4 full bands.)

Problem 3.9 Gap criterion **B · Guided calculation**

From the course notes, Chapter 3

Compute E_g/kT at 300 K for C, Si, Ge and GaAs (Table 3.2) and classify them with the criterion $E_g < 100\,kT$ / $E_g > 200\,kT$. Above which temperature would diamond satisfy $E_g < 100\,kT$?

(Answer: E_g/kT: C 212, Si 43, Ge 25, GaAs 55: Si, Ge, GaAs are semiconductors, diamond an insulator. $E_g < 100\,kT$ for diamond needs $T > 5.48/(100 \times 8.617 \times 10^{-5}) \approx 640$ K (gap held constant).)

Problem 3.10 Fermi–Dirac **B · Guided calculation**

From the course notes, Chapter 3

Find the probability that a state 0.1 eV above E_F is occupied at 300 K and at 600 K. What is the probability that a state 0.1 eV below E_F is empty? By how much does the Boltzmann approximation overestimate the occupation at $E_F + 3kT$?

(Answer: Above: $1/(1+\mathrm{e}^{0.1/0.0259}) = 0.020$ at 300 K and $1/(1+\mathrm{e}^{0.1/0.0517}) = 0.126$ at 600 K. Below, the probability of being empty is $1 - f = f(E_F + 0.1) = 0.020$ at 300 K (symmetry of the Fermi function). At $3kT$: $\mathrm{e}^{-3} = 0.0498$ against $1/(1+\mathrm{e}^{3}) = 0.0474$, 5 % too high.)

Problem 3.11 Effective densities of states **B · Guided calculation**

From the course notes, Chapter 3

From (3.16) compute N_C and N_V of silicon at 300 K with $m_n^* = 1.08\,m_0$ and $m_p^* = 0.56\,m_0$. What are they at 77 K and at 600 K?

(Answer: $N_C = 2.8 \times 10^{19}\,\mathrm{cm}^{-3}$, $N_V = 1.05 \times 10^{19}\,\mathrm{cm}^{-3}$ at 300 K. At 77 K, factor $(77/300)^{3/2} = 0.130$: 3.6×10^{18} and 1.4×10^{18}. At 600 K, factor 2.83: 7.9×10^{19} and $2.9 \times 10^{19}\,\mathrm{cm}^{-3}$.)

Problem 3.12 Intrinsic concentration and intrinsic level **B · Guided calculation**

From the course notes, Chapter 3

From Table 3.3 compute n_i of germanium and of gallium arsenide at 300 K and compare with the tabulated values. Compute the position of E_i with respect to midgap for Ge and GaAs from N_C and N_V. On which side of midgap is it, and why?

(Answer: Ge: $\sqrt{1.04 \times 10^{19} \times 6.0 \times 10^{18}}\ \mathrm{e}^{-0.66/(2\times0.0259)} = 7.9 \times 10^{18} \times 2.9 \times 10^{-6} = 2.3 \times 10^{13}$ (tabulated 2.4×10^{13}). GaAs: $1.8 \times 10^{18} \times 1.25 \times 10^{-12} = 2.3 \times 10^{6}$ (tabulated 1.8×10^{6}; n_i is very sensitive to the exact gap). $E_i - E_{\mathrm{mid}} = (kT/2)\ln(N_V/N_C)$: Ge $0.01295\ln(6.0/10.4) = -7.1$ meV (below midgap, because $N_V < N_C$); GaAs $0.01295\ln(7.0 \times 10^{18}/4.7 \times 10^{17}) = +35$ meV (above midgap, because m_n^* is small, so N_C is small).)

Problem 3.13 Temperature **B · Guided calculation**

From the course notes, Chapter 3

By what factor does n_i of silicon rise from 300 to 350 K (E_g held at 1.12 eV)? By what factor does the resistivity of the pure material fall if the mobility varies as $T^{-3/2}$?

(Answer: n_i is multiplied by $(350/300)^{3/2}\exp[(E_g/2k)(1/300-1/350)] = 1.26\times 22.1 = 27.8$. With $\mu \propto T^{-3/2}$, μ falls by 1.26, so σ rises by $27.8/1.26 = 22$ and ρ falls by 22.)

Problem 3.14 A donor; p-type samples

B · Guided calculation

From the course notes, Chapter 3

(i) For $N_D = 10^{16}\,\mathrm{cm}^{-3}$ in silicon at 300 K, the Fermi level is 0.206 eV below E_C and E_D is 45 meV below E_C. Compute N_D^+/N_D. Find the position of E_F with respect to E_D for which half the donors are ionised. (ii) Silicon doped with $N_A = 2\times 10^{16}\,\mathrm{cm}^{-3}$: find p_0, n_0, $E_i - E_F$ and $E_F - E_V$; is the nondegenerate assumption satisfied? (iii) Germanium doped with $N_A = 10^{17}\,\mathrm{cm}^{-3}$: find p_0 and n_0.

(Answer: (i) $E_F - E_D = -(0.206-0.045) = -0.161\,\mathrm{eV}$; $2\,\mathrm{e}^{-0.161/0.0259} = 2\times 2.0\times 10^{-3} = 4.0\times 10^{-3}$; $N_D^+/N_D = 1/1.004 = 0.996$. Half ionised: $2\,\mathrm{e}^{(E_F-E_D)/kT} = 1$, so $E_F = E_D - kT\ln 2$: E_F is $kT\ln 2 = 18\,\mathrm{meV}$ below the donor level. (ii) $p_0 = 2\times 10^{16}$; $n_0 = n_i^2/p_0 = 5\times 10^3\,\mathrm{cm}^{-3}$; $E_i - E_F = 0.0259\ln(2\times 10^6) = 0.376\,\mathrm{eV}$; $E_F - E_V = kT\ln(N_V/N_A) = 0.162\,\mathrm{eV}$, which is $6.3\,kT$: nondegenerate. (iii) $p_0 = 10^{17}$, $n_0 = (2.4\times 10^{13})^2/10^{17} = 5.8\times 10^9\,\mathrm{cm}^{-3}$.)

Problem 3.15 Temperature window

B · Guided calculation

From the course notes, Chapter 3

Take $n_i(T) = 10^{10}(T/300)^{3/2}\exp[-\frac{E_g}{2k}(\frac{1}{T}-\frac{1}{300})]\,\mathrm{cm}^{-3}$ with $E_g = 1.12\,\mathrm{eV}$ and $k = 8.617\times 10^{-5}$ eV/K. (i) Find the temperature at which $n_i = N_D/10$ for $N_D = 10^{14}$ and for $10^{16}\,\mathrm{cm}^{-3}$ (a few trials suffice). (ii) With (3.28), $N_C(T) = 2.8\times 10^{19}(T/300)^{3/2}$ and $E_C - E_D = 45\,\mathrm{meV}$, estimate n_0 at 50 K for $N_D = 10^{16}\,\mathrm{cm}^{-3}$ and check the condition of validity. (iii) Which sample keeps its doping-controlled behaviour to the higher temperature, and why?

(Answer: (i) $n_i = 10^{13}$ at about 425 K for $N_D = 10^{14}$; $n_i = 10^{15}$ at about 580 K for $N_D = 10^{16}$. (ii) $N_C(50\,\mathrm{K}) = 1.9\times 10^{18}$, $kT = 4.31\,\mathrm{meV}$: $n_0 \approx \sqrt{1.9\times 10^{18}\times 10^{16}/2}\,\mathrm{e}^{-5.22} = 5.3\times 10^{14}\,\mathrm{cm}^{-3}$; $n_0/N_D = 5\,\% \ll 1$, so the strongly frozen limit is valid (numerical solution: 5.1×10^{14}). (iii) The more heavily doped one: n_i must climb higher to reach its doping.)

Challenge: a new situation

These problems apply the method of the chapter to a device, a material or a situation that the chapter did not treat. The reasoning does not change: state the physical picture, choose the model, compute, check.

Problem 3.16 The limit of the Boltzmann approximation

C · Synthesis

Adapted from Neamen, Problem 4.32, p. 151

For the Boltzmann expression of the occupation of the donor levels to hold, the Fermi level must be at least $3kT$ below the donor level in n-type material, and at least $3kT$ above the acceptor level in p-type material.

(a) In silicon at 300 K with $E_C - E_D = 45$ meV, what is the highest electron density allowed? With $E_A - E_V = 45$ meV, the highest hole density?

(b) At this limit, what fraction of the donors is ionised? Is complete ionisation a good

assumption there?

(c) Compare with Problem 3.7. Why is the limit on holes lower than the limit on electrons?

(Answer: (a) $n_{max} = 2.5 \times 10^{17}$, $p_{max} = 9.1 \times 10^{16}\,\text{cm}^{-3}$; (b) 91 %.)

SOLUTION

(a) $E_C - E_F \geq 0.045 + 3 \times 0.0259 = 0.1227$ eV, so $n_0 \leq 2.8 \times 10^{19}\,\text{e}^{-4.74} = 2.5 \times 10^{17}\,\text{cm}^{-3}$. For holes, $E_F - E_V \geq 0.1227$ eV and $p_0 \leq 1.04 \times 10^{19}\,\text{e}^{-4.74} = 9.1 \times 10^{16}\,\text{cm}^{-3}$.

(b) $N_D^+/N_D = 1/(1 + 2\text{e}^{-3}) = 0.91$: 9 % of the donors are still neutral. Complete ionisation is a rough approximation at the limit; neutrality requires $N_D \simeq n_0/0.91 = 2.7 \times 10^{17}\,\text{cm}^{-3}$.

(c) Problem 3.7 found a 4 % error at $10^{17}\,\text{cm}^{-3}$, consistent with this limit. The limit on holes is lower because $N_V < N_C$: fewer valence states are available at the band edge, so a given hole density pushes E_F closer to E_V than the same electron density pushes it towards E_C.

Problem 3.17 A new semiconductor

C · Synthesis

Adapted from Neamen, Problems 4.53 and 4.61, pp. 153–154

(a) A semiconductor has $E_g = 1.50$ eV, $m_p^* = 10\,m_n^*$ and $n_i = 10^5\,\text{cm}^{-3}$ at 300 K. Where is E_i with respect to midgap? Impurities are added so that E_F is 0.45 eV below midgap: donors or acceptors? What net concentration?

(b) A new p-type material is to be designed, doped with $N_A = 5 \times 10^{15}\,\text{cm}^{-3}$ ($N_D = 0$). At 300 K, $N_C = 1.2 \times 10^{19}$ and $N_V = 1.8 \times 10^{19}\,\text{cm}^{-3}$, both proportional to $T^{3/2}$. A device made of it requires $p_0 \leq 5.08 \times 10^{15}\,\text{cm}^{-3}$ at 350 K. What is the smallest gap that meets the requirement?

HINT

(a) $E_i - E_{mid} = \frac{3}{4}kT\ln(m_p^*/m_n^*)$. (b) Neutrality gives $p_0(p_0 - N_A) = n_i^2$: find the largest allowed n_i, then the gap.

(Answer: (a) E_i is 44.7 meV above midgap; acceptors, $2 \times 10^{13}\,\text{cm}^{-3}$; (b) $E_g \geq 0.62$ eV.)

SOLUTION

(a) $E_i - E_{mid} = \frac{1}{2}kT\ln(N_V/N_C) = \frac{3}{4}kT\ln(m_p^*/m_n^*) = 0.75 \times 0.0259\ln 10 = 0.0447$ eV: E_i is above midgap, pushed away from the band with more states. With E_F 0.45 eV below midgap, $E_i - E_F = 0.495$ eV: the material is p-type, with $p_0 = n_i\,\text{e}^{0.495/0.0259} = 10^5 \times 1.97 \times 10^8 = 2.0 \times 10^{13}\,\text{cm}^{-3}$. Acceptors must be added, $N_A \approx 2 \times 10^{13}\,\text{cm}^{-3}$ (n_0 is negligible).

(b) Neutrality $p_0 = N_A + n_0$ with $n_0 = n_i^2/p_0$ gives $n_i^2 = p_0(p_0 - N_A) \leq 5.08 \times 10^{15} \times 0.08 \times 10^{15} = 4.06 \times 10^{29}$, so $n_i(350) \leq 6.4 \times 10^{14}\,\text{cm}^{-3}$. At 350 K, $kT = 0.03016$ eV and $N_C N_V = 2.16 \times 10^{38} \times (350/300)^3 = 3.43 \times 10^{38}\ \text{cm}^{-6}$. From $n_i^2 = N_C N_V\,\text{e}^{-E_g/kT}$: $E_g \geq kT\ln(N_C N_V/n_i^2) = 0.03016\ln(8.45 \times 10^8) = 0.62$ eV. A material with a smaller gap would create too many thermal pairs at 350 K, and the

> hole density would no longer be fixed by the doping.

SEMICONDUCTOR DEVICES PHYSICS · SOLUTIONS

CHAPTER 4

Carriers Out of Equilibrium

WHAT THIS SET USES

Equilibrium, steady state and transient; the excess densities Δn, Δp and the lifetime τ; the return to equilibrium $\mathrm{e}^{-t/\tau}$; absorption above E_g and the generation rate g_0; Case A (uniform light), $\Delta n = \tau g_0$ and $\Delta\sigma = q(\mu_n + \mu_p)\Delta n$; the continuity equations with generation and recombination; the diffusion length $L = \sqrt{D\tau}$; Case B (illuminated surface) and the internal field; the ambipolar coefficient $D_a = (n+p)/(n/D_p + p/D_n)$.

DATA

Silicon at 300 K: $n_i = 10^{10}\,\mathrm{cm}^{-3}$, $\mu_n = 1350$, $\mu_p = 460$ cm²/(V·s), $D_n = 35$, $D_p = 12$ cm²/s unless the problem gives other values; $hc = 1240$ eV·nm. Low-level injection unless stated otherwise.

Problem 4.1 The balance of a slice

A · Understanding

Adapted from Neamen, Problems 6.6 and 6.7, p. 234

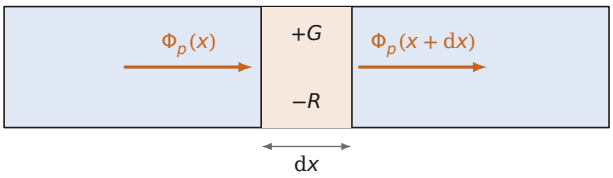
In one dimension the continuity equation for holes reads $\dfrac{\partial p}{\partial t} = -\dfrac{\partial \Phi_p}{\partial x} + G - R$, where Φ_p is the hole flux.

(a) Say in words what each term counts, for a thin slice of the sample.

(b) In a slice, holes are generated at $G = 10^{20}$ and recombine at $R = 2\times10^{19}$ $\mathrm{cm}^{-3}\mathrm{s}^{-1}$. What must $\partial\Phi_p/\partial x$ be for the hole density to stay constant? Do more holes leave or enter the slice?

(c) Repeat if the light is switched off ($G = 0$) while R keeps its value.

(d) Classify the following as equilibrium, steady state or transient: (i) a sample in the dark at constant temperature; (ii) a sample under constant light for a long time; (iii) the first microseconds after the light is switched on; (iv) a diode carrying a constant current.



A thin slice of the sample: what enters, what leaves, what is created and what disappears.

(Answer: (b) $+8\times10^{19}$ $\mathrm{cm}^{-3}\mathrm{s}^{-1}$, more holes leave; (c) -2×10^{19} $\mathrm{cm}^{-3}\mathrm{s}^{-1}$, more enter; (d) equilibrium; steady state; transient; steady state.)

SOLUTION

(a) $\partial p/\partial t$ is the rate of change of the number of holes per unit volume in the slice. $-\partial \Phi_p/\partial x$ is what transport brings in per unit volume: flux entering on one face minus flux leaving on the other, divided by the thickness. G and R are the holes created and destroyed inside the slice per unit volume and time.
(b) Constant p means $0 = -\partial \Phi_p/\partial x + G - R$, so $\partial \Phi_p/\partial x = G - R = 8 \times 10^{19}$ cm^{-3}s^{-1}. The flux increases across the slice: more holes leave than enter. The slice produces a net 8×10^{19} holes per cm³ and per second, and transport carries them away.
(c) $\partial \Phi_p/\partial x = -R = -2 \times 10^{19}$ cm^{-3}s^{-1}: more holes enter than leave; transport feeds the recombination of the slice.
(d) (i) Equilibrium: no light, no current, $G = R$ in every slice and every flux zero. (ii) Steady state: nothing changes in time, but $G \neq R$ and currents may flow; the sample is not in equilibrium. (iii) Transient. (iv) Steady state: the current and the densities are constant in time, but the device is far from equilibrium.

Problem 4.2 From watts of light to excess carriers

B · Guided calculation

Adapted from Neamen, Problem 6.4, p. 234

A silicon sample of 1 cm² cross-section and 0.1 cm thickness absorbs uniformly 1 W of red light at 630 nm. Each absorbed photon creates one electron–hole pair.

(a) Compute the photon energy and the number of photons absorbed per second.

(b) Deduce the generation rate g_0 per cm³ and per second.

(c) The lifetime of the excess carriers is 10 µs. What is the steady excess density (Case A)? Is low-level injection satisfied for a doping of 10^{16} cm^{-3}?

(d) What fraction of the light power is turned into heat at once, before any recombination? Explain where this energy goes.

HINT

$E = hc/\lambda$; the number of photons per second is the power divided by the energy of one photon.

(Answer: (a) 1.97 eV, 3.2×10^{18} s^{-1}; (b) 3.2×10^{19} cm^{-3}s^{-1}; (c) 3.2×10^{14} cm^{-3}, yes; (d) 43 %.)

SOLUTION

(a) $E = 1240/630 = 1.97$ eV $= 3.15 \times 10^{-19}$ J. Photons per second: $1/3.15 \times 10^{-19} = 3.2 \times 10^{18}$.
(b) The volume is 0.1 cm³, and the absorption is uniform: $g_0 = 3.2 \times 10^{18}/0.1 = 3.2 \times 10^{19}$ cm^{-3}s^{-1}.
(c) $\Delta n = \Delta p = \tau g_0 = 10^{-5} \times 3.2 \times 10^{19} = 3.2 \times 10^{14}$ cm^{-3}. With $N_D = 10^{16}$ cm^{-3} the excess is 3 % of the majority density: low-level injection holds.
(d) Each photon gives 1.97 eV; the pair needs $E_g = 1.12$ eV. The excess, 0.85 eV or 43 % of the energy, is carried by the electron and the hole as kinetic energy above the band edges, and is given to the lattice in a fraction of a picosecond by collisions: it becomes heat. This is one of the losses of a solar cell.

Problem 4.3 Light on, light off

B · Guided calculation

Adapted from Neamen, Problems 6.15 and 6.17, p. 235

An n-type silicon sample ($N_D = 10^{16}\,\text{cm}^{-3}$) has a hole lifetime $\tau_p = 0.5$ µs. At $t = 0$ a light is switched on that generates $g_0 = 5 \times 10^{20}\ \text{cm}^{-3}\text{s}^{-1}$ uniformly.

(a) Write the equation for $\Delta p(t)$ while the light is on, and solve it. What is the steady value?

(b) Find the times at which Δp reaches one quarter, one half, three quarters and 95 % of its steady value.

(c) The light is switched off at $t_1 = 0.5$ µs. Give $\Delta p(t_1)$ and $\Delta p(t)$ for $t > t_1$. Sketch $\Delta p(t)$.

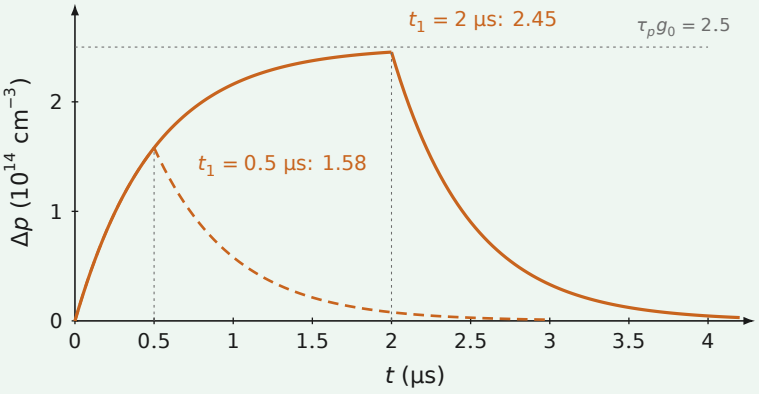
(d) Repeat (c) for $t_1 = 2$ µs. Why does the excess at the switch-off barely differ from the steady value?

> **HINT**
> $\mathrm{d}\Delta p/\mathrm{d}t = g_0 - \Delta p/\tau_p$.

(Answer: (a) $\Delta p = \tau_p g_0(1 - \mathrm{e}^{-t/\tau_p})$, $2.5 \times 10^{14}\,\text{cm}^{-3}$; (b) 0.14, 0.35, 0.69 and 1.50 µs; (c) $1.58 \times 10^{14}\,\text{cm}^{-3}$; (d) $2.45 \times 10^{14}\,\text{cm}^{-3}$.)

SOLUTION

(a) $\mathrm{d}\Delta p/\mathrm{d}t = g_0 - \Delta p/\tau_p$ with $\Delta p(0) = 0$ gives $\Delta p(t) = \tau_p g_0(1 - \mathrm{e}^{-t/\tau_p})$. Steady value $\tau_p g_0 = 5 \times 10^{-7} \times 5 \times 10^{20} = 2.5 \times 10^{14}\,\text{cm}^{-3}$ (2.5 % of N_D: low injection).
(b) $\Delta p/\tau_p g_0 = f$ gives $t = -\tau_p \ln(1 - f)$: $0.288\tau_p = 0.14$ µs, $0.693\tau_p = 0.35$ µs, $1.386\tau_p = 0.69$ µs, $3.0\tau_p = 1.50$ µs. Three lifetimes are needed to come within 5 % of the steady state.
(c) $\Delta p(t_1) = 2.5 \times 10^{14}(1 - \mathrm{e}^{-1}) = 1.58 \times 10^{14}\,\text{cm}^{-3}$. After the switch-off, $\mathrm{d}\Delta p/\mathrm{d}t = -\Delta p/\tau_p$: $\Delta p(t) = 1.58 \times 10^{14}\,\mathrm{e}^{-(t-t_1)/\tau_p}$. The sketch is a rising exponential up to 63 % of the steady value, then a decay with the same time constant.
(d) $\Delta p(t_1) = 2.5{\times}10^{14}(1{-}\mathrm{e}^{-4}) = 2.45{\times}10^{14}\,\text{cm}^{-3}$, then $\Delta p = 2.45{\times}10^{14}\,\mathrm{e}^{-(t-t_1)/\tau_p}$. After four lifetimes the transient is over (98 %): the same time constant governs the rise and the fall.



Light on at $t = 0$, off at t_1: the excess rises and falls with the same time constant τ_p.

Problem 4.4 A photoconductor switched off

B · Guided calculation

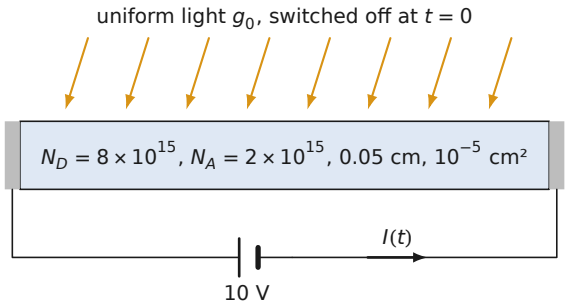
Adapted from Neamen, Problem 6.14, p. 234

A silicon bar is 0.05 cm long, with cross-section 10^{-5} cm², doped with $N_D = 8\times10^{15}$ and $N_A = 2\times10^{15}\,\text{cm}^{-3}$. A voltage of 10 V is applied along it. For $t < 0$ the bar has been illuminated for a long time, with a uniform generation rate $g_0 = 8\times10^{20}\ \text{cm}^{-3}\text{s}^{-1}$; the light is switched off at $t = 0$. The minority lifetime is $\tau_p = 0.5$ µs. Because of the compensating impurities, take $\mu_n = 1200$ and $\mu_p = 420$ cm²/(V·s).

(a) Find n_0, p_0 and the excess density before $t = 0$.

(b) Write the current $I(t)$ for $t \geq 0$ as a dark current plus a photocurrent, and evaluate both.

(c) By what fraction does the light raise the current? After how long is the photocurrent below 1 % of its initial value?



The photoconductor.

HINT

$I = \sigma\mathcal{E}A$ with $\sigma = q(n\mu_n + p\mu_p)$, $n = n_0 + \Delta n$, $p = p_0 + \Delta p$, $\Delta n = \Delta p$.

(Answer: (a) $n_0 = 6\times10^{15}\,\text{cm}^{-3}$, $p_0 \approx 1.7\times10^{4}\,\text{cm}^{-3}$, $\Delta p = 4\times10^{14}\,\text{cm}^{-3}$; (b) $I = 2.31 + 0.21\,\text{e}^{-t/\tau_p}$ mA; (c) 9 %, after 2.3 µs.)

SOLUTION

(a) $n_0 = N_D - N_A = 6\times10^{15}\,\text{cm}^{-3}$, $p_0 = 10^{20}/6\times10^{15} = 1.7\times10^{4}\,\text{cm}^{-3}$. Before $t = 0$, Case A: $\Delta p = \Delta n = \tau_p g_0 = 5\times10^{-7}\times8\times10^{20} = 4\times10^{14}\,\text{cm}^{-3}$, 7 % of n_0.

(b) The field is $\mathcal{E} = 10/0.05 = 200$ V/cm. For $t \geq 0$, $\Delta n = \Delta p = 4\times10^{14}\,\text{e}^{-t/\tau_p}$, and $I(t) = qA\mathcal{E}[n_0\mu_n + \Delta p(t)(\mu_n+\mu_p)]$. Dark current: $1.602\times10^{-19}\times10^{-5}\times200\times6\times10^{15}\times1200 = 2.31$ mA. Initial photocurrent: $1.602\times10^{-19}\times10^{-5}\times200\times4\times10^{14}\times1620 = 0.21$ mA. So $I(t) = 2.31 + 0.21\,\text{e}^{-t/0.5\ \mu\text{s}}$ mA.

(c) The light raises the current by $0.21/2.31 = 9$ %. The photocurrent falls below

1 % of its initial value when $e^{-t/\tau_p} = 0.01$: $t = 4.6\,\tau_p = 2.3$ µs. A measurement of this decay gives τ_p directly, which is the principle of the photoconductive decay measurement.

Problem 4.5 Light at the end of a bar

B · Guided calculation

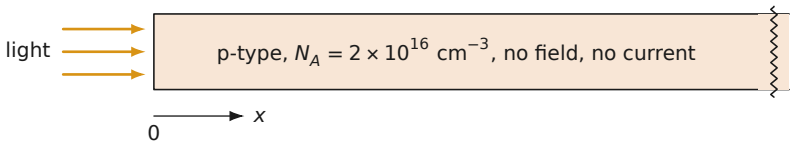
Adapted from Neamen, Problem 6.19, p. 235

A long p-type silicon bar ($N_A = 2\times10^{16}\,\text{cm}^{-3}$) is illuminated at its end $x = 0$ only, so that in steady state $\Delta n(0) = \Delta p(0) = 2\times10^{14}\,\text{cm}^{-3}$. No external field is applied and no current flows through the bar. The electron lifetime is $\tau_n = 1$ µs.

(a) Write the steady-state equation for the minority excess in the dark part, $x > 0$, and solve it. Compute L_n.

(b) Compute the electron and hole diffusion current densities at $x = 0$ and give their directions.

(c) The bar is open-circuited. Show that the total current would not be zero with diffusion alone, and find the small field that restores $J = 0$ at $x = 0$. Which carriers carry the drift current, and why is the field small?



Light absorbed at the end of a long bar. After Neamen, Fig. P6.19, p. 235.

HINT

In low injection the profile is set by the minority carriers; the majority excess follows it so that the sample stays neutral, $\Delta p \simeq \Delta n$.

(Answer: (a) $\Delta n = 2\times10^{14}\,e^{-x/L_n}$, $L_n = 59$ µm; (b) $J_n = -0.190$ A/cm², $J_p = +0.065$ A/cm²; (c) $\mathcal{E}(0) = 0.085$ V/cm.)

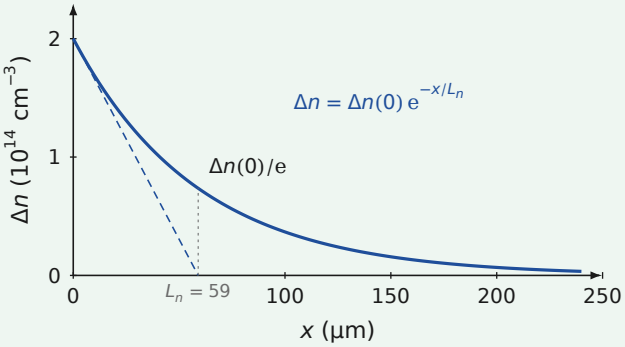
SOLUTION

(a) In the dark part, $D_n\,d^2\Delta n/dx^2 - \Delta n/\tau_n = 0$, whose solution that vanishes far from the end is $\Delta n = \Delta n(0)\,e^{-x/L_n}$, with $L_n = \sqrt{D_n\tau_n} = \sqrt{35\times10^{-6}} = 5.9\times10^{-3}$ cm = 59 µm.

(b) $J_n = qD_n\,d\Delta n/dx = -qD_n\Delta n(0)/L_n\,e^{-x/L_n}$: at $x = 0$, $-1.602\times10^{-19}\times35\times2\times10^{14}/5.9\times10^{-3} = -0.190$ A/cm². The electrons diffuse into the bar ($+x$), so their current flows towards $-x$. The holes have the same profile, $J_p = -qD_p\,d\Delta p/dx = +0.065$ A/cm², towards $+x$.

(c) Diffusion alone gives $J_n + J_p = -0.125$ A/cm² at $x = 0$, not zero: electrons diffuse faster than holes and would carry negative charge into the bar. A small field builds up and drives a drift current that cancels this difference, as in Case B of the chapter. The drift current is carried by the majority holes, which are very numerous:

$q\mu_p p_0 \mathcal{E} = 0.125$ A/cm² gives $\mathcal{E} = 0.125/(1.602\times10^{-19}\times460\times2\times10^{16}) = 0.085$ V/cm. Such a field changes the minority drift by only $q\mu_n \Delta n \mathcal{E} \approx 3\times10^{-4}$ A/cm², negligible against their diffusion current: the minority profile computed in (a) is unaffected.



The minority excess decays over one diffusion length; the tangent at the origin reaches zero at L_n.

Problem 4.6 Injected at one face, collected at the other

C · Synthesis

Adapted from Sze & Lee, Ch. 2, Problems 17 and 18, p. 80

Excess holes are injected at the face $x = 0$ of a thin n-type slab of thickness W and extracted at the opposite face, which holds them at equilibrium: $\Delta p(0) = \Delta p_0$, $\Delta p(W) = 0$. There is no field in the slab.

(a) Solve the steady-state equation $D_p\,\mathrm{d}^2\Delta p/\mathrm{d}x^2 = \Delta p/\tau_p$ with these conditions. Show that $\Delta p(x) = \Delta p_0\,\dfrac{\sinh[(W-x)/L_p]}{\sinh(W/L_p)}$.

(b) Compute the hole current densities at $x = 0$ and at $x = W$, and show that the fraction of the injected current that reaches the far face is $1/\cosh(W/L_p)$.

(c) Evaluate this fraction for $\tau_p = 50$ µs, $D_p = 50$ cm²/s and $W = 0.1$ mm. Then for $W = L_p$ and $W = 3L_p$.

(d) Show that for $W \ll L_p$ the profile tends to a straight line and the fraction to $1 - W^2/2L_p^2$. Where does this result reappear in the course?

HINT

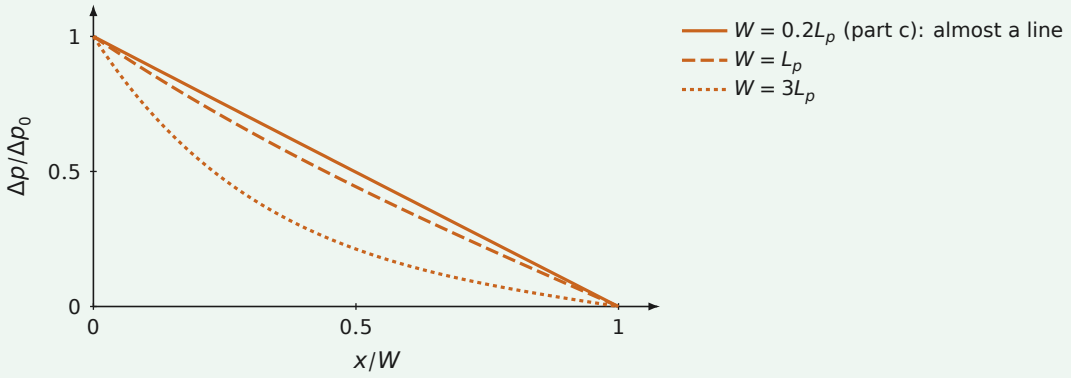
The general solution is $A\sinh(x/L_p) + B\cosh(x/L_p)$; it is convenient to write it as a combination of $\sinh[(W-x)/L_p]$ and $\sinh(x/L_p)$.

(Answer: (b) $J(0) = qD_p\Delta p_0 \coth(W/L_p)/L_p$, $J(W) = qD_p\Delta p_0/[L_p\sinh(W/L_p)]$; (c) L_p = 500 µm, 0.980; 0.648; 0.099.)

SOLUTION

(a) The function $\sinh[(W-x)/L_p]$ satisfies the equation, is zero at $x = W$ and equals $\sinh(W/L_p)$ at $x = 0$; dividing by this value gives the profile, which meets both conditions. The solution is unique.

(b) $J_p = -qD_p\,\mathrm{d}\Delta p/\mathrm{d}x = \dfrac{qD_p\Delta p_0}{L_p}\dfrac{\cosh[(W-x)/L_p]}{\sinh(W/L_p)}$. At $x = 0$: $J(0) = \dfrac{qD_p\Delta p_0}{L_p}\coth(W/L_p)$; at $x = W$: $J(W) = \dfrac{qD_p\Delta p_0}{L_p\sinh(W/L_p)}$. The ratio is $J(W)/J(0) = 1/\cosh(W/L_p)$. The difference is the holes that recombine inside the slab.
(c) $L_p = \sqrt{50\times5\times10^{-5}} = 0.05$ cm = 500 µm; $W/L_p = 0.2$ and $1/\cosh 0.2 = 0.980$: 98 % of the injected holes cross. For $W = L_p$: $1/\cosh 1 = 0.648$; for $W = 3L_p$: $1/\cosh 3 = 0.099$.
(d) For $W \ll L_p$, $\sinh u \simeq u$, so $\Delta p \simeq \Delta p_0(W-x)/W$: a straight line, as in a short region without recombination. And $1/\cosh u \simeq 1 - u^2/2$: the fraction is $1 - W^2/2L_p^2$. This is the **base transport factor** $\alpha_T = 1 - W_B^2/2L_n^2$ of the bipolar transistor (Chapter 8), and the straight-line profile is that of the short diode (Chapter 6).



For a thin slab the profile is a straight line and almost all the injected holes reach the far face.

Practice exercises from the course notes
These exercises closed Chapter 4 of the course notes; they now live here, with their answers. Tables, equations and sections cited by number are those of the course notes. Use the silicon parameters of the notation caption (Table 4.1), at 300 K, unless stated otherwise.

Problem 4.7 Reading a decay

B · Guided calculation

From the course notes, Chapter 4

In n-type silicon with $n_0 = 10^{16}\,\mathrm{cm}^{-3}$, a photoconductance measurement shows the excess falling from 10^{14} to $10^{13}\,\mathrm{cm}^{-3}$ in 23 µs. Find τ and the effective α. Check that low-level injection holds. How long until p is within a factor 2 of p_0?

(Answer: Δn falls by a factor 10 in 23 µs: $\tau = 23/\ln 10 = 10$ µs. $\alpha = 1/(\tau n_0) = 10^{-11}$ cm³/s. $\Delta n/n_0 \le 1\%$: low level. $p = 2p_0$ when $\Delta n = p_0$: $t = \tau\ln(10^{14}/10^4) = 230$ µs.)

Problem 4.8 Which light is absorbed? B · Guided calculation

From the course notes, Chapter 4

Compute the cut-off wavelength λ_g of Si, Ge and GaAs. Which of them absorbs the light of a 1.55 µm telecom laser? A red laser pointer (650 nm)?

(Answer: $\lambda_g = 1240/E_g$: Si 1.11 µm, Ge 1.88 µm, GaAs 0.87 µm. At 1.55 µm ($h\nu = 0.80$ eV) only Ge absorbs (this is why Ge and InGaAs, not Si, are used for telecom detectors). At 650 nm (1.91 eV) all three absorb.)

Problem 4.9 Where does the energy go? B · Guided calculation

From the course notes, Chapter 4

A blue photon (450 nm) is absorbed in silicon. Compute its energy, the energy that creates the pair, and the excess. What does the excess become, and in what form does it finally leave? Which part of your answer is outside the model of the chapter?

(Answer: $h\nu = 1240/450 = 2.76$ eV. The pair needs $E_g = 1.12$ eV; the excess 1.64 eV is kinetic energy of the electron and the hole (derived from the threshold picture). Both give it to the lattice by emitting phonons and settle at the band edges: the excess becomes heat. The time this takes, about a picosecond, is an experimental fact outside the model. The remaining 1.12 eV is stored in the pair until it recombines.)

Problem 4.10 Photoconductivity B · Guided calculation

From the course notes, Chapter 4

Pure silicon receives light that maintains $\Delta n = \Delta p = 10^{12}\,\mathrm{cm}^{-3}$ extra pairs. By what factor does its conductivity increase? Would the same light change much the conductivity of silicon doped with 10^{16} donors per cm³?

(Answer: Pure silicon: without light $\sigma_i = qn_i(\mu_n + \mu_p)$, with light $q(n_i + \Delta n)(\mu_n + \mu_p)$: factor $(10^{10} + 10^{12})/10^{10} = 101$. Doped silicon: $\sigma \approx qN_D\mu_n = q \times 10^{16} \times 1350$; the light adds $q \times 10^{12} \times 1810$, a relative change of 1.3×10^{-4}: negligible. Photoconductors use pure or lightly doped material.)

Problem 4.11 Case A without the approximation B · Guided calculation

From the course notes, Chapter 4

In Case A keep the term $\alpha\Delta n^2$ and find the exact plateau from $g_0 = \alpha(n_0+p_0)\Delta n+\alpha\Delta n^2$, with $\alpha = 1/(\tau n_0)$, $\tau = 10$ µs, $n_0 = 10^{16}\,\mathrm{cm}^{-3}$. Compare with τg_0 for $g_0 = 10^{19}$ and for $g_0 = 10^{22}$ $\mathrm{cm}^{-3}\mathrm{s}^{-1}$. What do you conclude?

(Answer: $\alpha = 10^{-11}$ cm³/s. The quadratic $\Delta n^2 + n_0\Delta n - g_0/\alpha = 0$ gives $\Delta n = [-n_0 + \sqrt{n_0^2 + 4g_0/\alpha}]/2$. For $g_0 = 10^{19}$: 9.90×10^{13}, 1% below $\tau g_0 = 10^{14}$. For $g_0 = 10^{22}$: 2.7×10^{16}, against $\tau g_0 = 10^{17}$: the linear result overestimates by a factor 3.7. The formula τg_0 is valid only while $\tau g_0 \ll n_0$ (low level).)

Problem 4.12 Colour of an LED

B · Guided calculation

From the course notes, Chapter 4

An InGaN alloy has a gap of 2.7 eV, an AlGaInP alloy 1.95 eV. Find the emitted wavelengths and colours. Why could not silicon, even with a suitable gap, replace them?

(Answer: InGaN: 1240/2.7 = 459 nm, blue. AlGaInP: 1240/1.95 = 636 nm, red. Silicon has an indirect gap: even with the right gap, an electron and a hole at the band extrema have different k and recombine without emitting light, through defects.)

Problem 4.13 Mechanisms

B · Guided calculation

From the course notes, Chapter 4

n-type silicon with $\tau_{SRH} = 50$ µs, $B = 10^{-14}$ cm³/s, $C_n = 3 \times 10^{-31}$ cm^6/s. Compute the three rates and the lifetime for $n_0 = 10^{17}$ and 10^{19} cm^{-3}. At which doping do Auger and SRH have equal weight?

(Answer: SRH: $1/\tau_{SRH} = 2 \times 10^4$ s^{-1} at any doping. At 10^{17}: radiative $Bn_0 = 10^3$, Auger $C_n n_0^2 = 3{\times}10^3$; total $2.4{\times}10^4$ s^{-1}, $\tau = 42$ µs (SRH dominant). At 10^{19}: radiative 10^5, Auger $3{\times}10^7$ s^{-1}; $\tau = 33$ ns (Auger dominant). Equal weight of Auger and SRH: $C_n n_0^2 = 2 \times 10^4$, $n_0 = 2.6 \times 10^{17}$ cm^{-3}.)

Problem 4.14 Two samples, one light

B · Guided calculation

From the course notes, Chapter 4

(in the spirit of the surface-illumination problems of Neamen, ch. 6). Two n-type silicon samples are identical except for their hole lifetime: 10 µs and 0.1 µs. Both receive $g_0 = 10^{19}$ cm^{-3}s^{-1} in their illuminated layer. For each, find $\Delta p(0)$, L_p, Δp at 50 µm into the dark bulk, and $J_p(0)$. By which factor does the lifetime change each quantity?

(Answer: $\tau = 10$ µs: $\Delta p(0) = \tau g_0/2 = 5 \times 10^{13}$, $L_p = 110$ µm, $\Delta p(50\ \mu\text{m}) = 5 \times 10^{13}\, e^{-0.456} = 3.2 \times 10^{13}$ cm^{-3}, $J_p(0) = q g_0 L_p/2 = 8.8$ mA/cm². $\tau = 0.1$ µs: $\Delta p(0) = 5 \times 10^{11}$, $L_p = 11$ µm, $\Delta p(50\ \mu\text{m}) = 5 \times 10^{11}\, e^{-4.56} = 5.2 \times 10^9$ cm^{-3}, $J_p(0) = 0.88$ mA/cm². A factor 100 on τ gives 100 on $\Delta p(0)$, 10 on L_p and $J_p(0)$, and about 6000 on Δp at 50 µm, where both effects combine.)

Problem 4.15 A conservation check

B · Guided calculation

From the course notes, Chapter 4

For the steady profile of Case B, compute (i) the number of excess holes per cm² of surface, (ii) the number recombining per cm² and per second, (iii) the hole flux entering at $x = 0$ by diffusion. Compare (ii) and (iii), and show that both equal $g_0 L_p/2$.

(Answer: (i) $\int \Delta p\, dx = (\tau g_0/2) L_p = 5.5 \times 10^{11}$ cm^{-2}. (ii) Recombination integrated over depth: $(\tau g_0/2) L_p/\tau = g_0 L_p/2 = 5.5 \times 10^{16}$ cm^{-2}s^{-1}. (iii) $D_p \Delta p(0)/L_p = D_p \tau g_0/(2L_p) = g_0 L_p/2 = 5.5 \times 10^{16}$ cm^{-2}s^{-1}, because $D_p \tau = L_p^2$: what diffuses in is exactly what recombines.)

Problem 4.16 The internal field

B · Guided calculation

From the course notes, Chapter 4

For the steady profile of Case B, compute $\mathcal{E}$ at $x = 0$ and at $x = L_p$ from (4.24), and the ratio of hole drift to hole diffusion current at both points. Then compute the distance $\mu_p \mathcal{E} \tau$ covered by drift in one lifetime in an applied field of 10 V/cm and compare it with L_p.

(Answer: At $x = 0$: $\mathcal{E} \approx (D_p - D_n)(-\Delta p(0)/L_p)/(\mu_n n_0) = 7.7$ mV/cm, pointing into the sample; at $x = L_p$ the gradient is divided by e: 2.8 mV/cm. Drift/diffusion $\approx (\Delta p/n_0)(1 - D_p/D_n)$: 0.33% at $x = 0$, 0.12% at $x = L_p$. Applied 10 V/cm: $\mu_p \mathcal{E} \tau = 460 \times 10 \times 10^{-5} = 0.046$ cm = 460 µm, about four times L_p: an imposed field can dominate, the internal one cannot.)

Problem 4.17 Challenge: the edge of the light, in steady state

B · Guided calculation

From the course notes, Chapter 4

The illuminated layer occupies $x < 0$ (with $g = g_0$) and is very thick; the dark bulk is $x > 0$. Solve the steady-state form of (4.26) on both sides (bounded solutions), and match Δp and $d\Delta p/dx$ at $x = 0$. Show that $\Delta p(0) = \tau g_0/2$, as (4.27) gives for $t \to \infty$, and sketch the whole profile.

(Answer: For $x < 0$: $\Delta p = \tau g_0 + A\,\mathrm{e}^{x/L}$; for $x > 0$: $\Delta p = B\,\mathrm{e}^{-x/L}$. Continuity of Δp: $\tau g_0 + A = B$; continuity of the slope (of the current): $A/L = -B/L$. Hence $B = \tau g_0/2$, $A = -\tau g_0/2$: $\Delta p = \tau g_0(1 - \frac{1}{2}\mathrm{e}^{x/L})$ in the light, $\frac{1}{2}\tau g_0\,\mathrm{e}^{-x/L}$ in the dark. The profile is symmetric about the value $\tau g_0/2$ at the edge and reaches τg_0 a few L inside the light.)

Problem 4.18 A new situation

B · Guided calculation

From the course notes, Chapter 4

Repeat the steady state of Case B for p-type silicon, $N_A = 10^{16}\,\mathrm{cm}^{-3}$, with the same g_0, $D_n = 35$ cm²/s and $\tau_n = 10$ µs. No formula of the chapter is specific to n-type material: say which quantity plays which role.

(Answer: Swap $n \leftrightarrow p$: the minority carriers are electrons, $D_a \to D_n$, $L_n = \sqrt{D_n \tau_n} = \sqrt{35 \times 10^{-5}} = 1.87 \times 10^{-2}$ cm = 187 µm, $\Delta n(x) = (\tau_n g_0/2)\,\mathrm{e}^{-x/L_n}$ with $\tau_n g_0/2 = 5 \times 10^{13}\,\mathrm{cm}^{-3}$, majority holes unchanged to within 0.5%. Current at the boundary $|J_n(0)| = q g_0 L_n/2 = 15$ mA/cm².)

Problem 4.19 Electrons and holes

B · Guided calculation

From the course notes, Chapter 4

(i) From $\mu_n = 1350$ and $\mu_p = 460$ cm²/(V·s) find D_n and D_p at 300 K. (ii) Compute D_a for $n = p$. (iii) In the n-type sample of Case B ($n_0 = 10^{16}\,\mathrm{cm}^{-3}$) the excess is raised to $\Delta n = 10^{17}\,\mathrm{cm}^{-3}$. Compute D_a, say whether low-level injection still holds and what that does to the diffusion length.

(Answer: (i) $D = \mu kT/q$: $D_n = 1350 \times 0.0259 = 35.0$ and $D_p = 460 \times 0.0259 = 11.9$ cm²/s. (ii) $n = p$: $D_a = 2 \times 35 \times 11.9/(35 + 11.9) = 17.8$ cm²/s. (iii) With $n = 10^{16} + 10^{17}$ and $p = 10^{17}$: $D_a = (n + p)/(n/D_p + p/D_n) = 17.4$ cm²/s, 46% above D_p. Low-level injection no longer holds

($\Delta n = 10\,n_0$), so the lifetime is also no longer constant (Figure 4.2) and $L = \sqrt{D_a \tau}$ loses its meaning as a fixed length.)

Problem 4.20 The diffusion front — B · Guided calculation

From the course notes, Chapter 4

In the sample of Case B ($\tau = 10$ µs, $D_p = 12$ cm²/s, $\tau g_0/2 = 5 \times 10^{13}$ cm^{-3}): (i) compute the edge value $\Delta p(0, t)$ at $t = 1$, 10 and 30 µs; (ii) with the edge held fixed, at what depth is the excess half its edge value at $t = 0.5$ µs? (Use the short-time limit of (4.29) and erfc(0.477) = 0.5.) (iii) After how long would pure diffusion carry the front to L_p? What do you conclude?

(Answer: (i) $\Delta p(0, t) = 5 \times 10^{13}(1 - \mathrm{e}^{-t/\tau})$: 4.8×10^{12} at 1 µs, 3.2×10^{13} at 10 µs, 4.8×10^{13} cm^{-3} (95%) at 30 µs. (ii) $x/(2\sqrt{D_p t}) = 0.477$, i.e. $x = 0.954\sqrt{D_p t} = 0.954 \times \sqrt{12 \times 5 \times 10^{-7}}$ cm = 23 µm, or 0.21 L_p; the full solution (4.29) gives the same value, because at $t = 0.05\,\tau$ recombination has removed almost nothing. (iii) $\sqrt{D_p t} = L_p$ for $t = L_p^2/D_p = \tau = 10$ µs: the front would reach L_p only when recombination has removed most of the carriers. That is why it stops near L_p and why the steady state settles in a few τ.)

Problem 4.21 How long until the steady state? — B · Guided calculation

From the course notes, Chapter 4

(i) Read from Figure 4.11b the time after which the excess at $x = L_p$ reaches 90% of its final value, first with the edge held fixed, then for the real switch-on; give them in µs for $\tau = 10$ µs. (ii) When does the edge itself reach 90% of $\tau g_0/2$? (iii) Why is the point $x = L_p$ slower than the edge in the real switch-on? What do these times become for $\tau = 1$ µs?

(Answer: (i) Edge held fixed: $1.1\,\tau = 11$ µs; real switch-on: $3.0\,\tau = 30$ µs. (ii) $1 - \mathrm{e}^{-t/\tau} = 0.9$ for $t = \tau \ln 10 = 2.3\,\tau = 23$ µs. (iii) The point $x = L_p$ waits for the edge to fill and for diffusion to bring the carriers; the two delays add up. All times are proportional to τ: for $\tau = 1$ µs, 1.1 µs, 3.0 µs and 2.3 µs. A shorter lifetime gives a steady state reached sooner, but a shorter reach.)

Problem 4.22 Challenge: checking the erfc solution — B · Guided calculation

From the course notes, Chapter 4

Let $a = x/(2\sqrt{D_p t})$, $b = \sqrt{t/\tau}$ and $L_p = \sqrt{D_p \tau}$. (i) Show that $T_- = \mathrm{e}^{-x/L_p}\,\mathrm{erfc}(a - b)$ and $T_+ = \mathrm{e}^{x/L_p}\,\mathrm{erfc}(a + b)$ each obey (4.28). (Use $d\,\mathrm{erfc}(z)/dz = -(2/\sqrt{\pi})\,\mathrm{e}^{-z^2}$ and $D_p/L_p^2 = 1/\tau$.) (ii) Check that $\frac{\Delta p_s}{2}(T_- + T_+)$ equals Δp_s at $x = 0$ and 0 at $t = 0$ for $x > 0$. (iii) Find its limit for $t \to \infty$.

(Answer: (i) For T_-, let $z = a - b$ and $E = \mathrm{e}^{-x/L_p}$, with $\partial z/\partial x = 1/(2\sqrt{D_p t})$, $\partial z/\partial t = -(a + b)/(2t)$ and $\gamma = (2/\sqrt{\pi})\mathrm{e}^{-z^2}$. Then $\partial T_-/\partial t = E\gamma\,(a + b)/(2t)$ and $\partial^2 T_-/\partial x^2 = T_-/L_p^2 + E\gamma/(L_p\sqrt{D_p t}) + E\gamma z/(2D_p t)$. With $D_p/L_p^2 = 1/\tau$ and $D_p/(L_p\sqrt{D_p t}) = b/t$: $D_p\,\partial^2 T_-/\partial x^2 - T_-/\tau = E\gamma\,[b/t + (a - b)/(2t)] = E\gamma\,(a + b)/(2t) = \partial T_-/\partial t$. The calculation for T_+ is the same with $L_p \to -L_p$ and $b \to -b$. (ii) At $x = 0$, $a = 0$: $\frac{1}{2}[\mathrm{erfc}(-b) + \mathrm{erfc}(b)] = 1$. For $t \to 0$ and $x > 0$: $a \to +\infty$, $b \to 0$, both erfc tend to 0. (iii) For $t \to \infty$: $a - b \to -\infty$ (erfc $\to 2$) and $a + b \to +\infty$ (erfc $\to 0$), hence $\Delta p \to \Delta p_s\,\mathrm{e}^{-x/L_p}$, the steady profile.)

Challenge: a new situation
These problems apply the method of the chapter to a device, a material or a situation that the chapter did not treat. The reasoning does not change: state the physical picture, choose the model, compute, check.

Problem 4.23 Ambipolar diffusion in germanium

C · Synthesis

Adapted from Neamen, Problem 6.10, p. 234

Germanium at 300 K ($n_i = 2.4 \times 10^{13}\,\text{cm}^{-3}$, $\mu_n = 3900$, $\mu_p = 1900$ cm²/(V·s)) is doped with $N_D = 4 \times 10^{13}\,\text{cm}^{-3}$ donors. The excess-carrier lifetime is 2 µs.

(a) Compute n_0 and p_0, without the majority-carrier approximation. Why is it not valid here?

(b) Compute D_n, D_p and the ambipolar coefficient D_a in low injection. Compare D_a with D_p, which it would equal in strongly n-type material.

(c) Compute the diffusion length of the excess carriers.

(d) Explain physically why the excess electrons and holes diffuse together with a coefficient that lies between D_n and D_p.

HINT
$n_0 = \frac{N_D}{2} + \sqrt{\frac{N_D^2}{4} + n_i^2}$ (Chapter 2).

(Answer: (a) $n_0 = 5.12 \times 10^{13}$, $p_0 = 1.12 \times 10^{13}\,\text{cm}^{-3}$; (b) $D_n = 101$, $D_p = 49.2$, $D_a = 54.2$ cm²/s; (c) $L = 104$ µm.)

SOLUTION
(a) N_D is comparable to n_i, so thermal pairs are not negligible. $n_0 = 2 \times 10^{13} + \sqrt{4 \times 10^{26} + 5.76 \times 10^{26}} = 5.12 \times 10^{13}\,\text{cm}^{-3}$, $p_0 = n_i^2/n_0 = 1.12 \times 10^{13}\,\text{cm}^{-3}$.
(b) $D_n = 3900 \times 0.0259 = 101$ cm²/s and $D_p = 1900 \times 0.0259 = 49.2$ cm²/s. In low injection $n \simeq n_0$, $p \simeq p_0$: $D_a = \dfrac{n_0 + p_0}{n_0/D_p + p_0/D_n} = \dfrac{6.24 \times 10^{13}}{1.04 \times 10^{12} + 1.1 \times 10^{11}} =$ 54.2 cm²/s, 10 % more than D_p. The minority carriers are not rare enough here for the majority carriers to follow them without effect.
(c) $L = \sqrt{D_a \tau} = \sqrt{54.2 \times 2 \times 10^{-6}} = 1.04 \times 10^{-2}$ cm = 104 µm.
(d) If electrons and holes diffused independently, the faster electrons would run ahead and separate charge; the resulting field slows the electrons and speeds up the holes until both move together, keeping the excess neutral. The common coefficient weights each carrier by the density of the *other*: when $p \ll n$, the many majority electrons screen at once and the pair diffuses at the speed of the minority holes, $D_a \to D_p$; when $n \simeq p$ both matter, and $D_a \to 2D_nD_p/(D_n + D_p)$.

Problem 4.24 Minority carriers in an applied field

C · Synthesis

Adapted from Neamen, Problem 6.22, p. 236

In n-type silicon, holes are generated in a thin plane at $x = 0$ and spread on both sides. A uniform field $\mathcal{E}_0$ is applied along $+x$. The hole lifetime is $\tau_p = 10$ µs.

(a) Write the steady-state continuity equation for Δp with diffusion, drift and recombination, and show that it has solutions e^{sx} with $s = \frac{1}{L_p}\left(\beta \pm \sqrt{1+\beta^2}\right)$, $\beta = \frac{\mu_p \mathcal{E}_0 L_p}{2D_p}$.

(b) Choose the root that keeps Δp finite on each side, and give the decay lengths downstream ($x > 0$) and upstream ($x < 0$).

(c) Evaluate them for $\mathcal{E}_0 = 0$ and for $\mathcal{E}_0 = 10$ V/cm. Sketch the profiles.

(d) Explain the result in terms of the distance drifted in one lifetime.

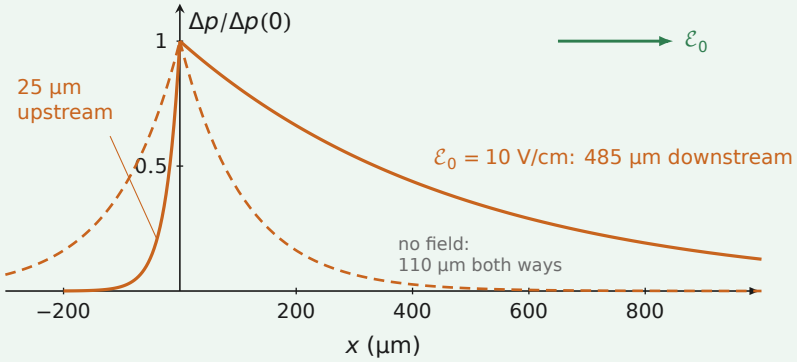
HINT

The hole flux is $\mu_p \mathcal{E}_0 p - D_p\, \mathrm{d}p/\mathrm{d}x$; in the n-type material the field does not change p_0.

(Answer: (c) $L_p = 110$ µm both ways for $\mathcal{E}_0 = 0$; for 10 V/cm, $\beta = 2.10$: 485 µm downstream, 25 µm upstream.)

SOLUTION

(a) $0 = -\frac{\mathrm{d}}{\mathrm{d}x}\left(\mu_p \mathcal{E}_0 \Delta p - D_p \frac{\mathrm{d}\Delta p}{\mathrm{d}x}\right) - \frac{\Delta p}{\tau_p}$, i.e. $D_p \Delta p'' - \mu_p \mathcal{E}_0 \Delta p' - \Delta p/\tau_p = 0$. With e^{sx}: $D_p s^2 - \mu_p \mathcal{E}_0 s - 1/\tau_p = 0$. Dividing by D_p and using $D_p \tau_p = L_p^2$ and $\mu_p \mathcal{E}_0 / D_p = 2\beta/L_p$: $s^2 - (2\beta/L_p)s - 1/L_p^2 = 0$, whose roots are $s = (\beta \pm \sqrt{1+\beta^2})/L_p$.
(b) The root with + is positive, the one with − negative. For $x > 0$ the solution must decay: $\Delta p \propto e^{-x/L_{down}}$ with $1/L_{down} = (\sqrt{1+\beta^2} - \beta)/L_p$. For $x < 0$: $\Delta p \propto e^{x/L_{up}}$ with $1/L_{up} = (\sqrt{1+\beta^2} + \beta)/L_p$.
(c) $L_p = \sqrt{12 \times 10^{-5}} = 110$ µm. Without field, $\beta = 0$ and both lengths equal L_p: a symmetric profile. With 10 V/cm, $\beta = 460 \times 10 \times 1.10 \times 10^{-2}/24 = 2.10$ and $\sqrt{1+\beta^2} = 2.33$: $L_{down} = 110/0.226 = 485$ µm and $L_{up} = 110/4.43 = 25$ µm. The profile is strongly skewed downstream.
(d) In one lifetime the holes drift $\mu_p \mathcal{E}_0 \tau_p = 460 \times 10 \times 10^{-5} = 0.046$ cm = 460 µm, more than four diffusion lengths. Downstream, drift carries them far before they recombine, and $L_{down} \approx \mu_p \mathcal{E}_0 \tau_p$ when $\beta \gg 1$; upstream, diffusion must fight the drift and they get only a short way. This is how a field collects minority carriers, as the field of a junction does in Chapter 6.



A field stretches the profile downstream and compresses it upstream.

SEMICONDUCTOR DEVICES PHYSICS · SOLUTIONS

CHAPTER 5

The PN Junction at Equilibrium

WHAT THIS SET USES

The constant Fermi level; the depletion layer and the depletion approximation; Poisson's equation, the triangular field and the parabolic potential; $V_d = \frac{kT}{q}\ln\frac{N_A N_D}{n_i^2}$; $W = \sqrt{\frac{2\varepsilon V_d}{q}\frac{N_A+N_D}{N_A N_D}}$, $N_A W_p = N_D W_n$, $\mathcal{E}_{max} = qN_D W_n/\varepsilon$; the band diagram and the carrier densities across the junction; N side grounded.

DATA

Silicon at 300 K: $n_i = 10^{10}\,\text{cm}^{-3}$, $kT/q = 25.9$ mV, $\varepsilon = 1.036\times10^{-12}$ F/cm, $N_C = 2.8\times10^{19}$, $N_V = 1.04\times10^{19}\,\text{cm}^{-3}$, $E_g = 1.12$ eV. Abrupt junctions with uniform doping on each side unless stated otherwise.

Problem 5.1 Reading a band diagram

B · Guided calculation

Adapted from Neamen, Problem 7.6, p. 269

A silicon junction at equilibrium is doped so that, in the neutral regions, $E_F - E_i = 0.365$ eV on the N side and $E_i - E_F = 0.330$ eV on the P side.

(a) Sketch the band diagram of the two pieces apart, then of the junction, with E_F flat. Mark qV_d.

(b) Deduce N_D and N_A.

(c) Read V_d on the diagram, and check it with the formula for V_d.

HINT

On the diagram, qV_d is the step of E_i between the two neutral regions, which is the distance between the two Fermi levels of the pieces apart.

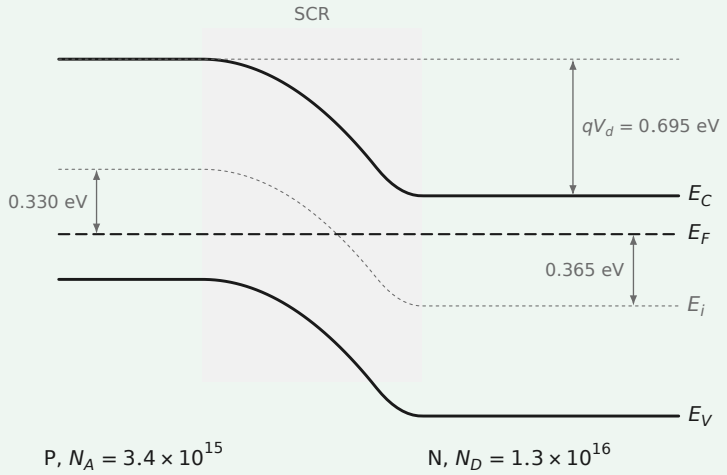
(Answer: (b) $N_D = 1.32\times10^{16}$, $N_A = 3.42\times10^{15}\,\text{cm}^{-3}$; (c) $V_d = 0.695$ V.)

SOLUTION

(a) Apart, the Fermi level of the N piece is 0.365 eV above its E_i, that of the P piece 0.330 eV below. In contact E_F is the same everywhere; the bands of the P side rise with respect to those of the N side by the difference of the two Fermi levels, which is qV_d.

(b) $N_D = n_0 = n_i\,\text{e}^{0.365/0.0259} = 10^{10}\times1.32\times10^{6} = 1.32\times10^{16}\,\text{cm}^{-3}$ and $N_A = p_0 = n_i\,\text{e}^{0.330/0.0259} = 3.42\times10^{15}\,\text{cm}^{-3}$.

(c) $qV_d = 0.365 + 0.330 = 0.695$ eV, so $V_d = 0.695$ V. The formula gives $V_d = 0.0259\ln(1.32\times10^{16}\times3.42\times10^{15}/10^{20}) = 0.0259\times26.84 = 0.695$ V: the same, since $\ln(N_A N_D/n_i^2) = \ln(N_D/n_i) + \ln(N_A/n_i)$.



The junction at equilibrium, with the flat Fermi level (energies to scale, distances not).

Problem 5.2 From the SCR back to the dopings

B · Guided calculation

Adapted from Neamen, Problem 7.8, p. 270

A silicon junction at equilibrium has $V_d = 0.710$ V, and 25 % of its SCR lies in the N region.

(a) Using the neutrality of the SCR, find the ratio N_A/N_D. Which side is more heavily doped?

(b) Find N_A and N_D.

(c) Compute W, W_n, W_p and $\mathcal{E}_{max}$.

(d) Sketch the charge density and the field. Where is the field largest?

HINT

$N_A W_p = N_D W_n$: the SCR extends further into the less doped side.

(Answer: (a) $N_A/N_D = 1/3$; (b) $N_A = 5.2 \times 10^{15}$, $N_D = 1.55 \times 10^{16}\,\mathrm{cm}^{-3}$; (c) $W = 0.486$ µm, $W_n = 0.122$ µm, $W_p = 0.365$ µm, $\mathcal{E}_{max} = 2.9 \times 10^4$ V/cm.)

SOLUTION

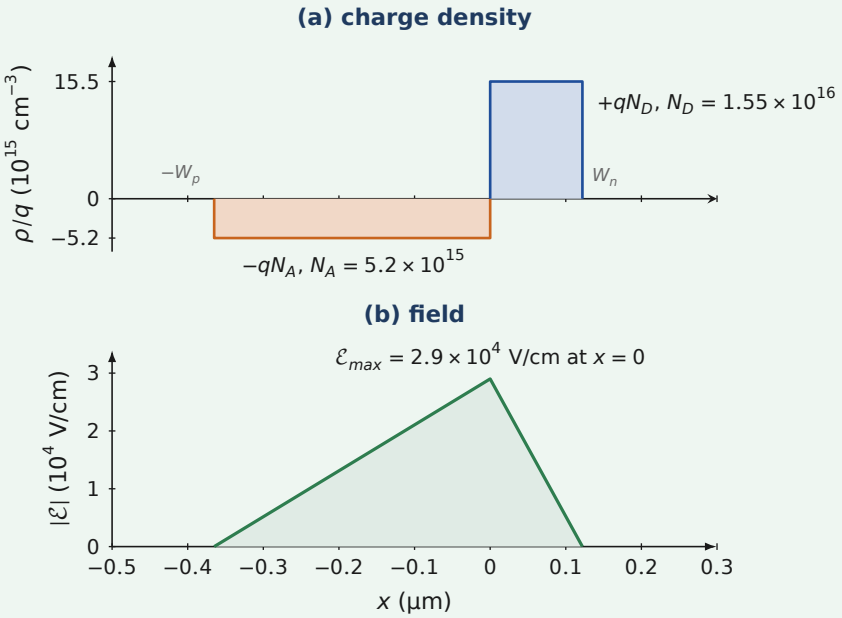
(a) $W_n = W/4$ and $W_p = 3W/4$. Neutrality of the SCR, $N_A W_p = N_D W_n$, gives $N_A/N_D = W_n/W_p = 1/3$: the N side is three times more heavily doped.

(b) $N_A N_D = n_i^2\, \mathrm{e}^{V_d/V_T} = 10^{20}\,\mathrm{e}^{27.41} = 8.04 \times 10^{31}$, and with $N_D = 3N_A$: $N_A = \sqrt{8.04 \times 10^{31}/3} = 5.2 \times 10^{15}\,\mathrm{cm}^{-3}$, $N_D = 1.55 \times 10^{16}\,\mathrm{cm}^{-3}$.

(c) $W = \sqrt{\frac{2 \times 1.036 \times 10^{-12} \times 0.710}{1.602 \times 10^{-19}} \cdot \frac{N_A + N_D}{N_A N_D}} = 4.86 \times 10^{-5}$ cm $= 0.486$ µm; $W_n = 0.122$ µm, $W_p = 0.365$ µm; $\mathcal{E}_{max} = qN_D W_n/\varepsilon = 1.602 \times 10^{-19} \times 1.55 \times 10^{16} \times 1.22 \times 10^{-5}/1.036 \times 10^{-12} = 2.9 \times 10^4$ V/cm.

(d) The charge is $-qN_A$ over W_p and $+qN_D$ over W_n: a lower, wider negative slab

and a taller, narrower positive one, of equal areas. The field is a triangle, maximal at the metallurgical junction, $x = 0$, and pointing from N to P.



Equal charges on the two sides: the lightly doped P side holds the wider part of the SCR. The field is largest at the metallurgical junction.

Problem 5.3 A junction with an almost pure side

B · Guided calculation

Adapted from Neamen, Problem 7.13, p. 270

An n region with $N_D = 10^{16}\,\text{cm}^{-3}$ meets a very lightly doped p region with $N_A = 10^{12}\,\text{cm}^{-3}$, a model for an n region next to nearly intrinsic silicon.

(a) Compute V_d. Why is it much smaller than that of the reference junction (0.835 V)?

(b) Compute W_n, W_p and $\mathcal{E}_{max}$. Sketch the field.

(c) Compare W_p with the reference junction. What would happen to the SCR if the p region were only 10 µm thick?

(Answer: (a) 0.477 V; (b) $W_n = 2.5 \times 10^{-3}$ µm, $W_p = 24.8$ µm, $\mathcal{E}_{max} = 384$ V/cm; (c) 75 times wider; it would reach the contact.)

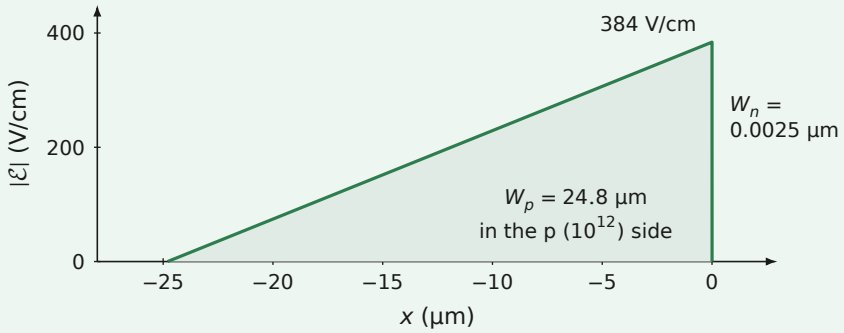
SOLUTION

(a) $V_d = 0.0259\ln(10^{12} \times 10^{16}/10^{20}) = 0.0259\ln 10^8 = 0.477$ V. Each decade of doping on either side adds $V_T \ln 10 = 59.6$ mV; the p side here has six decades less than the reference, which loses $6 \times 0.0596 = 0.36$ V.

(b) $W = \sqrt{2\varepsilon V_d/(qN_A)}$ to excellent accuracy, since $N_A \ll N_D$: $W = 24.8$ µm, almost all on the p side; $W_n = W_p N_A/N_D = 2.5 \times 10^{-3}$ µm. $\mathcal{E}_{max} = qN_A W_p/\varepsilon = 1.602 \times 10^{-19} \times 10^{12} \times 2.48 \times 10^{-3}/1.036 \times 10^{-12} = 384$ V/cm. The field drops almost vertically in the thin N slab and decreases linearly over 25 µm of the p side.

(c) The reference junction has $W = 0.33$ µm; this one is 75 times wider, because the

few acceptors must be uncovered over a long distance to balance the charge. A p region only 10 µm thick would be fully depleted before neutrality is reached; the field would then extend to the contact, as in a p–i–n diode, and the formulas of the abrupt junction with two neutral regions would no longer apply.



Almost the whole SCR lies in the nearly intrinsic side, where the field is small.

Problem 5.4 The built-in voltage and temperature — B · Guided calculation

Adapted from Sze & Lee, Ch. 3, Problem 3, p. 121

An abrupt silicon junction has $N_A = 10^{17}$ and $N_D = 10^{15}\,\text{cm}^{-3}$.

(a) Compute V_d at 250, 300, 350, 400, 450 and 500 K, using the expression of $n_i(T)$ of Problem 3.6 with $E_g = 1.12$ eV and $kT/q = 25.9\,(T/300)$ mV. Plot $V_d(T)$.

(b) Explain the trend with the band diagram: what happens to the Fermi levels of the two pieces apart as T rises?

(c) Up to which temperature are the dopings still the majority densities on both sides? What happens to V_d beyond?

(d) At 300 K compute W and $\mathcal{E}_{max}$.

(Answer: (a) 0.795, 0.716, 0.634, 0.550, 0.465, 0.379 V; (d) $W = 0.97$ µm, $\mathcal{E}_{max} = 1.5\times10^4$ V/cm.)

SOLUTION

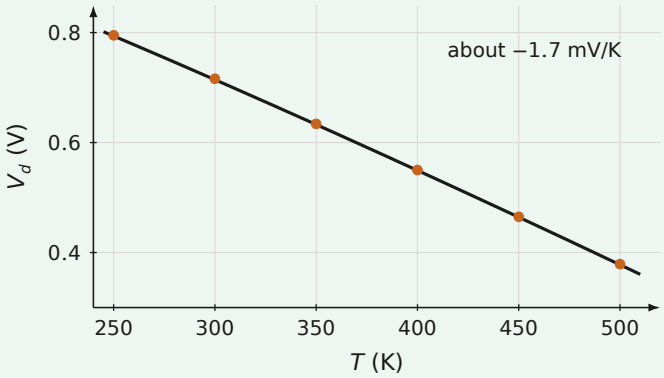
(a) $V_d = \frac{kT}{q}\ln\frac{N_A N_D}{n_i(T)^2}$. With $n_i = 1.0\times10^8$, 10^{10}, 2.8×10^{11}, 3.5×10^{12}, 2.5×10^{13} and $1.25\times10^{14}\,\text{cm}^{-3}$: $V_d = 0.795$, 0.716, 0.634, 0.550, 0.465 and 0.379 V. The decrease is almost linear, about 1.7 mV/K.

(b) As T rises, n_i grows quickly, and each Fermi level moves towards the intrinsic level of its piece: $E_F - E_i = kT\ln(N_D/n_i)$ decreases on the N side, $E_i - E_F$ on the P side. The step qV_d, which is the distance between the two Fermi levels apart, shrinks.

(c) The less doped side loses its extrinsic character first, when n_i approaches $N_D = 10^{15}\,\text{cm}^{-3}$; from Chapter 3 this happens around 520 K–600 K. Beyond, the N side becomes intrinsic, its Fermi level reaches E_i, and V_d falls to zero when both sides are intrinsic: the junction disappears.

(d) $W = \sqrt{2\varepsilon V_d(N_A + N_D)/(qN_A N_D)} = 9.7\times10^{-5}$ cm = 0.97 µm, almost all in the

N side ($W_n = 0.957$ µm), and $\mathcal{E}_{max} = qN_D W_n/\varepsilon = 1.5 \times 10^4$ V/cm.



The built-in voltage falls almost linearly with temperature.

Problem 5.5 Which side controls the field? B · Guided calculation

Adapted from Neamen, Problem 7.15, p. 271

Silicon junctions at equilibrium, 300 K.

(a) With $N_A = 10^{17}\,\text{cm}^{-3}$, compute V_d and $\mathcal{E}_{max}$ for $N_D = 10^{14}$, 10^{15}, 10^{16} and $10^{17}\,\text{cm}^{-3}$.

(b) Show that for $N_A \gg N_D$, $\mathcal{E}_{max} \simeq \sqrt{2qN_D V_d/\varepsilon}$. What does the field depend on?

(c) What can you say about the case $N_A = 10^{14}\,\text{cm}^{-3}$ and the same values of N_D?

(Answer: (a) V_d = 0.656, 0.716, 0.775, 0.835 V; $\mathcal{E}_{max} = 4.5 \times 10^3$, 1.5×10^4, 4.7×10^4, 1.1×10^5 V/cm.)

SOLUTION

(a) $V_d = 0.0259 \ln(N_A N_D/10^{20})$ and $\mathcal{E}_{max} = \sqrt{\frac{2qV_d}{\varepsilon}\frac{N_A N_D}{N_A+N_D}}$, the second form following from $\mathcal{E}_{max} = 2V_d/W$ for a triangular field. Results: $N_D = 10^{14}$: 0.656 V and 4.5×10^3 V/cm; 10^{15}: 0.716 V and 1.48×10^4 V/cm; 10^{16}: 0.775 V and 4.67×10^4 V/cm; 10^{17}: 0.835 V and 1.14×10^5 V/cm.
(b) When $N_A \gg N_D$, $N_A N_D/(N_A + N_D) \simeq N_D$: $\mathcal{E}_{max} \simeq \sqrt{2qN_D V_d/\varepsilon}$. The field depends on the lightly doped side, and on the heavy side only through the logarithm in V_d. From 10^{14} to 10^{16} the field grows almost as $\sqrt{N_D}$ (factor 10 for a factor 100).
(c) The roles are exchanged: with $N_A = 10^{14}$, the field is controlled by N_A as long as $N_D \gg N_A$, and stays near $\sqrt{2qN_A V_d/\varepsilon} \approx 5\times10^3$ V/cm, rising only slowly through V_d. Whenever one side is a hundred times more doped than the other, the junction is **one-sided**: everything (width, field) is fixed by the lightly doped side.

Problem 5.6 How sharp are the edges of the SCR? C · Synthesis

Adapted from Neamen, Problem 7.14, p. 270

The depletion approximation assumes that the carriers vanish abruptly at the edges of the SCR. In reality the transition from the SCR to the neutral region takes a few Debye lengths, $L_D = \sqrt{\varepsilon kT/(q^2 N_D)}$ on the N side.

(a) Show that L_D is a length, and interpret it: over L_D, the potential changes by kT/q when the charge density is qN_D.

(b) With $N_A = 8\times10^{17}\,\text{cm}^{-3}$, compute L_D, W_n and the ratio L_D/W_n for $N_D = 8\times10^{14}$, 2.2×10^{16} and $8\times10^{17}\,\text{cm}^{-3}$.

(c) The ratio barely changes with the doping. Show that $L_D/W_n \simeq \sqrt{\frac{kT/q}{2V_d}}$ for a one-sided junction and explain why. Is the depletion approximation acceptable?

HINT

For a slab of charge density qN_D and thickness L, Poisson gives a potential drop $qN_D L^2/(2\varepsilon)$.

(Answer: (b) L_D = 0.145, 0.028, 0.0046 µm; W_n = 1.11, 0.220, 0.028 µm; ratios 0.13, 0.13, 0.17.)

SOLUTION

(a) $\varepsilon kT/q^2 N_D$ has units (F/cm)(V)/(C·cm^{-3}) = (C/(V·cm))·V·cm³/C = cm²: L_D is a length. From Poisson, a charge density qN_D over a length L creates a potential drop of order $qN_D L^2/\varepsilon$; setting it equal to kT/q gives $L \sim L_D$. Over a Debye length the potential changes by a thermal voltage, which changes the carrier density by a factor e: this is the distance over which the edge of the SCR is 'blurred'.

(b) L_D = 0.145 µm, 0.028 µm and 0.0046 µm. W_n from the general formula: 1.11 µm, 0.220 µm and 0.0276 µm (in the symmetric case, half of W). Ratios: 0.13, 0.13 and 0.17.

(c) For a one-sided junction $W_n = \sqrt{2\varepsilon V_d/qN_D}$, so $L_D/W_n = \sqrt{\frac{\varepsilon kT/(q^2N_D)}{2\varepsilon V_d/(qN_D)}} = \sqrt{\frac{kT/q}{2V_d}}$: both lengths scale as $1/\sqrt{N_D}$, and the ratio depends only on V_d, which varies logarithmically. With $V_d \approx 0.75$ V, $\sqrt{0.0259/1.5} = 0.13$. The edge is blurred over about 13 % of the SCR width on each side: the depletion approximation is a good description of the SCR as a whole, and it improves under reverse bias, where V_d is replaced by $V_d - V$.

Practice exercises from the course notes

These exercises closed Chapter 5 of the course notes; they now live here, with their answers. Tables, equations and sections cited by number are those of the course notes. Silicon at 300 K, with the data of Table 5.1, unless stated otherwise.

Problem 5.7 Another junction

B · Guided calculation

From the course notes, Chapter 5

$N_A = 10^{17}\,\text{cm}^{-3}$, $N_D = 10^{15}\,\text{cm}^{-3}$. Compute the positions of E_F in the two pieces apart, V_d by both routes, W, W_n, W_p and $\mathcal{E}_{max}$. Compare with the reference junction.

(Answer: N piece: $E_F - E_i = 0.0259\ln(10^5) = 0.298$ eV; P piece: $E_i - E_F = 0.0259\ln(10^7) = 0.417$ eV; $V_d = 0.716$ V by both routes. $W = 0.97$ µm; $W_n = 0.96$ µm; $W_p = 9.6$ nm; $\mathcal{E}_{max} = 14.8$ kV/cm. Lower doping gives a smaller V_d, a wider layer and a weaker field.)

Problem 5.8 The hole current with E_F

B · Guided calculation

From the course notes, Chapter 5

Starting from $J_p = q\mu_p p\mathcal{E} - qD_p\,dp/dx$ and $p = n_i\,\text{e}^{(E_i - E_F)/kT}$, show that $J_p = \mu_p p\,dE_F/dx$.

(Answer: $dp/dx = (p/kT)(dE_i/dx - dE_F/dx) = (p/kT)(q\mathcal{E} - dE_F/dx)$. With $D_p = \mu_p kT/q$: $-qD_p\,dp/dx = -q\mu_p p\mathcal{E} + \mu_p p\,dE_F/dx$; adding $q\mu_p p\mathcal{E}$ leaves $J_p = \mu_p p\,dE_F/dx$.)

Problem 5.9 A bar and a battery

B · Guided calculation

From the course notes, Chapter 5

A bar of N silicon, $N_D = 10^{16}\,\text{cm}^{-3}$, 1 mm long, carries 10 mA/cm². (i) Its resistivity and the field? (ii) The slope of E_F in eV/cm, from $J_n = \mu_n n\,dE_F/dx$; compare with $q\mathcal{E}$. (iii) The total drop of E_F along the bar, and its relation to the voltage across it.

(Answer: (i) $\rho = 1/(qN_D\mu_n) = 0.46$ Ω·cm; $\mathcal{E} = \rho J = 4.6$ mV/cm. (ii) $dE_F/dx = J/(\mu_n N_D)$ in eV/cm when J is divided by q: $10^{-2}/(1.602\times10^{-19}\times1350\times10^{16}) = 4.6\times10^{-3}$ eV/cm, equal to $q\mathcal{E}$: E_F is parallel to the bands. (iii) 4.6×10^{-4} eV over 1 mm: q times the voltage across the bar, 0.46 mV.)

Problem 5.10 V_d and doping

B · Guided calculation

From the course notes, Chapter 5

(i) Show that V_d grows by 59.6 mV per decade of N_A or of N_D. (ii) Compute V_d for $N_D = 10^{16}$ and $N_A = 10^{15}$, 10^{16}, 10^{17}. (iii) Show that $qV_d = E_g - kT\ln(N_C/N_D) - kT\ln(N_V/N_A)$ and evaluate it with $N_C = 2.8\times10^{19}$, $N_V = 1.04\times10^{19}\,\text{cm}^{-3}$; why does it differ slightly from 0.835 eV?

(Answer: (i) $V_d = (kT/q)[\ln N_A + \ln N_D - \ln n_i^2]$: multiplying N_A by 10 adds $(kT/q)\ln 10 = 59.6$ mV. (ii) 0.716 V, 0.775 V, 0.835 V. (iii) $1.12 - 0.206 - 0.061 = 0.853$ eV; the difference comes from $n_i = 10^{10}$ being a rounded reference value (6.9×10^9 with these N_C, N_V; Chapter 3).)

Problem 5.11 Holes, and mass action

B · Guided calculation

From the course notes, Chapter 5

(i) Redo route 1 with $J_p = 0$ and recover (5.13). (ii) Show that $n(x)p(x) = n_i^2$ at every point of the SCR. (iii) Where is $n = p$? Compute its position from (5.15) and compare with Figure 5.8. Where is $E_F = E_i$ on the band diagram?

(Answer: (i) $J_p = 0$ gives $(1/p)\,dp/dx = -(q/kT)\,dV/dx$; integrating from x_p ($p = p_{p0}$, $V = -V_d$) to x_n ($p = p_{n0}$, $V = 0$): $\ln(p_{n0}/p_{p0}) = -qV_d/kT$. (ii) $n = n_{n0}\mathrm{e}^{qV/kT}$ and $p = p_{p0}\mathrm{e}^{-q(V+V_d)/kT}$; their product is $N_D N_A \mathrm{e}^{-qV_d/kT} = n_i^2$. (iii) $n = n_i$ where $V = -(kT/q)\ln(N_D/n_i) = -0.358$ V. On the N side, $V = -(qN_D/2\varepsilon)(x - x_n)^2$ gives $x_n - x = \sqrt{2\varepsilon \times 0.358/qN_D} = 0.215$ µm, so $x = 0.11$ µm. There $E_F = E_i$: the intrinsic level crosses the Fermi level.)

Problem 5.12 Poisson, step by step

B · Guided calculation

From the course notes, Chapter 5

(i) Integrate $d^2V/dx^2 = qN_A/\varepsilon$ on the P side with $\mathcal{E}(x_p) = 0$ and $V(x_p) = V_P$ and recover the first line of (5.15). (ii) Derive (5.16) and (5.17). (iii) Solve them for W_n and W_p.

(Answer: (i) $dV/dx = (qN_A/\varepsilon)(x - x_p)$ since $\mathcal{E}(x_p) = 0$; then $V = (qN_A/2\varepsilon)(x - x_p)^2 + V_P$. (ii) Equate dV/dx and V of the two arcs at $x = 0$. (iii) Substitute $W_p = N_D W_n/N_A$ into (5.17): $N_D W_n^2(1 + N_D/N_A) = 2\varepsilon V_d/q$, hence (5.18).)

Problem 5.13 A one-sided junction

B · Guided calculation

From the course notes, Chapter 5

For a $\mathrm{P^+N}$ junction ($N_A \gg N_D$), show that $W \simeq \sqrt{2\varepsilon V_d/qN_D}$ and $\mathcal{E}_{max} \simeq \sqrt{2qN_D V_d/\varepsilon}$. Compute both for $N_D = 10^{14}$, 10^{16}, $10^{18}\,\mathrm{cm^{-3}}$ (take V_d from (5.13) with $N_A = 10^{19}$, ignoring degeneracy).

(Answer: Neglecting $1/N_A$ in (5.18) and using (5.19). $V_d = 0.775$, 0.894, 1.013 V; $W = 3.2$ µm, 0.34 µm, 36 nm; $\mathcal{E}_{max} = 4.9$, 53, 560 kV/cm. At $N_D = 10^{18}$ the one-sided assumption and non-degeneracy are both at their limit.)

Problem 5.14 Band diagram to scale

B · Guided calculation

From the course notes, Chapter 5

Using (5.15) and $E_C(x) = E_{Cn} - qV(x)$, compute $E_C - E_F$ in the reference junction at $x = -0.1$, 0, 0.1, 0.2 and 0.327 µm (take E_i at mid-gap, so $E_C - E_F = 0.202$ eV in neutral N), and check your values against the top of Figure 5.8. Where is the step qV_d located, and where is $E_F = E_i$?

(Answer: $E_C - E_F = 0.202 - V(x)$ in eV: 1.037 eV at −0.1 µm (neutral P), 1.029 eV at 0 (the P side carries only 8 mV), 0.600 eV at 0.1 µm, 0.327 eV at 0.2 µm and 0.202 eV at x_n. Almost all the bending is on the N side, over 0.33 µm; $E_F = E_i$ at $x = 0.11$ µm. (With the course N_C and N_V instead of mid-gap E_i, $E_C - E_F = 0.206$ eV in neutral N and $E_F - E_V = 0.061$ eV in neutral P: the differences come from the rounded n_i, Section 5.8.))

Challenge: a new situation

These problems apply the method of the chapter to a device, a material or a situation that the chapter did not treat. The reasoning does not change: state the physical picture, choose the model, compute, check.

Problem 5.15 A junction without a change of type

C · Synthesis

Adapted from Neamen, Problem 7.12, p. 270

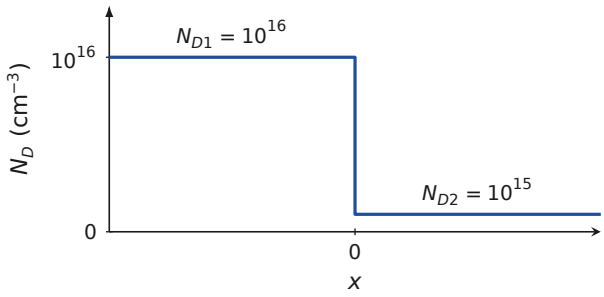
An ‘isotype’ junction joins two n regions of different doping: $N_{D1} = 10^{16}\,\text{cm}^{-3}$ for $x < 0$ and $N_{D2} = 10^{15}\,\text{cm}^{-3}$ for $x > 0$, at equilibrium.

(a) Sketch the band diagram of the two pieces apart and in contact. Find the built-in voltage between the two neutral regions. Which side is at the higher potential?

(b) Describe the charge near $x = 0$: on one side electrons have left, on the other they have accumulated. Why can there be no depletion approximation with a ‘sharp edge’ on the accumulation side?

(c) Estimate the width of the transition region with the Debye lengths of the two sides.

(d) Would this junction rectify? Use the barrier for the majority carriers to argue.



The isotype step. After Neamen, Fig. P7.12, p. 270.

HINT

The step of the bands is the difference of $E_C - E_F$ between the two pieces apart.

(Answer: (a) $V_{bi} = V_T \ln(N_{D1}/N_{D2}) = 59.6$ mV, the 10^{16} side positive; (c) about 0.04 to 0.13 µm; (d) no.)

SOLUTION

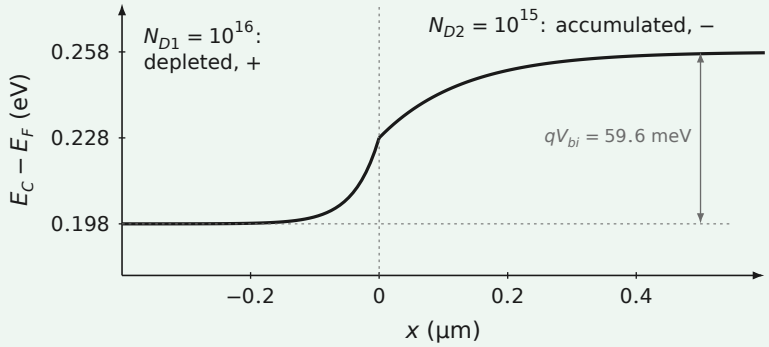
(a) Apart, $E_C - E_F = kT \ln(N_C/N_D)$: 0.146 eV on the 10^{16} side and 0.205 eV on the 10^{15} side. In contact E_F is flat, and the bands of the lightly doped side are higher by 0.0596 eV: $V_{bi} = V_T \ln(N_{D1}/N_{D2}) = 59.6$ mV, with the heavily doped side at the higher potential (its electron energy is lower). This is the result of Problem 1.7 for a step instead of an exponential.

(b) Electrons diffuse from the 10^{16} side into the 10^{15} side. They leave behind a thin layer of uncovered donors (positive) on the heavily doped side and pile up as an accumulation layer (negative) on the lightly doped side. In an accumulation layer the charge is carried by the free electrons themselves, whose density varies continuously and exponentially with the potential: there is no region empty of carriers, hence no sharp edge.

(c) $L_D = \sqrt{\varepsilon kT/q^2 N_D}$ is 0.042 µm on the 10^{16} side and 0.13 µm on the 10^{15} side; the transition region spans a few tenths of a micrometre at most.

(d) No. The barrier for the majority electrons is only 60 mV, about $2kT$, and the

majority carriers exist on both sides: a voltage of either sign simply moves electrons through a low barrier, almost like a resistor. A p–n junction rectifies because the carriers that would have to cross the high barrier in reverse are the rare minority carriers. Isotype junctions are used as ohmic contacts (an N^+N contact) and as 'back-surface fields' in solar cells.



The conduction-band edge: depletion on the heavily doped side, accumulation on the lightly doped one, with no sharp edge.

Problem 5.16 The linearly graded junction

C · Synthesis

Adapted from Sze & Lee, Ch. 3, Problem 6, p. 121

In a diffused junction the net doping varies linearly through the junction: $N_D - N_A = ax$, with $a = 10^{20}$ cm^{-4}. Assume the depletion approximation on $-W/2 < x < W/2$.

(a) Write the charge density, integrate Poisson's equation and show that the field is parabolic, with $\mathcal{E}_{max} = qaW^2/(8\varepsilon)$ at $x = 0$.

(b) Integrate again and show that $V_d = qaW^3/(12\varepsilon)$, i.e. $W = (12\varepsilon V_d/qa)^{1/3}$.

(c) At the edges of the SCR the doping is $\pm aW/2$. Show that $V_d = \frac{2kT}{q}\ln\frac{aW}{2n_i}$.

(d) Solve (b) and (c) together by iteration, starting from $V_d = 0.7$ V. Give V_d, W and $\mathcal{E}_{max}$.

HINT

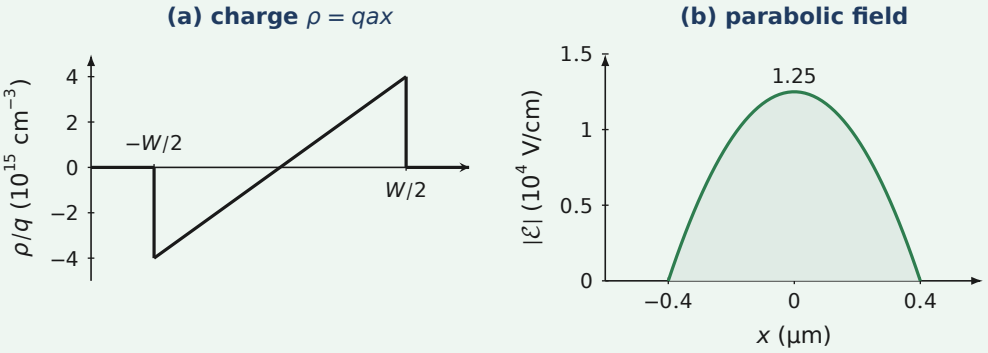
In (c), apply the formula of the abrupt junction to the dopings found at the two edges, $N_A = N_D = aW/2$.

(Answer: (d) $V_d = 0.668$ V, $W = 0.80$ µm, $\mathcal{E}_{max} = 1.25 \times 10^4$ V/cm.)

SOLUTION

(a) $\rho = qax$ in the SCR. Poisson: $\mathrm{d}\mathcal{E}/\mathrm{d}x = qax/\varepsilon$, with $\mathcal{E}(\pm W/2) = 0$: $\mathcal{E}(x) = \frac{qa}{2\varepsilon}\left(x^2 - \frac{W^2}{4}\right)$, negative (pointing from N to P, i.e. towards $-x$) and largest in magnitude at $x = 0$: $|\mathcal{E}_{max}| = qaW^2/(8\varepsilon)$.
(b) $V_d = -\int_{-W/2}^{W/2}\mathcal{E}\,\mathrm{d}x = \frac{qa}{2\varepsilon}\left[\frac{W^2}{4}W - \frac{W^3}{12}\right] = \frac{qaW^3}{12\varepsilon}$.
(c) In the neutral regions near the edges, $n = aW/2$ on the N side and $p = aW/2$ on the P side; the built-in voltage between them is $\frac{kT}{q}\ln\frac{(aW/2)^2}{n_i^2} = \frac{2kT}{q}\ln\frac{aW}{2n_i}$.

(d) Start with $V_d = 0.7$ V: $W = (12 \times 1.036 \times 10^{-12} \times 0.7/1.602 \times 10^{-19} \times 10^{20})^{1/3} = 8.16 \times 10^{-5}$ cm; then $V_d = 0.0518 \ln(10^{20} \times 8.16 \times 10^{-5}/2 \times 10^{10}) = 0.669$ V; again $W = 8.03 \times 10^{-5}$ cm and $V_d = 0.668$ V. Converged: $V_d = 0.668$ V, $W = 0.80$ µm, $\mathcal{E}_{max} = 1.602 \times 10^{-19} \times 10^{20} \times (8.03 \times 10^{-5})^2/(8 \times 1.036 \times 10^{-12}) = 1.25 \times 10^4$ V/cm. The iteration converges fast because V_d depends only logarithmically on W.



Linear charge, parabolic field.

SEMICONDUCTOR DEVICES PHYSICS · SOLUTIONS

CHAPTER 6

The PN Junction Out of Equilibrium

WHAT THIS SET USES

The applied voltage drops across the SCR; low injection and quasi-equilibrium; quasi-Fermi levels, $n = n_i\,\mathrm{e}^{(E_{Fn}-E_i)/kT}$, $p = n_i\,\mathrm{e}^{(E_i-E_{Fp})/kT}$; the edge densities $n_p(x_p) = n_{p0}\mathrm{e}^{V/V_T}$, $p_n(x_n) = p_{n0}\mathrm{e}^{V/V_T}$; diffusion in the neutral regions; the Shockley law $J = J_0(\mathrm{e}^{V/V_T} - 1)$ with $J_0 = qn_i^2\left(\frac{D_n}{N_A d_p} + \frac{D_p}{N_D d_n}\right)$ for short regions and $d \to L$ for long regions; who carries the current; the solar cell and the LED.

DATA

Silicon at 300 K, $n_i = 10^{10}\,\mathrm{cm}^{-3}$, $V_T = 25.9$ mV, $D_n = 35$, $D_p = 12$ cm²/s unless the problem gives other values. Reference diode (long): $N_A = 10^{18}$, $N_D = 10^{16}\,\mathrm{cm}^{-3}$, $\tau_n = 0.1$ µs, $\tau_p = 10$ µs, $L_n = 18.7$ µm, $L_p = 110$ µm, $J_0 = 2.1\times10^{-12}$ A/cm². Short example of the course (Section 6.11): same doping and lifetimes, $d_p = 2$ µm, $d_n = 10$ µm, $J_0 = 2.2\times10^{-11}$ A/cm². Low injection unless stated otherwise.

Problem 6.1 Reading the characteristic

A · Understanding

Adapted from Neamen, Problems 8.1 and 8.9, pp. 323–324, and Sze & Lee, Ch. 3, Problem 10, p. 121

An ideal diode follows $I = I_0(\mathrm{e}^{V/V_T} - 1)$ at 300 K.

(a) In forward bias ($V \gg V_T$), by how much must V increase to multiply the current by 10? By 100?

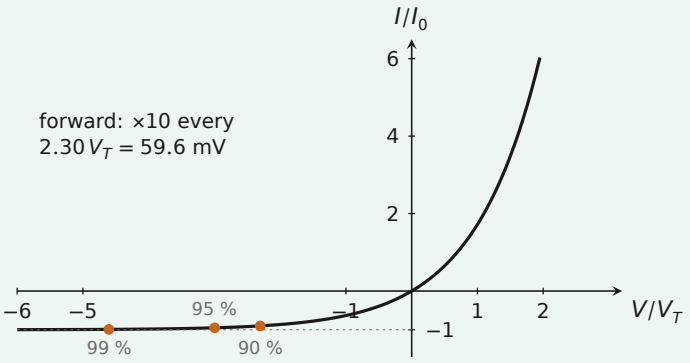
(b) At what reverse voltage does the reverse current reach 90 % of I_0? 95 %? 99 %?

(c) Explain physically, with the minority densities at the edges of the SCR, why the reverse current saturates after a few V_T, whatever the reverse voltage.

(Answer: (a) 59.6 mV and 119 mV; (b) −59.6, −77.6, −119 mV.)

SOLUTION

(a) $I_2/I_1 = \mathrm{e}^{\Delta V/V_T}$, so $\Delta V = V_T \ln(I_2/I_1)$: $25.9\ln 10 = 59.6$ mV per decade, 119 mV for two decades.
(b) $-I/I_0 = 1 - \mathrm{e}^{V/V_T} = 0.9$ gives $\mathrm{e}^{V/V_T} = 0.1$, $V = V_T \ln 0.1 = -59.6$ mV. For 95 %, $V = V_T \ln 0.05 = -77.6$ mV; for 99 %, −119 mV.
(c) In reverse bias the edge densities are $p_{n0}\mathrm{e}^{V/V_T}$ and $n_{p0}\mathrm{e}^{V/V_T}$: after a few V_T they are practically zero. The gradient of the minority density across a neutral region then goes from its equilibrium value at the contact to zero at the edge, and no longer changes: the reverse current is the diffusion of the few minority carriers towards the SCR, which sweeps them across. A larger reverse voltage cannot extract carriers that are not there.



The ideal characteristic in reduced units: saturation in reverse bias after a few V_T, a decade every 59.6 mV in forward bias.

Problem 6.2 Minority densities at the edges

B · Guided calculation

Adapted from Neamen, Problem 8.2, p. 324

A silicon junction has $N_D = 2\times10^{15}$ and $N_A = 8\times10^{15}\,\mathrm{cm}^{-3}$.

(a) Compute the equilibrium minority densities n_{p0} and p_{n0}.

(b) Compute the minority densities at the edges of the SCR for $V = 0.45$ V, 0.55 V and -0.55 V.

(c) At 0.55 V, compare $p_n(x_n)$ with N_D. Is low injection still satisfied?

(d) Check that at each edge $np = n_i^2\,\mathrm{e}^{V/V_T}$. What is the separation $E_{Fn} - E_{Fp}$ there?

(Answer: (a) $n_{p0} = 1.25\times10^4$, $p_{n0} = 5\times10^4\,\mathrm{cm}^{-3}$; (b) 0.45 V: 4.4×10^{11} and 1.8×10^{12}; 0.55 V: 2.1×10^{13} and 8.3×10^{13}; -0.55 V: 7×10^{-6} and $3\times10^{-5}\,\mathrm{cm}^{-3}$; (c) 4 % of N_D: yes; (d) qV.)

SOLUTION

(a) $n_{p0} = n_i^2/N_A = 10^{20}/8\times10^{15} = 1.25\times10^4\,\mathrm{cm}^{-3}$, $p_{n0} = n_i^2/N_D = 5\times10^4\,\mathrm{cm}^{-3}$.

(b) $\mathrm{e}^{V/V_T} = 3.5\times10^7$ at 0.45 V, 1.67×10^9 at 0.55 V, 6×10^{-10} at -0.55 V. Hence $n_p(x_p) = 4.4\times10^{11}$, 2.1×10^{13} and $7.5\times10^{-6}\,\mathrm{cm}^{-3}$; $p_n(x_n) = 1.8\times10^{12}$, 8.3×10^{13} and $3.0\times10^{-5}\,\mathrm{cm}^{-3}$.

(c) $8.3\times10^{13}/2\times10^{15} = 4$ %: the majority density is barely changed, low injection holds. It fails first on the lightly doped side, where p_{n0} is largest.

(d) At x_n: $n \simeq N_D$ and $p = p_{n0}\mathrm{e}^{V/V_T}$, so $np = N_D p_{n0}\mathrm{e}^{V/V_T} = n_i^2\mathrm{e}^{V/V_T}$; the same at x_p. Since $np = n_i^2\mathrm{e}^{(E_{Fn}-E_{Fp})/kT}$, the separation of the quasi-Fermi levels is qV, as quasi-equilibrium says.

Problem 6.3 The limit of low injection

B · Guided calculation

Adapted from Neamen, Problem 8.4, p. 324

A silicon diode has $N_D = 5\times10^{15}$ and $N_A = 5\times10^{16}\,\text{cm}^{-3}$. Low injection is taken to hold as long as the minority density at each edge does not exceed 10 % of the majority density on that side.

(a) Find the largest forward voltage allowed by each side. Which side sets the limit, and why?

(b) The neutral regions are short, $d_p = 2$ µm and $d_n = 10$ µm. Compute J_0 and the current density at the limit voltage.

(c) What changes in the diode when this limit is exceeded? Name two consequences.

> **HINT**
> On the N side, $p_{n0}\text{e}^{V/V_T} \le 0.1\,N_D$.

(Answer: (a) N side 0.620 V, P side 0.739 V: the N side; (b) $J_0 = 9.5\times10^{-11}$ A/cm², $J = 2.4$ A/cm².)

> **SOLUTION**
> (a) N side: $p_{n0} = 2\times10^4\,\text{cm}^{-3}$, $\text{e}^{V/V_T} \le 0.1\times5\times10^{15}/2\times10^4 = 2.5\times10^{10}$, $V \le 0.620$ V. P side: $n_{p0} = 2\times10^3\,\text{cm}^{-3}$, $\text{e}^{V/V_T} \le 5\times10^{15}/2\times10^3 = 2.5\times10^{12}$, $V \le 0.739$ V. The lightly doped N side sets the limit: its minority density starts higher, so it reaches a given fraction of its (smaller) majority density sooner.
> (b) $J_0 = qn_i^2\left(\frac{D_n}{N_A d_p} + \frac{D_p}{N_D d_n}\right) = 16.02\times\left(\frac{35}{5\times10^{16}\times2\times10^{-4}} + \frac{12}{5\times10^{15}\times10^{-3}}\right) = 16.02\times(3.5\times10^{-12} + 2.4\times10^{-12}) = 9.5\times10^{-11}$ A/cm². At 0.620 V, $J = J_0\times2.5\times10^{10} = 2.4$ A/cm².
> (c) The majority density rises in the neutral region to keep it neutral, so the edge density no longer follows $p_{n0}\text{e}^{V/V_T}$ but grows more slowly (as $\text{e}^{V/2V_T}$ in high injection); the current rises more slowly than the Shockley law predicts. At the same time, the current is large enough for the ohmic drop in the neutral regions and contacts to take part of the applied voltage. Both effects bend the characteristic away from the ideal exponential.

Problem 6.4 A long diode

B · Guided calculation

Adapted from Sze & Lee, Ch. 3, Problems 12 and 13, pp. 121–122

An ideal silicon N^+P diode has $N_D = 10^{18}$ and $N_A = 10^{16}\,\text{cm}^{-3}$, lifetimes $\tau_n = \tau_p = 1$ µs, and area 1.2×10^{-5} cm². Both neutral regions are much longer than the diffusion lengths. Take $D_n = 35$ cm²/s for the electrons in the P region and $D_p = 5$ cm²/s for the holes in the heavily doped N region.

(a) Compute L_n and L_p, and the minority profiles on each side in forward bias. Why do they decay exponentially, as in the reference diode, and not linearly as in the short example of the course?

(b) Compute J_0, I_0, and the share of the current carried by electrons.

(c) Compute the current at +0.7 V and −0.7 V, and the voltage for 1 mA.

(d) If the P region were only 2 µm long (short), by what factor would the electron part of J_0 change? Explain with the gradient of the electron profile.

(Answer: (a) L_n = 59 µm, L_p = 22 µm; (b) $J_0 = 9.5\times10^{-12}$ A/cm², $I_0 = 1.1\times10^{-16}$ A, 99.6 % electrons; (c) 62 µA and -1.1×10^{-16} A, 0.772 V; (d) ×29.6.)

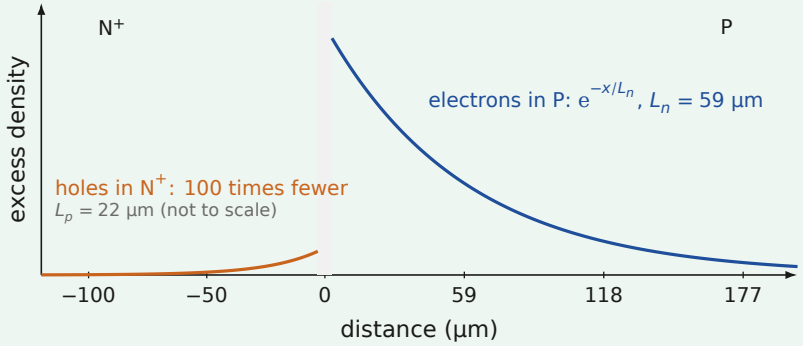
SOLUTION

(a) $L_n = \sqrt{35\times10^{-6}} = 59$ µm and $L_p = \sqrt{5\times10^{-6}} = 22$ µm. In a long region the injected carriers recombine before reaching the contact: $\Delta n(x) = \Delta n(x_p)\,\mathrm{e}^{-(x_p-x)/L_n}$ on the P side, and similarly for holes. In the reference diode the regions are short ($d \ll L$): carriers cross before recombining, and the profile is a straight line to the contact.

(b) $J_0 = qn_i^2\left(\frac{D_n}{N_A L_n}+\frac{D_p}{N_D L_p}\right) = 16.02\times(5.92\times10^{-13}+2.24\times10^{-15}) = 9.5\times10^{-12}$ A/cm², and $I_0 = J_0\times1.2\times10^{-5} = 1.1\times10^{-16}$ A. Electron share: 5.92/(5.92 + 0.022) = 99.6 %: the current is carried by the minority carriers of the lightly doped side.

(c) $I(0.7) = I_0(\mathrm{e}^{27.03}-1) = 1.1\times10^{-16}\times5.5\times10^{11} = 62$ µA; $I(-0.7) = -I_0 = -1.1\times10^{-16}$ A. For 1 mA: $V = V_T\ln(10^{-3}/1.1\times10^{-16}) = 0.772$ V.

(d) The electron term becomes $D_n/(N_A d_p)$ instead of $D_n/(N_A L_n)$: a factor $L_n/d_p = 59/2 = 29.6$ larger. The same edge density falls to zero over 2 µm instead of over 59 µm: the gradient, hence the diffusion current, is 30 times larger.



Long regions: the minority excesses decay exponentially over the diffusion lengths (generic picture, linear axes).

Problem 6.5 Choosing who carries the current

B · Guided calculation

Adapted from Neamen, Problem 8.11, p. 324

A silicon diode has the geometry of the short example of the course (Section 6.11), d_p = 2 µm and d_n = 10 µm, both short.

(a) Show that $J_n/J_p = \dfrac{D_n}{D_p}\dfrac{N_D}{N_A}\dfrac{d_n}{d_p}$. Evaluate it for the short example of the course and check the share of 87 % for holes given there.

(b) What ratio N_D/N_A makes electrons carry 90 % of the current?

(c) What ratio N_D/N_A makes holes carry 80 % of the current?

(d) Which design choice of the bipolar transistor of Chapter 8 uses this result?

> **HINT**
> Each term of J_0 is the current of the minority carriers of one side.

(Answer: (a) 0.146, holes 87.3 %; (b) 0.62; (c) 0.017.)

> **SOLUTION**
> (a) $J_n/J_p = \frac{D_n/(N_A d_p)}{D_p/(N_D d_n)} = \frac{D_n}{D_p}\frac{N_D}{N_A}\frac{d_n}{d_p}$. Reference: $\frac{35}{12}\times\frac{10^{16}}{10^{18}}\times\frac{10}{2} = 0.146$; holes carry $1/1.146 = 87.3$ %.
> (b) $J_n/J_p = 9$: $N_D/N_A = 9\times\frac{12}{35}\times\frac{2}{10} = 0.62$.
> (c) $J_n/J_p = 0.25$: $N_D/N_A = 0.25\times\frac{12}{35}\times\frac{2}{10} = 0.017$, i.e. $N_A \approx 58\,N_D$.
> (d) The emitter of a bipolar transistor must inject almost only the carriers that cross the base: in an NPN transistor the emitter is N and much more heavily doped than the base, so that the electron current from emitter to base dominates the hole current from base to emitter. This is the emitter efficiency γ of Chapter 8, and the factor N_B/N_E appears in it exactly as N_A/N_D does here.

Problem 6.6 The diode as a thermometer

B · Guided calculation

Adapted from Neamen, Problems 8.21 and 8.26, pp. 326–327

Assume that J_0 depends on temperature only through $n_i^2 \propto T^3 e^{-E_g/kT}$, with $E_g = 1.12$ eV constant.

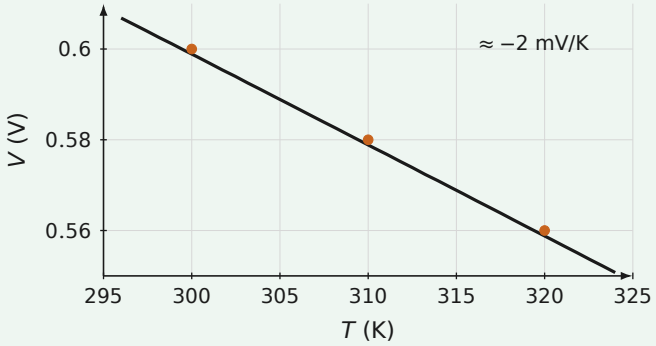
(a) By what factor does J_0 increase from 300 to 310 K? From 300 to 400 K?

(b) A silicon diode is biased at a constant current; at 300 K its voltage is 0.600 V. Show that $V(T) = \frac{kT}{q}\left[\frac{0.600}{0.0259} - \ln\frac{J_0(T)}{J_0(300)}\right]$ and compute V at 310 K and 320 K.

(c) Deduce the temperature coefficient in mV/K. Why does the voltage decrease although kT/q increases?

(Answer: (a) 4.5 and 1.2×10^5; (b) 0.580 and 0.560 V; (c) about −2.0 mV/K.)

> **SOLUTION**
> (a) $\frac{J_0(T)}{J_0(300)} = \left(\frac{T}{300}\right)^3 \exp\left[-\frac{E_g}{k}\left(\frac{1}{T}-\frac{1}{300}\right)\right]$. At 310 K: $1.104\times\exp(1.12\times1.075\times10^{-4}/8.617\times10^{-5}) = 1.104\times4.04 = 4.5$. At 400 K: $2.37\times5.1\times10^4 = 1.2\times10^5$. The reverse current of a silicon diode roughly doubles every 5 K near room temperature.
> (b) At constant $I \gg I_0$: $V = \frac{kT}{q}\ln\frac{I}{I_0(T)} = \frac{kT}{q}\left[\ln\frac{I}{I_0(300)} - \ln\frac{I_0(T)}{I_0(300)}\right]$, and $\ln[I/I_0(300)] = 0.600/0.0259 = 23.17$. At 310 K: $0.02676\times(23.17-1.50) = 0.580$ V. At 320 K: $0.02763\times(23.17-2.91) = 0.560$ V.
> (c) About −2.0 mV/K. Two effects compete: the factor kT/q increases (+), but the saturation current increases much faster (exponentially, through n_i^2), so less voltage is needed for the same current. The second wins. This linear, reproducible drift is used in temperature sensors and in the reference circuits of integrated circuits.



At constant current the forward voltage falls by about 2 mV per kelvin.

Problem 6.7 Quasi-Fermi levels in a short region

C · Synthesis

Adapted from Neamen, Problem 6.37, p. 238

Consider the N region of the short example of the course ($N_D = 10^{16}\,\text{cm}^{-3}$, $d_n = 10\,\mu\text{m}$, short) at $V = +0.5$ V. Let y be the distance from the edge x_n, $0 \le y \le d_n$.

(a) Write the hole density $p(y)$, from the edge value to the equilibrium value at the contact.

(b) Show that $E_{Fn} - E_{Fp} = kT\ln[p(y)/p_{n0}]$ in this region, and compute it at $y = 0$, $d_n/2$, $0.9\,d_n$ and $0.99\,d_n$. Sketch $E_{Fp}(y)$.

(c) The hole current is $J_p = \mu_p p\,\mathrm{d}E_{Fp}/\mathrm{d}y$. Using your profile, explain why the slope of E_{Fp} is small near the edge and large near the contact, although J_p is the same everywhere.

(d) Why is E_{Fn} practically flat in the same region?

HINT

In the N region E_{Fn} stays at the level of the majority electrons; $p = n_i\,\mathrm{e}^{(E_i - E_{Fp})/kT}$ and $n = N_D = n_i\,\mathrm{e}^{(E_{Fn} - E_i)/kT}$.

(Answer: (b) 0.500, 0.482, 0.440, 0.381 eV.)

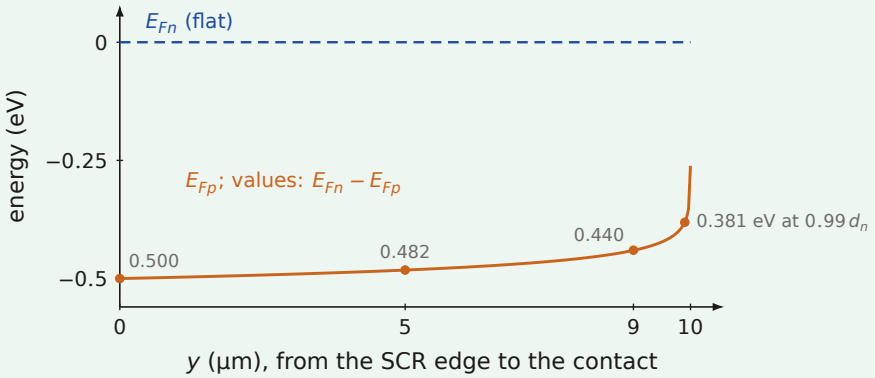
SOLUTION

(a) $p_{n0} = 10^4\,\text{cm}^{-3}$ and $\mathrm{e}^{0.5/0.0259} = 2.4\times10^8$. The short region gives a straight line of the excess: $p(y) = p_{n0} + p_{n0}(\mathrm{e}^{V/V_T} - 1)(1 - y/d_n)$.

(b) $np = n_i^2\,\mathrm{e}^{(E_{Fn}-E_{Fp})/kT}$ with $n = N_D$ gives $\mathrm{e}^{(E_{Fn}-E_{Fp})/kT} = N_D p/n_i^2 = p/p_{n0}$. At $y = 0$: $kT\ln(2.4\times10^8) = 0.500$ eV $= qV$. At $d_n/2$: the excess is halved, $0.500 - kT\ln 2 = 0.482$ eV. At $0.9d_n$: $0.500 - kT\ln 10 = 0.440$ eV. At $0.99d_n$: $0.500 - kT\ln 100 = 0.381$ eV. E_{Fp} stays within a few kT of its edge value over most of the region and drops to E_{Fn} only in the last fraction of a micrometre before the contact.

(c) $\mathrm{d}E_{Fp}/\mathrm{d}y = J_p/(\mu_p p)$: with the same J_p everywhere, the slope is inversely proportional to p. Near the edge p is 2.4×10^8 times p_{n0} and the slope is tiny; near the contact p falls towards p_{n0} and the slope becomes large. The logarithm of a straight

line looks like a plateau ending in a cliff.
(d) $J_n = \mu_n n \, dE_{Fn}/dy$ with $n = 10^{16}\,\text{cm}^{-3}$, a density far larger than p almost everywhere: any current requires only a tiny slope of E_{Fn}. The majority quasi-Fermi level stays flat, as the chapter states.



The hole quasi-Fermi level stays nearly flat near the edge and drops steeply near the contact, where the holes are few.

Practice exercises from the course notes
These exercises closed Chapter 6 of the course notes; they now live here, with their answers. Tables, equations and sections cited by number are those of the course notes. Silicon at 300 K, with the data of Table 6.1, unless stated otherwise.

Problem 6.8 The junction in reverse bias

B · Guided calculation

From the course notes, Chapter 6

For the reference junction, compute W and $\mathcal{E}_{max}$ at -1 V and -10 V. Silicon breaks down (avalanche, outside the model) when the field approaches 3×10^5 V/cm at this doping: at about what reverse voltage?

(Answer: -1 V: $W = 0.49$ µm, $\mathcal{E}_{max} = 75$ kV/cm. -10 V: $W = 1.19$ µm, $\mathcal{E}_{max} = 182$ kV/cm. From $\mathcal{E}_{max}^2 = (2q/\varepsilon)(V_d - V)\, N_A N_D/(N_A + N_D)$: $\mathcal{E}_{max} = 3 \times 10^5$ V/cm for $V_d - V = 29.4$ V, that is, $V \approx -29$ V.)

Problem 6.9 Holes at the N edge

B · Guided calculation

From the course notes, Chapter 6

Using $p = N_V \, e^{-(E_{Fp} - E_V)/kT}$, the flat E_{Fp} across the SCR and $E_{Vp} - E_{Vn} = q(V_d - V)$, show that $p_n(x_n) = p_{n0} e^{qV/kT}$. Check that $np = n_i^2 e^{qV/kT}$ at x_n.

(Answer: $p_n(x_n) = N_V e^{-(E_{Fp} - E_{Vn})/kT} = N_V e^{-(E_{Fp} - E_{Vp})/kT} e^{-(E_{Vp} - E_{Vn})/kT}$, that is, $p_n(x_n) = N_A e^{-q(V_d - V)/kT} = p_{n0} e^{qV/kT}$. At x_n, $n = N_D$: $np = N_D p_{n0} e^{qV/kT} = n_i^2 e^{qV/kT}$.)

Problem 6.10 The saturation current

B · Guided calculation

From the course notes, Chapter 6

(i) Compute the two terms of J_0 for the reference diode and the share of the holes. (ii) What do they become if τ_p is divided by 4 (a poorer N region)? (iii) If N_D is raised to $10^{17}\,\mathrm{cm}^{-3}$ (keeping D_p and τ_p)?

(Answer: (i) Hole term $qn_i^2 D_p/(N_D L_p) = 1.75\times10^{-12}$, electron term $qn_i^2 D_n/(N_A L_n) = 3.0\times10^{-13}$ A/cm²; $J_0 = 2.1\times10^{-12}$ A/cm²; holes 85 %. (ii) $L_p = \sqrt{D_p\tau_p}$ is halved, so the hole term doubles: 3.5×10^{-12}; $J_0 = 3.8\times10^{-12}$ A/cm², holes 92 %. (iii) Hole term divided by 10: 1.75×10^{-13}; $J_0 = 4.8\times10^{-13}$ A/cm², holes 37 %: the junction is no longer one-sided.)

Problem 6.11 Between short and long

B · Guided calculation

From the course notes, Chapter 6

(i) Solve (6.6) with $\Delta p(x_n) = \Delta p_0$ and $\Delta p(x_d) = 0$, and show that

$$J_p(x_n) = \frac{qD_p\Delta p_0}{L_p\tanh(d_n/L_p)}.$$

(ii) Check the two limits. (iii) For the short diode of Section 6.11 ($d_n = 10$ µm, $L_p = 110$ µm), by how much does the exact value differ from the short-diode formula?

(Answer: (i) $\Delta p = \Delta p_0\,\sinh[(x_d - x)/L_p]/\sinh(d_n/L_p)$;
its slope at x_n is $-\Delta p_0/[L_p\tanh(d_n/L_p)]$. (ii) $\tanh u \approx u$ for $u \ll 1$ gives d_n; $\tanh u \to 1$ gives L_p. (iii) $u = 0.091$, $\tanh u/u = 0.997$: the short-diode formula is 0.3 % low.)

Problem 6.12 Where does the voltage go?

B · Guided calculation

From the course notes, Chapter 6

Suppose the N region of a diode with the same doping is 200 µm thick. (i) Compute its resistivity and its resistance for a 1 cm² cross-section. (ii) What voltage drops across it at 10 mA/cm², 1 A/cm² and 100 A/cm²? (iii) What do you conclude about the statement “the voltage drops across the SCR”?

(Answer: (i) $\rho = 1/(qN_D\mu_n) = 0.46$ Ω·cm; $R = \rho\times0.02$ cm = 9.2 mΩ for 1 cm². (ii) 0.09 mV, 9.2 mV, 0.92 V. (iii) True at modest currents; at 100 A/cm² the neutral region would take more voltage than V_d: any extra voltage drops there, and the ideal law no longer applies to the terminal voltage. (For the 10 µm region of the short diode of Section 6.11, the drops are twenty times smaller.))

Problem 6.13 The characteristic

B · Guided calculation

From the course notes, Chapter 6

With $J_0 = 2.1\times10^{-12}$ A/cm² (long reference diode): (i) at what voltages is $J = 1$ µA/cm², 1 mA/cm², 1 A/cm²? (ii) Ratio of the currents at +0.5 V and −0.5 V? (iii) A 1 mm² diode is biased at 0.65 V: what current? (iv) By how much must V rise to multiply the current by 10?

(Answer: (i) $V = (kT/q)\ln(J/J_0+1)$: 0.34 V, 0.52 V, 0.70 V. (ii) $(e^{19.3}-1)/(1-e^{-19.3}) = 2.4\times10^8$. (iii) $J = 2.1\times10^{-12}\times e^{25.1} = 0.16$ A/cm², $I = 1.6$ mA for 10^{-2} cm². (iv) $(kT/q)\ln 10 = 59.6$ mV.)

Problem 6.14 A solar cell

B · Guided calculation

From the course notes, Chapter 6

$J_{ph} = 35$ mA/cm², $J_0 = 10^{-12}$ A/cm². (i) Compute V_{oc} and J_{sc}. (ii) What do they become at 0.1 and 2 suns? (iii) A poorer material gives $J_0 = 10^{-11}$ A/cm²: new V_{oc}? (iv) Write the power $P(V) = -VJ$ and find its maximum numerically. (v) What J_{ph} would give $V_{oc} = V_d$?

(Answer: (i) $V_{oc} = 0.63$ V, $J_{sc} = -35$ mA/cm². (ii) 0.1 sun: 0.57 V and −3.5 mA/cm²; 2 suns: 0.65 V and −70 mA/cm². (iii) $V_{oc} = 0.57$ V: 59.6 mV less per decade of J_0. (iv) Maximum near $V = 0.55$ V, $J = -33.4$ mA/cm², $P = 18.3$ mW/cm²; $P/(V_{oc}|J_{sc}|) = 0.83$. (v) $J_{ph} = J_0 e^{qV_d/kT} = 100$ A/cm², about 3000 suns; long before that, low injection and the model fail.)

Problem 6.15 An LED

B · Guided calculation

From the course notes, Chapter 6

A GaAs LED ($E_g = 1.42$ eV) carries 20 mA. (i) Emitted wavelength? (ii) How many electrons pass through it per second? (iii) If half the recombinations are radiative and all the photons get out, what light power? (iv) The electron diffusion length on the P side is 2 µm and the SCR 0.1 µm: where is the light emitted?

(Answer: (i) $\lambda = 1240/1.42 = 873$ nm, infrared. (ii) $0.02/1.602 \times 10^{-19} = 1.25 \times 10^{17}$ electrons/s. (iii) 6.2×10^{16} photons/s of $1.42 \times 1.602 \times 10^{-19}$ J: 14 mW. (iv) In the P region, within about 2 µm of the edge of the SCR, where the injected electrons recombine; the SCR, twenty times thinner and crossed in a few picoseconds, emits only a small part of it.)

Challenge: a new situation

These problems apply the method of the chapter to a device, a material or a situation that the chapter did not treat. The reasoning does not change: state the physical picture, choose the model, compute, check.

Problem 6.16 A germanium diode

C · Synthesis

Adapted from Neamen, Problem 8.7, p. 324

An ideal germanium diode at 300 K has $N_A = 4 \times 10^{15}$ and $N_D = 2 \times 10^{17}$ cm^{-3}, $D_n = 90$ and $D_p = 48$ cm²/s, $\tau_n = \tau_p = 2$ µs, area 10^{-4} cm², and long neutral regions. For germanium $n_i = 2.4 \times 10^{13}$ cm^{-3}.

(a) Compute L_n, L_p, J_0 and I_0.

(b) Compute the current at +0.25 V and −0.25 V.

(c) Compare J_0 with that of a silicon diode of the same design. Explain the ratio with n_i and the gap, and say why germanium diodes conduct at lower voltage but leak more.

(Answer: (a) $L_n = 134$ µm, $L_p = 98$ µm, $J_0 = 1.6 \times 10^{-4}$ A/cm², $I_0 = 1.6 \times 10^{-8}$ A; (b) 0.24 mA and −16 nA; (c) $(2.4 \times 10^{13}/10^{10})^2 = 5.8 \times 10^6$ times larger.)

SOLUTION

(a) $L_n = \sqrt{90 \times 2 \times 10^{-6}} = 1.34 \times 10^{-2}$ cm, $L_p = \sqrt{48 \times 2 \times 10^{-6}} = 0.98 \times 10^{-2}$ cm. $J_0 = qn_i^2 \left(\frac{D_n}{N_A L_n} + \frac{D_p}{N_D L_p} \right) = 1.602 \times 10^{-19} \times 5.76 \times 10^{26} \times (1.68 \times 10^{-12} + 2.45 \times 10^{-14}) = 1.6 \times 10^{-4}$ A/cm², and $I_0 = 1.6 \times 10^{-8}$ A.

(b) $e^{0.25/0.0259} = 1.56 \times 10^4$: $I(0.25) = 0.24$ mA, and $I(-0.25) = -I_0(1 - e^{-9.65}) = -16$ nA.

(c) Everything else equal, $J_0 \propto n_i^2$: the germanium diode has a saturation current $(2.4 \times 10^{13}/10^{10})^2 = 5.8 \times 10^6$ times larger. For the same current a germanium diode needs $V_T \ln(5.8 \times 10^6) = 0.40$ V less than silicon: it conducts at about 0.2–0.3 V instead of 0.6–0.7 V. The price is a reverse current millions of times larger, and a stronger sensitivity to temperature, because the smaller gap gives far more thermal pairs.

Problem 6.17 Open-circuit voltage of an illuminated diode

C · Synthesis

Adapted from Neamen, Problem 8.36, p. 328

A long silicon diode has $N_D = 10^{18}$, $N_A = 3 \times 10^{16}\,\text{cm}^{-3}$, $\tau_n = \tau_p = 0.1$ µs, $D_n = 18$ and $D_p = 6$ cm²/s. Light absorbed in and near the SCR produces a photocurrent density $J_{ph} = 25$ mA/cm². The diode is open-circuited.

(a) Explain why the photocurrent forward biases the junction when no current can leave the device.

(b) Compute J_0, then V_{oc} from $J = J_0(e^{V/V_T} - 1) - J_{ph} = 0$.

(c) With the device connected to a load, the power delivered is $P = V(J_{ph} - J_0(e^{V/V_T} - 1))$ per cm². Find its maximum numerically, and the fill factor $P_{max}/(V_{oc} J_{ph})$.

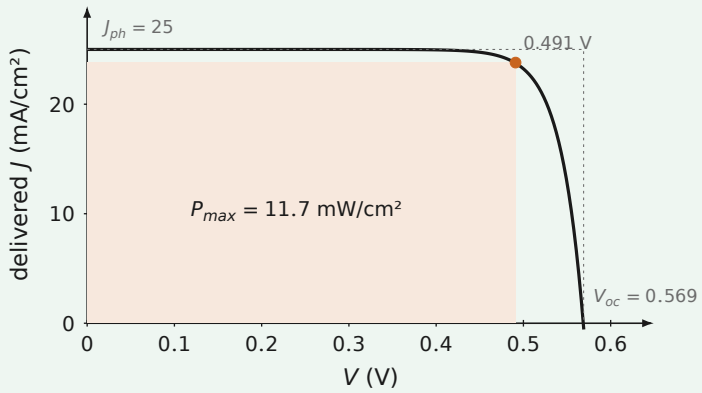
(Answer: (b) $J_0 = 7.3 \times 10^{-12}$ A/cm², $V_{oc} = 0.569$ V; (c) $P_{max} = 11.7$ mW/cm² at 0.491 V, fill factor 0.82.)

SOLUTION

(a) The field of the SCR sweeps photogenerated electrons to the N side and holes to the P side. In open circuit they accumulate there: the N side becomes more negative and the P side more positive, which is a forward bias. The forward bias lowers the barrier and lets an injection current flow back, opposite to the photocurrent, until the two cancel.

(b) $L_n = \sqrt{18 \times 10^{-7}} = 13.4$ µm, $L_p = \sqrt{6 \times 10^{-7}} = 7.7$ µm. $J_0 = 16.02 \times \left(\frac{18}{3 \times 10^{16} \times 1.34 \times 10^{-3}} + \frac{6}{10^{18} \times 7.7 \times 10^{-4}} \right) = 7.3 \times 10^{-12}$ A/cm². $V_{oc} = V_T \ln(1 + J_{ph}/J_0) = 0.0259 \ln(3.4 \times 10^9) = 0.569$ V.

(c) Computing $P(V)$ on a fine grid gives a maximum of 11.7 mW/cm² at $V = 0.491$ V, where $J = 23.8$ mA/cm². The fill factor is $11.7/(0.569 \times 25) = 0.82$. The maximum lies a few V_T below V_{oc}, where the forward current starts to eat into the photocurrent.



The illuminated diode as a generator; the shaded rectangle is the maximum power.

SEMICONDUCTOR DEVICES PHYSICS · SOLUTIONS

CHAPTER 7

The Capacitance Effects of the Junction

WHAT THIS SET USES

The SCR under reverse bias, $W = \sqrt{\frac{2\varepsilon(V_d - V)}{q}\frac{N_A + N_D}{N_A N_D}}$; the transition capacitance $C_T = \varepsilon S/W$; the $1/C_T^2$ plot; the varicap; the stored minority charge $Q = \bar{\tau}_{tr} I$ (transit time $d^2/2D$ for a short region, lifetime for a long one); the diffusion capacitance $C_D = (q/kT)\,\bar{\tau}_{tr} I$; the small-signal diode, $r_d = kT/qI$ in parallel with C_T and C_D; the storage time on switching off, of the order of $\bar{\tau}_{tr} I_F / I_R$.

DATA

Silicon at 300 K, $n_i = 10^{10}\,\text{cm}^{-3}$, $V_T = 25.9$ mV, $\varepsilon = 1.036 \times 10^{-12}$ F/cm, $D_p = 12$ cm²/s. Reference diode: P^+N, $N_A = 10^{18}$, $N_D = 10^{16}\,\text{cm}^{-3}$, $S = 10^{-4}$ cm², $V_d = 0.835$ V, $C_{T0} = 3.15$ pF.

Problem 7.1 Capacitance of a reverse-biased junction — B · Guided calculation

Adapted from Neamen, Problem 7.24, p. 272

A silicon junction has $N_A = 2 \times 10^{15}$, $N_D = 4 \times 10^{16}\,\text{cm}^{-3}$ and area 5×10^{-4} cm².

(a) Compute V_d.

(b) Compute W and C_T at $V = 0$ and at $V = -5$ V.

(c) By what factor does C_T fall? Why is it not $\sqrt{5/0.71}$?

(d) Explain, with the picture of the charges at the edges of the SCR, why $C_T = \varepsilon S/W$ although the charge is spread over the whole SCR.

(Answer: (a) 0.710 V; (b) 0.694 µm, 7.46 pF; 1.97 µm, 2.63 pF; (c) 2.84.)

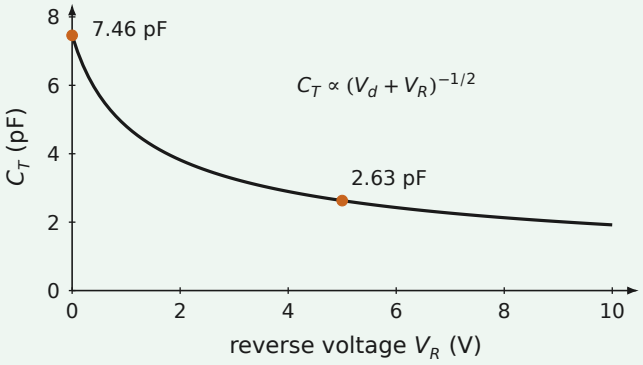
SOLUTION

(a) $V_d = 0.0259 \ln(2 \times 10^{15} \times 4 \times 10^{16}/10^{20}) = 0.710$ V.

(b) At $V = 0$: $W = \sqrt{\frac{2\times1.036\times10^{-12}\times0.710}{1.602\times10^{-19}} \times \frac{4.2\times10^{16}}{8\times10^{31}}} = 6.94 \times 10^{-5}$ cm and $C_T = \varepsilon S/W = 1.036 \times 10^{-12} \times 5 \times 10^{-4}/6.94 \times 10^{-5} = 7.46$ pF. At -5 V the voltage across the SCR is $V_d + 5 = 5.71$ V: $W = 1.97$ µm and $C_T = 2.63$ pF.

(c) $7.46/2.63 = 2.84 = \sqrt{5.71/0.710}$. W grows as $\sqrt{V_d - V}$, and the built-in voltage is part of the voltage across the SCR: it must be added to the applied reverse voltage.

(d) A small change $\mathrm{d}V$ moves only the edges of the SCR: it uncovers (or covers) a thin slab of ions $\mathrm{d}Q$ at each edge, $+\mathrm{d}Q$ on the N side and $-\mathrm{d}Q$ on the P side. The charge inside does not change. Two thin sheets of opposite charge a distance W apart, separated by a dielectric: a parallel-plate capacitor of thickness W.



The transition capacitance falls as the SCR widens.

Problem 7.2 Halving the capacitance

B · Guided calculation

Adapted from Neamen, Problem 7.23, p. 272

A silicon junction has $N_A = 2\times10^{16}$ and $N_D = 5\times10^{15}\,\text{cm}^{-3}$. At a reverse voltage of 0.5 V its capacitance is 1.5 times its value at a larger reverse voltage V_{R2}.

(a) Show that $C_T(V_{R1})/C_T(V_{R2}) = \sqrt{(V_d+V_{R2})/(V_d+V_{R1})}$.

(b) Compute V_d and V_{R2}.

(c) What reverse voltage would halve the capacitance obtained at 0.5 V? Comment on the cost, in volts, of tuning by a factor 2.

(Answer: (b) $V_d = 0.716$ V, $V_{R2} = 2.02$ V; (c) 4.15 V.)

SOLUTION

(a) $C_T = \varepsilon S/W$ with $W \propto \sqrt{V_d+V_R}$: the ratio of capacitances is the inverse ratio of widths.
(b) $V_d = 0.0259\ln(10^{32}/10^{20}) = 0.716$ V. $(V_d+V_{R2}) = 2.25(V_d+0.5) = 2.736$ V, so $V_{R2} = 2.02$ V.
(c) Factor 2 on C_T needs factor 4 on V_d+V_R: $4\times1.216 - 0.716 = 4.15$ V. Because $C_T \propto (V_d+V_R)^{-1/2}$, each halving costs four times the voltage: an abrupt junction is a poor tuning element over a wide range, which is why varicaps use special doping profiles (hyperabrupt).

Problem 7.3 Tuning a resonant circuit

B · Guided calculation

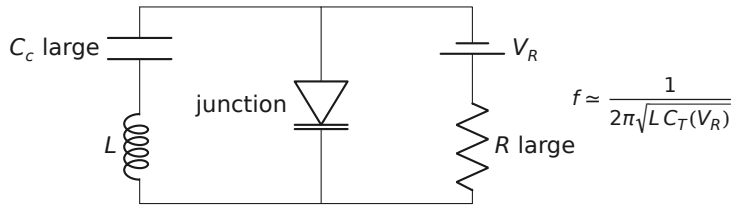
Adapted from Neamen, Problem 7.25, p. 272

An abrupt silicon junction ($N_A = 2\times10^{17}$, $N_D = 5\times10^{15}\,\text{cm}^{-3}$, area 8×10^{-4} cm²) is placed in parallel with an inductance L. At a reverse voltage of 10 V the resonant frequency is 1.25 MHz.

(a) Compute C_T at 10 V and deduce L.

(b) Compute the resonant frequency at reverse voltages of 1 V and 5 V.

(c) Over the range 1–10 V, by what factor does the frequency change? Express the dependence of f on V_R for this junction.



A varicap tuning circuit (the capacitor C_c blocks the bias, the resistor isolates the source).

(Answer: (a) 4.90 pF, $L = 3.31$ mH; (b) 0.796 MHz and 1.07 MHz; (c) 1.57, $f \propto (V_d + V_R)^{1/4}$.)

SOLUTION

(a) $V_d = 0.0259\ln(10^{33}/10^{20}) = 0.775$ V. At 10 V, $W = 1.69$ µm and $C_T = 1.036 \times 10^{-12} \times 8 \times 10^{-4}/1.69 \times 10^{-4} = 4.90$ pF. $f = 1/(2\pi\sqrt{LC})$ gives $L = 1/[(2\pi f)^2 C] = 1/[(7.85 \times 10^6)^2 \times 4.90 \times 10^{-12}] = 3.31$ mH.
(b) At 1 V: $W = 0.686$ µm, $C_T = 12.1$ pF, $f = 0.796$ MHz. At 5 V: $W = 1.24$ µm, $C_T = 6.70$ pF, $f = 1.07$ MHz.
(c) $1.25/0.796 = 1.57$. $f \propto C_T^{-1/2} \propto (V_d + V_R)^{1/4}$: a tenfold voltage range moves the frequency by only about 60 %.

Problem 7.4 Measuring a junction with $1/C^2$ — B · Guided calculation

Adapted from Neamen, Problem 7.30, p. 273

A silicon junction has $N_A = 2 \times 10^{17}$, $N_D = 2 \times 10^{15}$ cm^{-3} and area 10^{-5} cm².

(a) Compute V_d and C_T at reverse voltages of 1, 3 and 5 V.

(b) Compute $1/C_T^2$ at these voltages. Show that it is a straight line in V_R and give its slope.

(c) Pretend that you only know the measured points. Deduce N_D from the slope (one-sided junction) and V_d from the intercept with the voltage axis. Compare with the true values.

HINT

$1/C_T^2 = W^2/(\varepsilon S)^2 = \dfrac{2(V_d + V_R)}{q\varepsilon S^2}\dfrac{N_A + N_D}{N_A N_D}$.

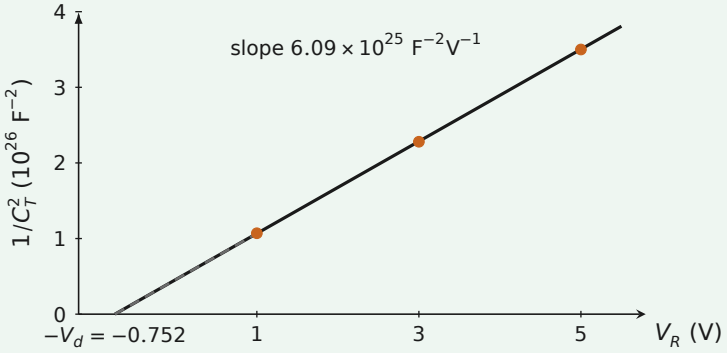
(Answer: (a) 0.752 V; 96.9, 66.2, 53.5 fF; (b) $1/C^2 = 1.07$, 2.28, 3.50×10^{26} F^{-2}, slope 6.09×10^{25} F^{-2}V^{-1}; (c) $N_D = 1.98 \times 10^{15}$ cm^{-3}, intercept -0.752 V.)

SOLUTION

(a) $V_d = 0.0259\ln(4 \times 10^{32}/10^{20}) = 0.752$ V. $C_T = 96.9$, 66.2 and 53.5 fF at 1, 3 and 5 V.
(b) $1/C_T^2 = 1.07 \times 10^{26}$, 2.28×10^{26} and 3.50×10^{26} F^{-2}. The increments are equal for equal steps of V_R: a straight line, of slope $(3.50 - 1.07) \times 10^{26}/4 = 6.09 \times 10^{25}$ F^{-2}V^{-1},

as the formula of the hint predicts: $\frac{2}{q\varepsilon S^2}\frac{N_A+N_D}{N_A N_D}$.

(c) For a one-sided junction the slope is $2/(q\varepsilon S^2 N_D)$, so $N_D = 2/(q\varepsilon S^2 \times \text{slope}) = 1.98 \times 10^{15}\,\text{cm}^{-3}$, 1 % below the true value because the heavy side contributes $N_D/N_A = 1$ %. The line crosses $1/C^2 = 0$ at $V_R = -V_d$, i.e. at −0.752 V. The intercept is an extrapolation, sensitive to errors on each point, which is why V_d is less accurately measured than N_D.



The $1/C^2$ plot: the slope gives the doping of the light side, the intercept gives $-V_d$.

Problem 7.5 Charge stored in a thin neutral region

B · Guided calculation

Adapted from Sze & Lee, Ch. 3, Problem 15, p. 122

An abrupt silicon P^+N junction has $N_D = 10^{16}\,\text{cm}^{-3}$. It is forward biased at 0.7 V. The neutral N region is 1 µm long and the hole diffusion length is 5 µm.

(a) Is the N region short or long? Write the hole excess at the edge and sketch its profile.

(b) Compute the stored hole charge per unit area with the straight-line profile.

(c) The exact profile is $\Delta p(y) = \Delta p_0 \sinh[(d-y)/L_p]/\sinh(d/L_p)$. Integrate it and compare with (b).

(d) Compute the hole current density and the ratio Q/J. Check that it is the transit time $d^2/2D_p$.

(Answer: (a) short; (b) 4.38×10^{-8} C/cm²; (c) 0.3 % less; (d) $J = 105$ A/cm², 0.42 ns.)

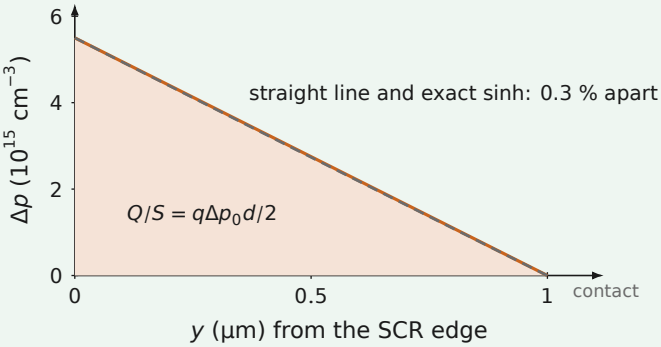
SOLUTION

(a) $d/L_p = 0.2$: short. $\Delta p_0 = p_{n0}(e^{V/V_T} - 1) = 10^4 \times 5.5 \times 10^{11} = 5.5 \times 10^{15}\,\text{cm}^{-3}$, a straight line down to zero at the contact. (Low injection is marginal: Δp_0 is 55 % of N_D; the problem keeps the ideal model.)

(b) The area of the triangle: $Q/S = q\Delta p_0 d/2 = 1.602 \times 10^{-19} \times 5.5 \times 10^{15} \times 10^{-4}/2 = 4.38 \times 10^{-8}$ C/cm².

(c) $\int_0^d \Delta p\,\mathrm{d}y = \Delta p_0 L_p \frac{\cosh(d/L_p)-1}{\sinh(d/L_p)} = \Delta p_0 L_p \tanh\frac{d}{2L_p}$, which gives 4.36×10^{-8} C/cm², 0.3 % less: recombination removes a few carriers on the way, and the straight line is an excellent approximation.

(d) $J = qD_p \Delta p_0/d = 1.602\times10^{-19}\times12\times5.5\times10^{15}/10^{-4} = 105$ A/cm². $Q/J = 4.17\times10^{-10}$ s $= d^2/2D_p = 10^{-8}/24$: the stored charge is the current times the transit time, $Q = \tau_{tr} J$.



The stored charge is the area of the triangle.

Problem 7.6 Diffusion capacitance and the small-signal impedance calculation B · Guided

Adapted from Neamen, Problems 8.39 and 8.41, p. 328

For a forward-biased diode the measured diffusion capacitance grows linearly with the current, with a slope $\mathrm{d}C_D/\mathrm{d}I = 2.5\times10^{-6}$ F/A.

(a) With $C_D = (q/kT)\,\bar{\tau}_{tr} I$, find $\bar{\tau}_{tr}$.

(b) At $I = 1$ mA, compute C_D and r_d. Neglect C_T.

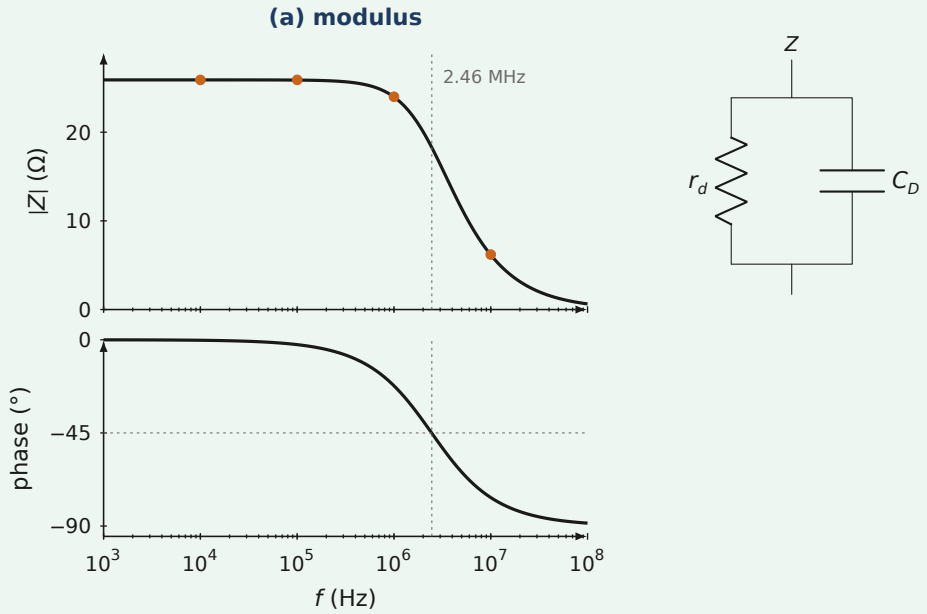
(c) Compute the modulus and phase of the impedance of $r_d \parallel C_D$ at 10 kHz, 100 kHz, 1 MHz and 10 MHz. Above which frequency is the diode mainly capacitive?

(d) Show that this corner frequency, $1/(2\pi r_d C_D)$, depends only on $\bar{\tau}_{tr}$, not on the current.

(Answer: (a) 65 ns; (b) 2.5 nF, 25.9 Ω; (c) 25.9 Ω/−0.2°, 25.9 Ω/−2.3°, 24.0 Ω/−22°, 6.2 Ω/−76°; corner 2.46 MHz.)

SOLUTION

(a) $\bar{\tau}_{tr} = V_T\,\mathrm{d}C_D/\mathrm{d}I = 0.0259\times2.5\times10^{-6} = 6.5\times10^{-8}$ s $= 65$ ns.
(b) $C_D = 2.5\times10^{-6}\times10^{-3} = 2.5$ nF, $r_d = V_T/I = 25.9$ Ω.
(c) $Z = r_d/(1 + j\omega r_d C_D)$ with $r_d C_D = 65$ ns: at 10 kHz, $\omega r_d C_D = 4\times10^{-3}$, $|Z| = 25.9$ Ω, −0.2°; at 100 kHz, 25.9 Ω, −2.3°; at 1 MHz, $\omega r_d C_D = 0.41$, 24.0 Ω, −22°; at 10 MHz, 6.2 Ω, −76°. The capacitive current exceeds the resistive one above $f_c = 1/(2\pi r_d C_D) = 2.46$ MHz.
(d) $r_d C_D = \frac{kT}{qI}\cdot\frac{q}{kT}\bar{\tau}_{tr} I = \bar{\tau}_{tr}$: the corner frequency is $1/(2\pi\bar{\tau}_{tr})$, a property of the stored charge alone. (Neamen's small-signal analysis of a long diode gives half of this capacitance; the course explains, in Chapter 7, that the quasi-static value is the usual engineering formula and that the exact factor is between 1/2 and 2/3.)



The small-signal diode: resistive below $1/(2\pi\bar{\tau}_{tr})$, capacitive above.

Problem 7.7 How good is the storage-time estimate? C · Synthesis

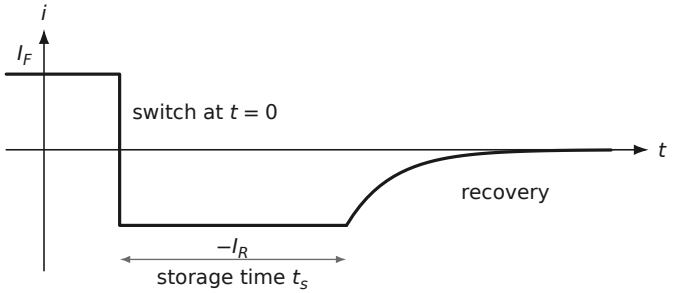
Adapted from Neamen, Problem 8.47, p. 328

A long diode with hole lifetime τ_p carries a forward current I_F and is switched to a reverse current I_R. The course estimates the storage time by $t_s \approx \tau_p I_F/I_R$, an upper bound. A full solution of the diffusion equation for a long region gives $\operatorname{erf}\sqrt{t_s/\tau_p} = I_F/(I_F + I_R)$.

(a) Explain why $\tau_p I_F/I_R$ is an upper bound: which ways out of the neutral region does it ignore?

(b) Compute t_s/τ_p from the exact relation for $I_R/I_F = 0.2$ and $I_R/I_F = 1$, and compare with the estimate.

(c) A rectifier with $\tau_p = 10$ µs is switched with $I_R = I_F$. Estimate its storage time. What could be done to reduce it by a factor 1000?



Switching a diode off: the current stays at $-I_R$ while the stored charge is removed, then recovers.

HINT

erf(0.975) = 0.833 and erf(0.477) = 0.500.

(Answer: (b) 0.95 against 5; 0.23 against 1; (c) about 2.3 µs.)

SOLUTION

(a) The estimate assumes that the whole stored charge $\tau_p I_F$ leaves through the junction at the rate I_R. In reality the charge also decreases by recombination during the storage time, and the junction voltage changes sign as soon as the excess *at the edge* vanishes, while carriers further away remain. Both effects shorten t_s.
(b) $I_R/I_F = 0.2$: $\operatorname{erf} x = 1/1.2 = 0.833$, $x = 0.975$, $t_s = 0.95\,\tau_p$, against $5\tau_p$ estimated. $I_R/I_F = 1$: $\operatorname{erf} x = 0.5$, $x = 0.477$, $t_s = 0.23\,\tau_p$, against τ_p. The estimate gives the right scale but overstates the time, more so when I_R is small, because recombination then has time to act.
(c) $t_s \approx 0.23 \times 10 = 2.3$ µs. Reducing it a thousand times requires a lifetime a thousand times shorter (recombination centres such as gold or platinum), or a short neutral region whose transit time is in the nanosecond range, or a Schottky diode, which stores almost no minority charge.

Practice exercises from the course notes
These exercises closed Chapter 7 of the course notes; they now live here, with their answers. Tables, equations and sections cited by number are those of the course notes. Silicon at 300 K, the reference diode of Table 7.1, unless stated otherwise.

Problem 7.8 Transition capacitance

B · Guided calculation

From the course notes, Chapter 7

(i) Compute W and C_T at −2 V and −20 V. (ii) By what factor must the reverse voltage grow to halve C_T, starting from $V = 0$? (iii) Compare $C_T(0)$ with a parallel-plate capacitor of the same area made of a 1 µm layer of silicon dioxide ($\varepsilon_r = 3.9$).

(Answer: (i) −2 V: $W = 0.61$ µm, $C_T = 1.71$ pF; −20 V: $W = 1.65$ µm, $C_T = 0.63$ pF. (ii) C_T is halved when $1 - V/V_d = 4$: $V = -3V_d = -2.5$ V. (iii) $\varepsilon_{ox} S/1\ \mu\text{m} = 3.9 \times 8.854 \times 10^{-14} \times 10^{-4}/10^{-4} = 3.5 \times 10^{-13}$ F = 0.35 pF: the junction, 0.33 µm of silicon, is nine times larger: thinner, and with three times the permittivity.)

Problem 7.9 A measurement

B · Guided calculation

From the course notes, Chapter 7

A P^+N diode of area 10^{-4} cm² gives $C = 2.13$ pF at −1 V and $C = 1.19$ pF at −5 V. (i) Compute $1/C^2$ at both voltages, the slope, and deduce N_D. (ii) Deduce V_d from the intercept. (iii) Compare with the values of the reference diode, and say why the result for V_d is less accurate than for N_D.

(Answer: (i) $1/C^2 = 0.2204$ and 0.7062 pF^{-2}; slope 0.1214 pF^{-2}/V $= 1.214 \times 10^{23}$ F^{-2}/V; $N_D = 2/(q\varepsilon S^2 \times \text{slope}) = 9.9 \times 10^{15}$ cm^{-3}. (ii) $V_d = -1 + 0.2204/0.1214 = 0.82$ V. (iii) The reference values are 10^{16} and 0.835 V. V_d comes from an extrapolation over more than one volt beyond the

measured points, so the three-digit rounding of C weighs more on it.)

Problem 7.10 A varicap

B · Guided calculation

From the course notes, Chapter 7

The reference diode is used with $L = 10$ µH. (i) Compute f_0 at -1 V and -10 V. (ii) What range of capacitance is needed to tune from 88 to 108 MHz? (iii) What exponent m makes f_0 linear in the tuning voltage, and what doping profile gives it?

(Answer: (i) 34.5 MHz and 53.8 MHz. (ii) $C = 1/(4\pi^2 f^2 L)$: from 0.33 pF (88 MHz) to 0.22 pF (108 MHz) with 10 µH; the reference diode would need -77 V and -175 V, far beyond its breakdown, so a smaller L is used. (iii) $m = 2$: a hyperabrupt profile, whose doping decreases away from the junction.)

Problem 7.11 Stored charge

B · Guided calculation

From the course notes, Chapter 7

At $I = 10$ µA: (i) compute the stored charge in the N region and in the P region; (ii) compute C_D and compare it with C_T at the same voltage. (iii) Check (7.14) against (7.13).

(Answer: (i) $I_p = 0.854 \times 10 = 8.54$ µA, $Q_p = 10$ µs $\times$ 8.54 µA = 85.4 pC; $I_n = 1.46$ µA, $Q_n = 0.1$ µs $\times$ 1.46 µA = 0.15 pC. (ii) $C_D = (Q_p + Q_n)/(kT/q) = 3.3$ nF at $V = 0.637$ V, where $C_T = 3.15/\sqrt{1 - 0.637/0.835} = 6.5$ pF: 500 times smaller. (iii) $(q^2 n_i^2/kT)(L_p/N_D)e^{qV/kT}$: the hole term equals $(q/kT)\tau_p J_p$, since $\tau_p D_p/L_p = L_p$ gives $\tau_p J_p = q n_i^2 L_p e^{qV/kT}/N_D$; same for the electron term.)

Problem 7.12 Long or short

B · Guided calculation

From the course notes, Chapter 7

Take the short diode of Chapter 6 (Section 6.11: same doping and lifetimes, $d_p = 2$ µm, $d_n = 10$ µm; $J_0 = 2.2 \times 10^{-11}$ A/cm², holes 87 %). (i) Compute the transit times $d_n^2/2D_p$ and $d_p^2/2D_n$ and the mean $\bar{\tau}_{tr}$. (ii) What is the stored charge at 0.1 mA, and by what factor is C_D smaller than for the long diode at the same current? (iii) What does this imply for a power rectifier, which needs thick regions?

(Answer: (i) $(10^{-3})^2/(2 \times 12) = 41.7$ ns and $(2 \times 10^{-4})^2/(2 \times 35) = 0.57$ ns; $\bar{\tau}_{tr} = 0.873 \times 41.7 + 0.127 \times 0.57 = 36.4$ ns. (ii) $Q = 36.4$ ns $\times$ 0.1 mA = 3.6 pC, against 0.86 nC for the long diode: C_D (141 pF) is 8.56 µs/36.4 ns = 235 times smaller. (iii) A rectifier with thick, long regions stores a lot of charge and switches slowly: its lifetime is reduced on purpose, at the cost of a larger J_0.)

Problem 7.13 Small signal

B · Guided calculation

From the course notes, Chapter 7

At $I = 0.1$ mA: (i) compute r_d, C_D and the impedance of C_D at 10 kHz and at 100 kHz. (ii) At which frequency do the resistive and capacitive currents become equal? (iii) Does it change at 10 µA?

(Answer: (i) $r_d = 259$ Ω, $C_D = 33$ nF; $1/(2\pi f C_D) = 482$ Ω at 10 kHz and 48 Ω at 100 kHz. (ii) At $f = 1/(2\pi r_d C_D) = 1/(2\pi\bar{\tau}) = 18.6$ kHz. (iii) No: r_d is 10 times larger and C_D 10 times smaller.)

Problem 7.14 Switching

B · Guided calculation

From the course notes, Chapter 7

The reference diode carries $I_F = 0.1$ mA and is switched to a reverse current $I_R = 0.05$ mA. (i) Solve the charge-control equation (7.18) and find the time t_Q at which the stored charge vanishes. (ii) The full solution gives $\operatorname{erf}\sqrt{t_s/\bar{\tau}} = I_F/(I_F + I_R)$; compute t_s ($\operatorname{erf} 0.685 = 2/3$). (iii) How does gold doping, which reduces the lifetimes a thousandfold, change the result? (iv) Compare with the short diode of Problem 7.13.

(Answer: (i) $Q(t) = -\bar{\tau} I_R + \bar{\tau}(I_F + I_R)\mathrm{e}^{-t/\bar{\tau}}$ vanishes at $t_Q = \bar{\tau}\ln(1 + I_F/I_R) = 8.56 \times \ln 3 = 9.4$ µs. (ii) $\sqrt{t_s/\bar{\tau}} = 0.685$, $t_s = 0.47\,\bar{\tau} = 4.0$ µs. (iii) Both times scale with $\bar{\tau}$: a thousand times shorter, a few nanoseconds, at the price of a larger leakage current ($J_0 \propto 1/L \propto 1/\sqrt{\tau}$). (iv) The short diode stores $\bar{\tau}_{tr} I_F = 3.6$ pC; at most $Q/I_R = 73$ ns.)

Problem 7.15 A junction in a chip

B · Guided calculation

From the course notes, Chapter 7

The drain of a MOS transistor is a $\mathrm{N^+P}$ junction of area 1 µm × 1 µm on a substrate with $N_A = 10^{17}\,\mathrm{cm^{-3}}$, at $V_d - V = 2$ V. (i) Compute its transition capacitance. (ii) A logic node has a total capacitance of 20 fF. How long does a current of 50 µA take to change it by 1 V? (iii) What fraction of that time is due to this junction?

(Answer: (i) $C_T = S\sqrt{\varepsilon q N_A/2(V_d - V)} = 10^{-8}\sqrt{1.036 \times 10^{-12} \times 1.602 \times 10^{-19} \times 10^{17}/4} = 0.64$ fF. (ii) $t = 20$ fF × 1 V/50 µA = 0.4 ns. (iii) 0.64/20, about 3 %: in a modern chip the wires and gates dominate, but every junction adds its share.)

Challenge: a new situation

These problems apply the method of the chapter to a device, a material or a situation that the chapter did not treat. The reasoning does not change: state the physical picture, choose the model, compute, check.

Problem 7.16 When the SCR reaches the contact

C · Synthesis

Adapted from Sze & Lee, Ch. 3, Problem 16, p. 122

A silicon $\mathrm{P^+N}$ diode has $N_A = 10^{19}$ and $N_D = 10^{15}\,\mathrm{cm^{-3}}$, but its N region is only 5 µm thick before a heavily doped $\mathrm{N^+}$ contact layer. Area 10^{-4} cm².

(a) At what reverse voltage does the SCR reach the $\mathrm{N^+}$ layer ('punch-through' of the N region)? What is then the maximum field?

(b) Sketch $C_T(V_R)$ below and above this voltage. What is the capacitance above it?

(c) Above punch-through, the extra reverse voltage cannot widen the SCR. Show that it adds a uniform field $\Delta V/5\ \mu$m, and draw the field profile. Why does this design raise the voltage a diode can block for a given maximum field?

(d) In which device of Chapter 8 is punch-through of a region a failure mode, and why?

(Answer: (a) $V_R = 18.5$ V, $\mathcal{E}_{max} = 7.7 \times 10^4$ V/cm; (b) constant, 0.207 pF.)

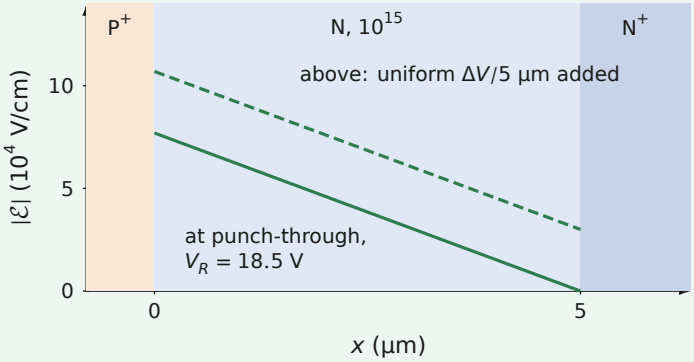
SOLUTION

(a) The SCR is almost entirely in the N region: $V_d - V = qN_D W^2/(2\varepsilon)$ with $W = 5$ µm gives $1.602\times10^{-19}\times10^{15}\times2.5\times10^{-7}/(2.07\times10^{-12}) = 19.3$ V. With $V_d = 0.0259\ln(10^{34}/10^{20}) = 0.835$ V, $V_R = 18.5$ V. The field is a triangle of height $\mathcal{E}_{max} = 2\times19.3/5\times10^{-4} = 7.7\times10^4$ V/cm.

(b) Below punch-through, $C_T = \varepsilon S/W$ decreases as $(V_d+V_R)^{-1/2}$. Above, W is fixed at 5 µm: $C_T = \varepsilon S/W = 1.036\times10^{-12}\times10^{-4}/(5\times10^{-4}) = 2.07\times10^{-13}$ F = 0.207 pF, constant.

(c) The N^+ layer is too heavily doped to be depleted further; the extra charge appears as thin sheets at the two edges, and Gauss's law gives a uniform extra field $\Delta V/W$ added to the triangle: the profile becomes a trapezoid. For a given maximum field (set by avalanche breakdown), a trapezoid holds more voltage than a triangle of the same height: this is the principle of the p–i–n power diode.

(d) In the bipolar transistor the BC SCR widens into the base as V_{CB} increases (the Early effect). If it crosses the whole base and meets the EB SCR, the base no longer separates emitter and collector: the collector current is no longer controlled by the base. This is base punch-through, one reason the base cannot be made arbitrarily thin or lightly doped.



Before punch-through the field is a triangle; above, the extra voltage adds a uniform field and the profile becomes a trapezoid.

SEMICONDUCTOR DEVICES PHYSICS · SOLUTIONS

CHAPTER 8

The Bipolar Junction Transistor

WHAT THIS SET USES

The two junctions sharing a short base; the straight-line electron profile and $I_C = qSD_n n(0)/W_B = I_S e^{V_{BE}/V_T}$ with $I_S = qSD_n n_i^2/(N_B W_B)$; the emitter efficiency γ, the transport factor $\alpha_T = 1 - W_B^2/2L_n^2$ (expansion of $1/\cosh(W_B/L_n)$), $\alpha_F = \gamma\alpha_T$, $1/\beta_F = \frac{D_p}{D_n}\frac{N_B}{N_E}\frac{W_B}{d_E} + \frac{W_B^2}{2L_n^2}$; the stored charge $Q_B = \tau_B I_C$ with $\tau_B = W_B^2/2D_n$; the four operating regions and the Ebers–Moll superposition; the voltages V_{BE} and V_{CB} (BC forward when $V_{CB} < 0$), the edge densities $n(0) = n_{B0}e^{V_{BE}/V_T}$, $n(W_B) = n_{B0}e^{-V_{CB}/V_T}$; base-width modulation, $V_A \simeq W_B/\delta = Q_{B,maj}/C_{jc}$, $r_o \simeq V_A/I_C$; the diffusion capacitance $C_D = \tau_F I_C/V_T$ and the transition capacitance C_{jc}.

DATA

Silicon at 300 K, $n_i = 10^{10}\,\text{cm}^{-3}$, $V_T = 25.9$ mV, $\varepsilon = 1.036 \times 10^{-12}$ F/cm. Reference NPN transistor: $N_E = 10^{18}$, $N_B = 10^{16}$, $N_C = 10^{15}\,\text{cm}^{-3}$; $d_E = 0.5$ µm; $W_B = 1$ µm at $V_{CB} = 5$ V; $S = 10^{-4}$ cm²; $D_n = 35$, $D_p = 12$ cm²/s, $\tau_n = 0.1$ µs; $I_S = 5.6 \times 10^{-14}$ A, $\beta_F = 121$, $\beta_R = 2.9$, $\tau_B = 0.143$ ns. Where a problem gives its own diffusivities or lifetimes (those of the textbook), use them.

Problem 8.1 The bands of a PNP transistor in its four regions A · Understanding

Adapted from Neamen, Problems 12.1 and 12.2, pp. 560–561

The course develops the NPN transistor. Consider instead a uniformly doped P^+NP transistor (emitter P^+, base N, collector P, more lightly doped than the base). Sketch, one below the other and on the same horizontal axis, the band diagram (E_C, E_V and the Fermi level or the two quasi-Fermi levels) and the field $\mathcal{E}(x)$:

(a) at thermal equilibrium;

(b) in forward-active operation ($V_{EB} > 0$, $V_{CB} < 0$);

(c) in inverse-active operation ($V_{EB} < 0$, $V_{CB} > 0$);

(d) in cut-off, both junctions reverse biased.

(e) In each case, say which carriers cross the base and in which direction, and compare with the NPN of the course: what changes, and what does not?

HINT

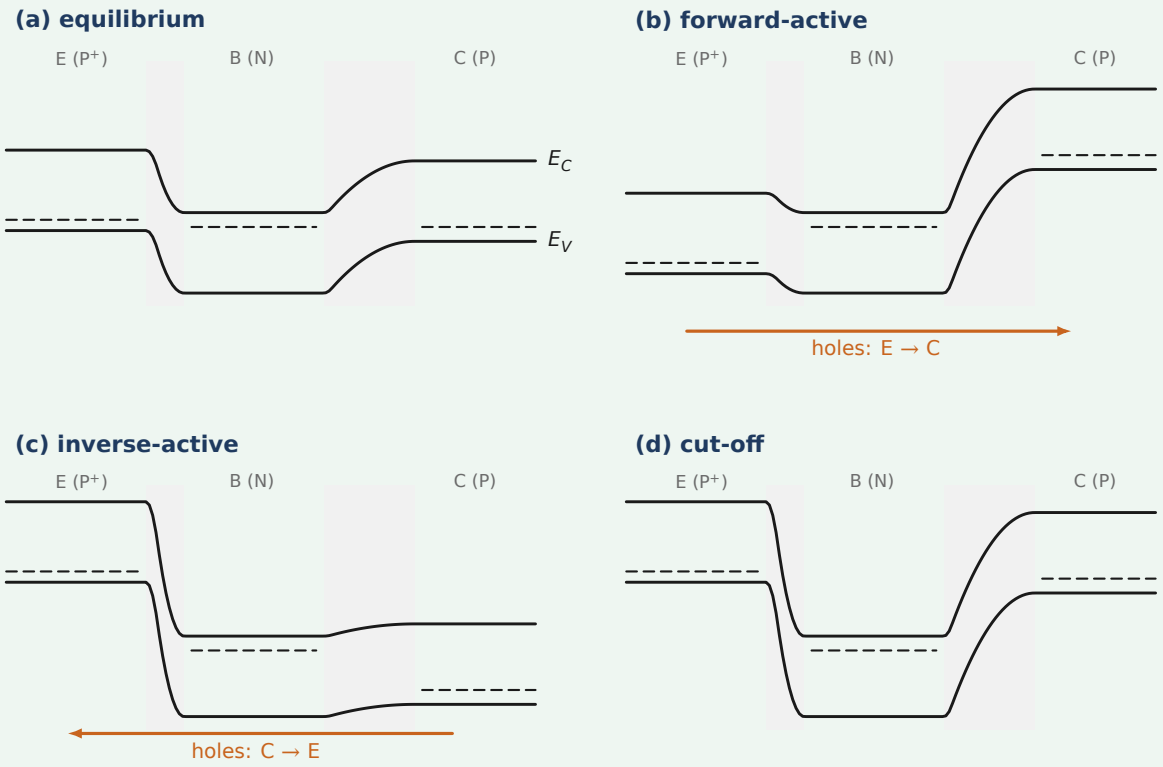
Start from the equilibrium picture: E_F is flat, close to E_V in the P regions and close to E_C in the N base. Then remember Chapter 6: a forward bias V lowers a barrier by qV, a reverse bias raises it.

(Answer: The base bands lie lower than those of the emitter and collector (a well for electrons, a hill for holes); forward-active: low EB barrier, high BC barrier, holes cross the base from emitter

to collector; inverse-active: the roles of the junctions are exchanged; cut-off: both barriers high, only leakage currents. The field points from the base towards each outer region at equilibrium.)

SOLUTION

(a) **Equilibrium.** E_F is flat. In the P^+ emitter it lies just above E_V, in the P collector a little higher above E_V (lighter doping), in the N base near E_C. The base bands are therefore *lower* than those of the emitter and collector: the base is a valley for electrons, a hill for holes (hole energy increases downwards). The step is qV_{dE} at the EB junction and $qV_{dC} < qV_{dE}$ at the BC junction. Each SCR holds positive donors on the base side and negative acceptors on the outer side, so the field points from the base towards the emitter in the EB SCR ($\mathcal{E} < 0$ if x runs from emitter to collector) and from the base towards the collector in the BC SCR ($\mathcal{E} > 0$). Each field is a triangle; the field is zero in the neutral regions. The EB SCR is narrow and lies mostly in the base; the BC SCR is wide and lies mostly in the collector.
(b) **Forward-active.** The EB step shrinks to $q(V_{dE} - V_{EB})$, the BC step grows to $q(V_{dC}+|V_{CB}|)$. The quasi-Fermi levels separate in the base: E_{Fp} (holes) stays at the emitter level up to the EB junction and drops across the base to reach the collector level. The EB field triangle is smaller, the BC triangle larger and wider. Holes are injected from the emitter into the base, diffuse across it and are swept by the BC field into the collector; a small electron current flows from the base into the emitter.
(c) **Inverse-active.** The opposite: the BC step is lowered, the EB step raised. Holes are injected by the collector into the base and collected by the emitter. Because the collector is more lightly doped than the base, the BC junction also injects many electrons from the base into the collector: the inverse gain is poor, as for the NPN ($\beta_R = 2.9$).
(d) **Cut-off.** Both steps are raised, both field triangles grow, the quasi-Fermi levels in the base lie below the equilibrium level (a deficit of minority holes at both edges). Only the small reverse (generation) currents of the two junctions flow.
(e) Everything is mirrored: holes play the role of electrons, the voltages and currents change sign, and the band diagram is the NPN one turned upside down. The physics, injection by a forward-biased junction, diffusion across a short base, collection by the field of a reverse-biased junction, does not change. Note that the field profile of the PNP is the NPN one with the opposite sign.



Electron energies of the PNP transistor in its four regions (generic picture, not to scale); dashed: the Fermi level of each neutral region.

Problem 8.2 Gains and the open-base current

A · Understanding

Adapted from Sze & Lee, Ch. 4, Problems 1 and 2, p. 157

An NPN transistor has a transport factor $\alpha_T = 0.998$ and an emitter efficiency $\gamma = 0.997$. When the emitter is left open, the reverse-biased BC junction carries a leakage current $I_{CBO} = 10$ nA (the reverse current of Chapter 6, generation in the BC SCR included).

(a) Compute α_F and β_F.

(b) In forward-active operation the collector receives the fraction α_F of the emitter current *plus* the leakage: $I_C = \alpha_F I_E + I_{CBO}$. The base is now left open ($I_B = 0$). Show that $I_E = I_C = I_{CBO}/(1 - \alpha_F)$ and compute it. Explain physically why the base must be forward biased although no current enters it.

(c) A second transistor has $\gamma = 0.999$, $\alpha_T \approx 1$ and $I_{CBO} = 10$ µA (a power device at high temperature). With $I_B = 0$, compute I_E and the part of it carried by holes injected into the emitter. Compare this hole current with I_{CBO} and explain the coincidence.

(d) Why must the base of a transistor never be left floating in a circuit?

> **HINT**
>
> With $I_B = 0$, every hole that leaves the base must be supplied from inside the device. Which holes enter the base, and which ones leave it?

(Answer: (a) $\alpha_F = 0.9950$, $\beta_F = 199$; (b) $I_E = 2.0$ µA $= (\beta_F + 1) I_{CBO}$; (c) $I_E = 10$ mA, holes 10 µA $= I_{CBO}$.)

SOLUTION

(a) $\alpha_F = \gamma\alpha_T = 0.997 \times 0.998 = 0.9950$ and $\beta_F = \alpha_F/(1-\alpha_F) = 199$.
(b) $I_B = 0$ means $I_E = I_C$, so $I_E = \alpha_F I_E + I_{CBO}$ and $I_E = I_{CBO}/(1-\alpha_F) = (\beta_F+1)I_{CBO} = 10$ nA/0.005 = 2.0 µA. The leakage of the BC junction brings holes into the base (the holes generated in the BC SCR and on the collector side are swept towards the base). With the base open, these holes cannot leave through the base contact: they accumulate until the EB junction becomes forward biased enough for the holes to leave by injection into the emitter and by recombination. That forward bias also injects electrons, and the transistor amplifies its own leakage by $\beta_F + 1$.
(c) $\alpha_F = 0.999$: $I_E = 10$ µA/0.001 = 10 mA. The hole part is $(1-\gamma)I_E = 10$ µA, exactly I_{CBO}. This is not a coincidence: with $\alpha_T = 1$ there is no recombination in the base, so the only exit for holes is injection into the emitter, and with $I_B = 0$ this exit must carry exactly the hole supply, I_{CBO}.
(d) With an open base the collector current is $(\beta_F+1)I_{CBO}$; I_{CBO} roughly doubles every 10 K, the current heats the device, which raises I_{CBO} further: a risk of thermal runaway, and a transistor that is not really 'off'. A resistor from base to emitter gives the holes another exit.

Problem 8.3 The saturation current from the base parameters **B · Guided calculation**

Adapted from Neamen, Problems 12.3 and 12.4, p. 561

The base of a silicon NPN transistor has $D_n = 18$ cm²/s, $n_{B0} = 4\times10^3$ cm^{-3} and a neutral width $W_B = 0.80$ µm; the EB area is $S = 5\times10^{-5}$ cm².

(a) Compare $I_C = qSD_n n(0)/W_B$ with $I_C = I_S e^{V_{BE}/V_T}$ and compute I_S. What is the base doping, with the course value of n_i?

(b) Compute I_C at $V_{BE} = 0.58$, 0.65 and 0.72 V. By how much must V_{BE} increase to multiply I_C by 10?

(c) For each point, compute $n(0)/N_B$. Up to which point is the low-injection model valid?

(d) A second transistor has $D_n = 22$ cm²/s, $W_B = 0.80$ µm and $n_{B0} = 2\times10^4$ cm^{-3}. What area gives $I_C = 2$ mA at $V_{BE} = 0.60$ V? With this area, what V_{BE} gives 5 mA?

(Answer: (a) $I_S = 7.2\times10^{-15}$ A, $N_B = 2.5\times10^{16}$ cm^{-3}; (b) 38.3 µA, 0.572 mA, 8.53 mA; 59.6 mV per decade; (c) 0.09 %, 1.3 %, 19 %; (d) $S = 1.97\times10^{-4}$ cm², 0.624 V.)

SOLUTION

(a) $n(0) = n_{B0}e^{V_{BE}/V_T}$, so $I_S = qSD_n n_{B0}/W_B = 1.602\times10^{-19}\times5\times10^{-5}\times18\times4\times10^3/0.8\times10^{-4} = 7.2\times10^{-15}$ A. $N_B = n_i^2/n_{B0} = 2.5\times10^{16}$ cm^{-3}.
(b) $e^{0.58/0.0259} = 5.3\times10^9$: $I_C = 38.3$ µA; at 0.65 V, 0.572 mA; at 0.72 V, 8.53 mA. A factor 10 needs $V_T \ln 10 = 59.6$ mV.
(c) $n(0) = n_{B0}e^{V_{BE}/V_T} = 2.1\times10^{13}$, 3.2×10^{14} and 4.7×10^{15} cm^{-3}: 0.09 %, 1.3 % and 19 % of N_B. The last point violates $n(0) \ll N_B$: at about 10 mA the transistor enters high injection, and the ideal law overestimates I_C.
(d) $S = I_C W_B/[qD_n n_{B0}e^{V_{BE}/V_T}] = 2\times10^{-3}\times0.8\times10^{-4}/(1.602\times10^{-19}\times22\times2\times$

$10^4 \times 1.16 \times 10^{10}) = 1.97 \times 10^{-4}$ cm². Then $V_{BE} = 0.60 + V_T \ln(5/2) = 0.624$ V. (The area changes only I_S; the 60 mV per decade law does not depend on the geometry.)

Problem 8.4 The current components of a PNP transistor B · Guided calculation

Adapted from Sze & Lee, Ch. 4, Problems 3, 4 and 5, pp. 157–158

A silicon PNP transistor has $N_E = 5 \times 10^{18}$, $N_B = 2 \times 10^{17}$ and $N_C = 10^{16}\,\text{cm}^{-3}$, a metallurgical base width of 1.0 µm and an area $S = 0.2$ mm². The EB junction is forward biased at $V_{EB} = 0.5$ V and the BC junction reverse biased at 5 V. The minority-carrier diffusivities given by the textbook are 52, 40 and 115 cm²/s in the emitter, base and collector, and the lifetimes 10^{-8}, 10^{-7} and 10^{-6} s. The emitter and the collector are thick: treat them as long regions.

(a) Compute V_{dE}, V_{dC}, the widths of the two SCRs and the part of each in the base. Deduce the neutral base width W_B.

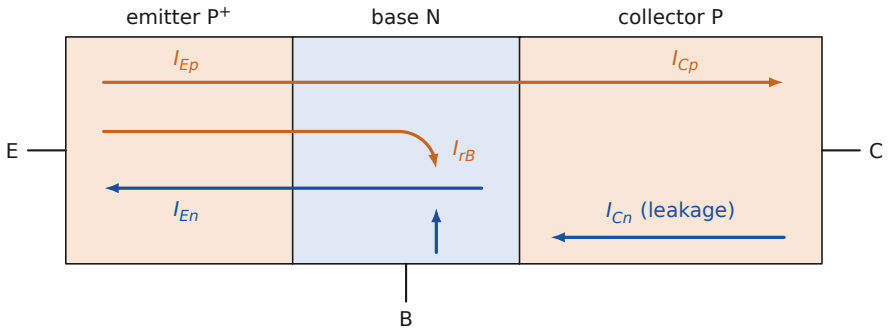
(b) Compute p_{B0}, n_{E0}, n_{C0} and the hole density $p(0)$ at the emitter edge of the neutral base.

(c) Compute the three diffusion lengths. Is the base short? Sketch the minority-carrier profiles in the three regions.

(d) Compute the current components: I_{Ep} (holes injected into the base), I_{rB} (holes recombining in the base, with Q_B/τ), $I_{Cp} = I_{Ep} - I_{rB}$, I_{En} (electrons injected into the emitter) and I_{Cn} (electrons extracted from the collector by the reverse-biased BC junction).

(e) Deduce I_E, I_C, I_B, then γ, α_T, α_F and β_F.

(f) Which loss limits the gain? Propose two changes of design that would improve it.



The current components of a PNP transistor in forward-active operation. After Sze & Lee, Ch. 4, Fig. 5, p. 128.

HINT

For a PNP, $p(0) = p_{B0}\text{e}^{V_{EB}/V_T}$. In a long emitter, the injected electron current is $qSD_E n_{E0}(\text{e}^{V_{EB}/V_T} - 1)/L_E$ (Chapter 6). I_{Cn} is the reverse current of a long N…P side: $qSD_C n_{C0}/L_C$.

(Answer: (a) $V_{dE} = 0.954$ V, $V_{dC} = 0.793$ V; 0.0553 µm (0.053 in the base), 0.887 µm (0.042 in the base); $W_B = 0.905$ µm; (b) $p_{B0} = 500$, $n_{E0} = 20$, $n_{C0} = 10^4$, $p(0) = 1.21 \times 10^{11}\,\text{cm}^{-3}$; (c) 7.2, 20,

107 µm, $W_B/L_B = 0.045$; (d) $I_{Ep} = 17.15$ µA, $I_{rB} = 17.5$ nA, $I_{Cp} = 17.13$ µA, $I_{En} = 0.112$ µA, $I_{Cn} = 3.4\times10^{-14}$ A; (e) $I_E = 17.26$ µA, $I_C = 17.13$ µA, $I_B = 0.129$ µA, $\gamma = 0.9935$, $\alpha_T = 0.99898$, $\alpha_F = 0.9925$, $\beta_F = 132$.)

SOLUTION

(a) $V_{dE} = 0.0259\ln(5\times10^{18}\times2\times10^{17}/10^{20}) = 0.954$ V and $V_{dC} = 0.0259\ln(2\times10^{33}/10^{20}) = 0.793$ V. EB SCR, under $V_{dE} - 0.5 = 0.454$ V: $W_E = 0.0553$ µm, of which the fraction $N_E/(N_E+N_B) = 0.96$, i.e. 0.053 µm, lies in the base. BC SCR, under 0.793+5 = 5.79 V: $W_C = 0.887$ µm, of which $N_C/(N_B+N_C) = 1/21$, 0.042 µm, lies in the base. $W_B = 1 - 0.053 - 0.042 = 0.905$ µm.

(b) $p_{B0} = n_i^2/N_B = 500$, $n_{E0} = 20$, $n_{C0} = 10^4\,\text{cm}^{-3}$. $\text{e}^{0.5/0.0259} = 2.42\times10^8$, so $p(0) = 1.21\times10^{11}\,\text{cm}^{-3}$, far below N_B: low injection.

(c) $L_E = \sqrt{52\times10^{-8}} = 7.2$ µm, $L_B = \sqrt{40\times10^{-7}} = 20$ µm, $L_C = \sqrt{115\times10^{-6}} = 107$ µm. $W_B/L_B = 0.045$: the base is short and its hole profile is a straight line from $p(0)$ to almost zero at the BC edge. In the emitter, the electron excess decays exponentially over L_E from $n_{E0}\text{e}^{V_{EB}/V_T} = 4.8\times10^9\,\text{cm}^{-3}$; in the collector, the electron density is pulled below n_{C0} near the BC SCR and recovers over L_C.

(d) $I_{Ep} = qSD_Bp(0)/W_B = 1.602\times10^{-19}\times2\times10^{-3}\times40\times1.21\times10^{11}/9.05\times10^{-5} = 17.15$ µA. $Q_B = \frac{1}{2}qSp(0)W_B = 1.75\times10^{-15}$ C, so $I_{rB} = Q_B/\tau_B = 1.75\times10^{-15}/10^{-7} = 17.5$ nA and $I_{Cp} = 17.13$ µA. $I_{En} = qSD_En_{E0}(\text{e}^{V_{EB}/V_T} - 1)/L_E = 1.602\times10^{-19}\times2\times10^{-3}\times52\times20\times2.42\times10^8/7.2\times10^{-4} = 0.112$ µA. $I_{Cn} = qSD_Cn_{C0}/L_C = 3.4\times10^{-14}$ A, a pure leakage, negligible here. (The exact sinh solution of the base gives $I_{Ep} = 17.16$ and $I_{Cp} = 17.15$ µA: the straight line is accurate to 0.1 %.)

(e) $I_E = I_{Ep}+I_{En} = 17.26$ µA, $I_C = I_{Cp}+I_{Cn} = 17.13$ µA, $I_B = I_E - I_C = 0.129$ µA. $\gamma = I_{Ep}/I_E = 0.9935$; $\alpha_T = I_{Cp}/I_{Ep} = 0.99898$, which is indeed $1 - W_B^2/2L_B^2$; $\alpha_F = \gamma\alpha_T = 0.9925$ and $\beta_F = 132$.

(f) The emitter loss $1-\gamma = 6.5\times10^{-3}$ is six times the base loss $1-\alpha_T = 1.0\times10^{-3}$. To raise γ: increase N_E/N_B, by doping the emitter more or the base less (the latter also lowers the Early voltage and raises the base resistance), or narrow the base, since $I_{Ep} \propto 1/W_B$. Thinning the emitter below L_E would do the opposite: the electron profile in the emitter would end at the contact and its gradient, hence I_{En}, would increase. To raise α_T: a narrower base. (The textbook diffusivities are larger than the course values for holes in a base doped at $2\times10^{17}\,\text{cm}^{-3}$; the method does not depend on them.)

Problem 8.5 The base as a charge reservoir — B · Guided calculation

Adapted from Sze & Lee, Ch. 4, Problems 8 and 9, p. 158

Use the PNP transistor of Problem 8.4 in forward-active operation.

(a) With the straight-line hole profile, show that the stored excess charge is $Q_B = \frac{1}{2}qSp(0)W_B$, the area of a triangle. Compute it.

(b) Show that $I_C \simeq 2D_BQ_B/W_B^2$, check it numerically, and deduce the transit time $\tau_B = Q_B/I_C$.

(c) Show that the recombination current is $I_{rB} = Q_B/\tau$ and that $\alpha_T = 1 - \tau_B/\tau$. Give a

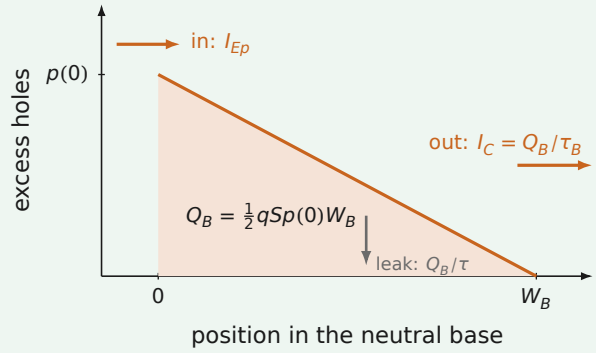
picture of the base that makes these two results obvious.

(d) The collector current is to be switched off in a time much shorter than τ_B. What base current must flow briefly, and why can it not be obtained by simply opening the base?

(Answer: (a) 1.75×10^{-15} C; (b) 17.1 µA, $\tau_B = 0.102$ ns; (c) $\alpha_T = 1 - 1.0\times10^{-3}$.)

SOLUTION

(a) $\int_0^{W_B} p(0)(1 - x/W_B)\,\mathrm{d}x = p(0)W_B/2$, so $Q_B = \frac{1}{2}qSp(0)W_B = 1.75\times10^{-15}$ C.
(b) $I_C \simeq qSD_Bp(0)/W_B$ and $p(0) = 2Q_B/(qSW_B)$ give $I_C = 2D_BQ_B/W_B^2 = 2\times40\times1.75\times10^{-15}/(9.05\times10^{-5})^2 = 17.1$ µA, the value of Problem 8.4. $\tau_B = Q_B/I_C = W_B^2/2D_B = 1.02\times10^{-10}$ s. The base behaves as a reservoir that empties through the collector in τ_B.
(c) Each excess hole recombines at the rate $1/\tau$, so the reservoir loses Q_B/τ per second by recombination, which the base contact must replace: $I_{rB} = Q_B/\tau = 17.5$ nA. The fraction lost is $I_{rB}/I_{Ep} \simeq (Q_B/\tau)/(Q_B/\tau_B) = \tau_B/\tau$, so $\alpha_T = 1 - \tau_B/\tau = 1 - 1.0\times10^{-3}$. Picture: a bucket filled at the emitter, emptied at the collector in the time τ_B, leaking slowly through recombination with time constant τ. The leak is small because the bucket is emptied a thousand times faster than it leaks.
(d) The stored charge must be removed: a reverse base current $I_B \approx -Q_B/t_{off}$ pulls the holes out through the base contact (for an NPN, electrons out). With an open base, the charge can only leave through the collector (time τ_B, while $n(0)$ is still maintained) and by recombination (time τ): the turn-off is slow, the same physics as the storage time of Chapter 7.



The base as a reservoir: filled at the emitter, emptied at the collector in τ_B, leaking slowly by recombination.

Problem 8.6 Which loss limits the gain? A · Understanding

Adapted from Neamen, Problem 12.27, p. 564

The two losses of the course are the transport loss in the base and the injection loss into the emitter.

(a) Compute the transport factor $\alpha_T = 1/\cosh(W_B/L_n)$ for $W_B/L_n = 0.01$, 0.1, 1 and 10, and compare with $1 - W_B^2/2L_n^2$. Taking $\gamma = 1$, deduce β_F in each case.

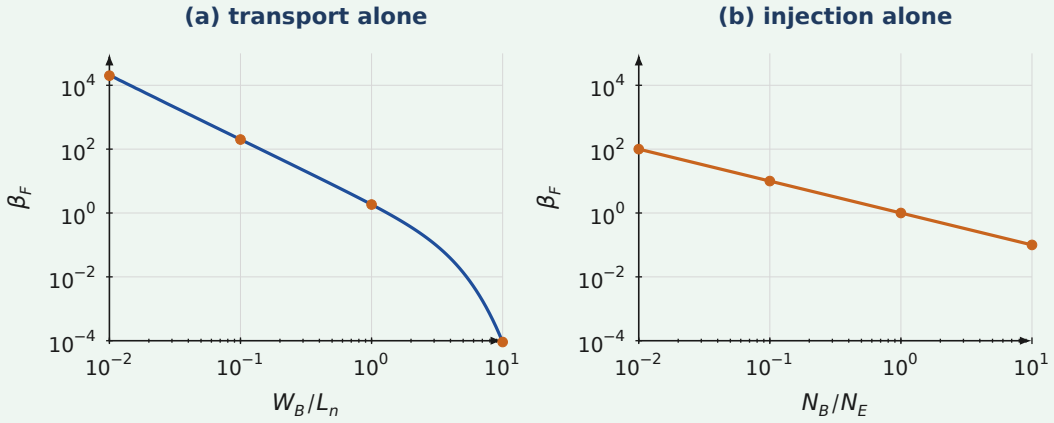
(b) Take a transistor with $D_E = D_B$ and $d_E = W_B$ (short regions), so that $\gamma = 1/(1 + N_B/N_E)$. Compute γ for $N_B/N_E = 0.01$, 0.1, 1 and 10 and, taking $\alpha_T = 1$, β_F.

(c) What do you conclude: which ratio controls each loss, and how large must it be for a useful gain of about 100?

(Answer: (a) α_T = 0.99995, 0.9950, 0.648, 9.1×10^{-5}; β_F = 2.0×10^{4}, 200, 1.84, 9×10^{-5}; (b) γ = 0.990, 0.909, 0.5, 0.091; β_F = 100, 10, 1, 0.1.)

SOLUTION

(a) $W_B/L_n = 0.01$: $\alpha_T = 0.99995$ (approximation identical), $\beta_F = 2.0\times10^{4}$. 0.1: $\alpha_T = 0.9950$ (approximation 0.995), $\beta_F = 200$. 1: $\alpha_T = 0.648$, while the approximation gives 0.5, no longer valid; $\beta_F = 1.84$. 10: $\alpha_T = 9.1\times10^{-5}$ (the approximation is meaningless): $\beta_F \approx 9\times10^{-5}$, two separate diodes, no transistor action (Chapter 8, 'How short must the base be?').
(b) $\gamma = 0.990$, 0.909, 0.5 and 0.091; $\beta_F = \gamma/(1-\gamma) = N_E/N_B$: 100, 10, 1 and 0.1.
(c) With α_T alone, $\beta_F \simeq 2L_n^2/W_B^2$; with γ alone, $\beta_F \simeq (D_B/D_E)(N_E/N_B)(d_E/W_B)$. A gain of 100 requires $W_B < L_n/7$ and an emitter about 100 times more doped than the base. The base transport is easy to make excellent (L_n is tens of microns, W_B about 1 µm); the injection ratio is limited because the emitter cannot be doped much above 10^{19} cm^{-3}. This is why, in modern transistors, the emitter efficiency limits the gain, as for the reference transistor of the course.



The gain allowed by each loss alone. Transport is easy to make excellent; injection is limited by the emitter doping.

Problem 8.7 Designing for a current gain of 150 B · Guided calculation

Adapted from Neamen, Problem 12.29, p. 565

A silicon NPN transistor is to have $\beta_F = 150$. The data are $D_B = 23$ and $D_E = 8$ cm²/s, the lifetimes $\tau_B = 2\times10^{-7}$ s in the base and $\tau_E = 8\times10^{-8}$ s in the emitter, $N_B = 2\times10^{16}$ cm^{-3} and an emitter width $d_E = 0.35$ µm. The technology cannot make a neutral base narrower than 0.80 µm. The course model neglects recombination in the EB SCR; here it is not negligible, and it is described by a third factor $\delta = 0.9975$, so that $\alpha_F = \gamma\,\alpha_T\,\delta$.

(a) Compute L_B and L_E. Are the base and the emitter short?

(b) Which base width should be chosen, and why? Compute α_T.

(c) Compute the required α_F, then the required γ, and the minimum emitter doping N_E.

(d) Check your design by recomputing β_F with N_E rounded to $5\times10^{18}\,\text{cm}^{-3}$. Which of the three losses is now the largest?

(Answer: (a) L_B = 21.4 µm, L_E = 8.0 µm, both short; (b) W_B = 0.80 µm, α_T = 0.99930; (c) α_F = 0.99338, γ = 0.99656, $N_E \geq 4.6\times10^{18}\,\text{cm}^{-3}$; (d) β_F = 156.)

SOLUTION

(a) $L_B = \sqrt{23\times2\times10^{-7}} = 21.4$ µm and $L_E = \sqrt{8\times8\times10^{-8}} = 8.0$ µm. $W_B/L_B = 0.037$ and $d_E/L_E = 0.044$: both regions are short, and the straight-line formulas of the course apply.

(b) The narrowest base, 0.80 µm, gives the best transport and the best injection ratio (the injected electron current is inversely proportional to W_B): $\alpha_T = 1/\cosh(0.8/21.4) = 0.99930$.

(c) $\alpha_F = \beta_F/(\beta_F+1) = 150/151 = 0.99338$. Then $\gamma = \alpha_F/(\alpha_T\delta) = 0.99338/(0.99930\times0.9975) = 0.99656$, so

$$\frac{D_E}{D_B}\frac{N_B}{N_E}\frac{W_B}{d_E} = \frac{1}{\gamma} - 1 = 3.45\times10^{-3},$$

so that

$$N_E = \frac{8}{23}\times2\times10^{16}\times\frac{0.80}{0.35}\Big/3.45\times10^{-3} = 4.6\times10^{18}\,\text{cm}^{-3}.$$

(d) With $N_E = 5\times10^{18}$: $\gamma = 0.99683$, $\alpha_F = 0.99365$ and $\beta_F = 156$, slightly above specification. The losses are $1-\gamma = 3.2\times10^{-3}$, $1-\delta = 2.5\times10^{-3}$ and $1-\alpha_T = 0.7\times10^{-3}$: the emitter injection is still the largest, but recombination in the EB SCR is close, and it would dominate at low current, where it is relatively larger.

Problem 8.8 The transistor used backwards

B · Guided calculation

Adapted from Neamen, Problem 12.19, p. 563

A silicon NPN transistor has $N_E = 10^{19}$, $N_B = 10^{17}$, $N_C = 7\times10^{15}\,\text{cm}^{-3}$ and a metallurgical base width of 1.2 µm. It operates in the inverse-active region with $V_{BE} = -2$ V and $V_{CB} = -0.565$ V (BC junction forward biased).

(a) Sketch the minority-carrier densities through the device.

(b) Compute the minority-carrier densities at the two edges of the neutral base, at the edge of the neutral collector and at the edge of the neutral emitter.

(c) Compute the two SCR widths, the part of each in the base and the neutral base width.

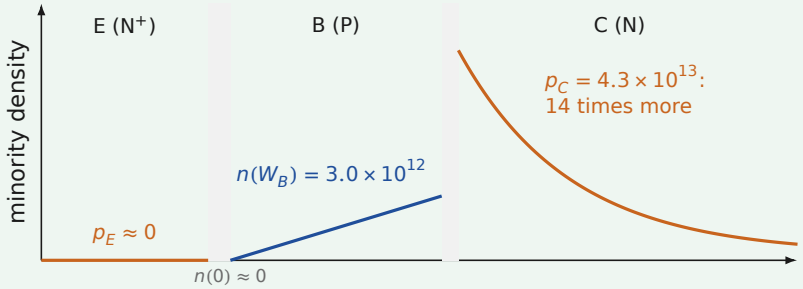
(d) Compare the electron density injected into the base by the BC junction with the hole density injected into the collector. Use this to explain why β_R is small, and why it cannot be improved without spoiling the forward operation.

(Answer: (b) $n(W_B) = 3.0\times10^{12}$, $n(0)\approx0$, $p_C = 4.3\times10^{13}$, $p_E\approx0\,\text{cm}^{-3}$; (c) EB 0.196 µm (0.195 in the base), BC 0.199 µm (0.013 in the base), W_B = 0.99 µm; (d) ratio 14.)

SOLUTION

(a) In inverse-active operation the BC junction injects, the EB junction collects. The electron profile in the base is a straight line from $n(W_B)$ at the collector side down

to zero at the emitter side; holes are injected into the collector, decaying from the BC edge; at the reverse-biased EB junction the hole density of the emitter is pulled to zero.
(b) $n_{B0} = 10^{20}/10^{17} = 10^3$, $p_{C0} = 1.43\times10^4$, $p_{E0} = 10\,\text{cm}^{-3}$. With $e^{0.565/0.0259} = 2.98\times10^9$: $n(W_B) = 3.0\times10^{12}\,\text{cm}^{-3}$ and $p_C = 4.3\times10^{13}\,\text{cm}^{-3}$ at the collector edge. At the EB side $e^{-2/0.0259} \approx 3\times10^{-34}$: $n(0)$ and p_E are practically zero.
(c) $V_{dE} = 0.954$ V; under $0.954 + 2 = 2.95$ V, $W_E = 0.196$ µm, almost all in the base (fraction $N_E/(N_E + N_B) = 0.99$): 0.195 µm. $V_{dC} = 0.766$ V; under $0.766 + V_{CB} = 0.766-0.565 = 0.201$ V, $W_C = 0.199$ µm, of which $N_C/(N_B+N_C) = 0.065$, 0.013 µm, in the base. $W_B = 1.2 - 0.195 - 0.013 = 0.99$ µm. Note that here the reverse-biased EB SCR eats most of the base.
(d) The injected densities are in the ratio $p_{C0}/n_{B0} = N_B/N_C = 14$: the BC junction injects into the collector 14 times more holes (per unit density) than electrons into the base, the inverse of the situation at the EB junction. The 'inverse emitter efficiency' is poor, and β_R is of order 1. Improving it would need a heavily doped collector, which would narrow the BC SCR, increase C_μ, reduce the breakdown voltage and the Early voltage: the asymmetry is the price of good forward operation.



Inverse-active operation (generic picture, linear axis): the BC junction injects many more holes into the collector than electrons into the base.

Problem 8.9 Charges stored in saturation

B · Guided calculation

Adapted from Neamen, Problem 12.18, p. 563

A silicon NPN transistor has $N_E = 10^{18}$, $N_B = 5\times10^{16}$ and $N_C = 10^{15}\,\text{cm}^{-3}$, $D_n = 25$ cm²/s in the base, a neutral base width $W_B = 0.70$ µm much shorter than L_n, and a long collector with hole diffusion length $L_C = 35$ µm. It operates in saturation with an electron current density $J_n = 125$ A/cm² across the base and $V_{BE} = 0.72$ V. (With the course value of n_i, the textbook value $V_{BE} = 0.70$ V could not carry this current at all, even with $V_{CB} = 0$: check it.)

(a) In saturation both junctions inject. Write the straight-line profile between $n(0)$ and $n(W_B)$ and show that $J_n = \frac{qD_n n_{B0}}{W_B}\left(e^{V_{BE}/V_T} - e^{-V_{CB}/V_T}\right)$. Compute V_{CB}.

(b) Compute V_{CE}.

(c) Compute the number of excess electrons per cm² in the base, and compare with forward-active operation at the same V_{BE}.

(d) Compute the hole density injected at the edge of the neutral collector and the number

of excess holes per cm² stored in the long collector. Is low injection valid there?

(e) Compare (c) and (d). What does this imply for the time needed to switch the transistor off?

(Answer: (a) $V_{CB} = -0.654$ V; (b) $V_{CE} = 66$ mV; (c) 8.9×10^{10} cm^{-2} (8.3×10^{10} forward-active); (d) 9.1×10^{15} cm^{-3}, 3.2×10^{13} cm^{-2}, no; (e) about 350 times more charge in the collector.)

SOLUTION

Check first: $qD_n n_{B0}/W_B = 1.602\times10^{-19}\times25\times2000/0.7\times10^{-4} = 1.14\times10^{-10}$ A/cm², and at 0.70 V the maximum, $V_{CB} = 0$, is $1.14\times10^{-10}\times e^{27.0} = 62.5$ A/cm² < 125. At 0.72 V the forward-active current is 135 A/cm².

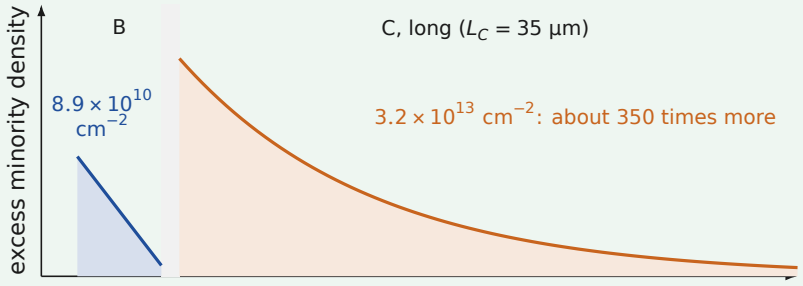
(a) $n(0) = n_{B0}e^{V_{BE}/V_T}$, $n(W_B) = n_{B0}e^{-V_{CB}/V_T}$, and the current is qD_n times the slope $[n(0)-n(W_B)]/W_B$. $e^{-V_{CB}/V_T} = 1.183\times10^{12}-125/1.144\times10^{-10} = 9.1\times10^{10}$, so $V_{CB} = -0.654$ V: the BC junction is forward biased.

(b) $V_{CE} = V_{CB} + V_{BE} = -0.654 + 0.72 = 66$ mV.

(c) $n(0) = 2.37 \times 10^{15}$ and $n(W_B) = 1.8 \times 10^{14}$ cm^{-3}; the trapezoid holds $\frac{1}{2}[n(0) + n(W_B)]W_B = 8.9 \times 10^{10}$ electrons per cm², against 8.3×10^{10} in forward-active operation (triangle). The base charge hardly changes.

(d) $p_{C0} = 10^5$ cm^{-3}, $p_C = p_{C0}e^{-V_{CB}/V_T} = 9.1 \times 10^{15}$ cm^{-3}. In a long region the excess is $p_C e^{-x/L_C}$, which holds $p_C L_C = 3.2 \times 10^{13}$ holes per cm². But p_C is nine times N_C: the collector is in high injection, and the ideal model is only an order-of-magnitude estimate.

(e) The lightly doped collector stores about 350 times more charge than the base. To switch the transistor off this charge must be removed first, by recombination (τ_p of the collector, microseconds) or by a reverse base current: the storage time of Chapter 7 again. This is why fast logic avoids deep saturation (Schottky-clamped transistors).



Saturation (generic picture, not to scale): the base charge barely changes, but the collector stores far more.

Problem 8.10 Early voltage: from data and from the physics calculation — B · Guided

Adapted from Neamen, Problems 12.35 and 12.37, p. 566

(a) A transistor carries $I_C = 1.2$ mA at $V_{CE} = 2$ V, and its Early voltage (the extrapolated intercept of the output line) is $V_A = 120$ V. Compute r_o and $g_o = 1/r_o$ with $r_o = V_A/I_C$, and the collector current at $V_{CE} = 4$ V. Repeat for $I_C = 0.25$ mA and $V_A =$

160 V.

(b) A silicon NPN transistor has $N_E = 2\times10^{18}$, $N_B = 2\times10^{16}$, $N_C = 2\times10^{15}\,\text{cm}^{-3}$, a metallurgical base width of 0.85 µm and $D_n = 25$ cm²/s. $V_{BE} = 0.650$ V. Compute the part of the EB SCR in the base, then, for $V_{CB} = 4$, 8 and 12 V, the part of the BC SCR in the base, W_B and $J_C = qD_n n(0)/W_B$.

(c) Estimate the Early voltage from the points at 4 and 12 V (extend the line to $J_C = 0$).

(d) Compute $V_{A,loc} = Q_{B,maj}/C_{jc}$ at $V_{CB} = 8$ V and compare. Explain the difference with the relation $V_{A,loc} = V_A + V_{CE}$ of the course.

(Answer: (a) 100 kΩ, 10 µS, 1.22 mA; 640 kΩ, 1.56 µS, 0.253 mA; (b) 0.119 µm; W_B = 0.565, 0.505, 0.458 µm; J_C = 28.1, 31.4, 34.7 A/cm²; (c) $V_A \approx 30$ V; (d) 38.8 V ≈ 30 + 8.65.)

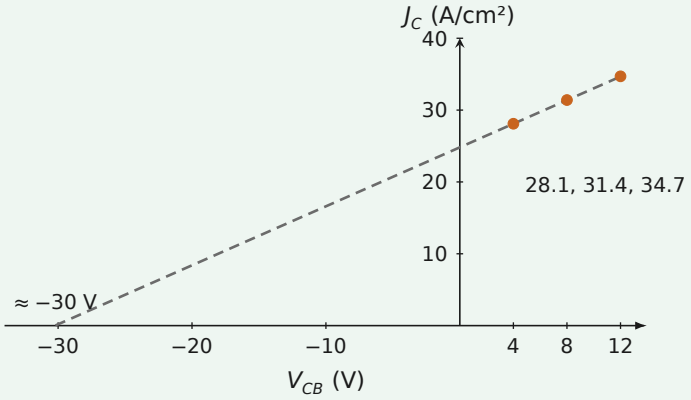
SOLUTION

(a) $r_o = 120/1.2\times10^{-3} = 100$ kΩ, $g_o = 10$ µS. In the linear model $I_C \propto V_A + V_{CE}$: $I_C(4\text{ V}) = 1.2\times122^{-1}\times124 = 1.22$ mA. For the second transistor, $r_o = 640$ kΩ, $g_o = 1.56$ µS and $I_C(4\text{ V}) = 0.253$ mA. (Strictly $r_o = (V_A + V_{CE})/I_C$, 1.7 % larger here; V_A/I_C assumes $V_{CE} \ll V_A$.)

(b) $V_{dE} = 0.871$ V; under $0.871 - 0.650 = 0.221$ V the EB SCR is 0.120 µm wide, 0.119 µm in the base. $V_{dC} = 0.692$ V. At 4, 8 and 12 V the BC SCR is 1.83, 2.49 and 3.00 µm wide, of which $N_C/(N_B + N_C) = 1/11$ lies in the base: 0.166, 0.226 and 0.273 µm. W_B = 0.565, 0.505 and 0.458 µm. With $n_{B0} = 10^{20}/2\times10^{16} = 5\times10^3$ and $n(0) = 5\times10^3\,\text{e}^{0.65/0.0259} = 3.9\times10^{14}\,\text{cm}^{-3}$, J_C = 28.1, 31.4 and 34.7 A/cm².

(c) Slope $(34.7 - 28.1)/8 = 0.82$ A/cm²/V; the line reaches $J_C = 0$ at $V_{CB} = 4 - 28.1/0.82 = -30$ V, i.e. $V_{CE} \approx -30 + 0.65$: $V_A \approx 30$ V.

(d) At 8 V: $Q_{B,maj}/S = qN_B W_B = 1.62\times10^{-7}$ C/cm² and $C_{jc}/S = \varepsilon/W_{BC} = 4.17\times10^{-9}$ F/cm², so $V_{A,loc} = 38.8$ V. The local value is measured from the operating point, the intercept from $V_{CE} = 0$: $V_{A,loc} = V_A + V_{CE} = 30 + 8.65 \approx 39$ V, consistent. The curve is not straight (the SCR widens more slowly at higher voltage), so the intercept depends on the two points chosen. A lightly doped base (2×10^{16}) and a thin one make this Early voltage small.



The collector current rises as the neutral base narrows; the line through the points meets $J_C = 0$ near −30 V.

Problem 8.11 Base width and speed

B · Guided calculation

Adapted from Sze & Lee, Ch. 4, Problems 23 and 24, p. 159

The emitter and collector delays are neglected, so that $\tau_F \simeq \tau_B = W_B^2/2D$.

(a) A silicon PNP transistor has $D_p = 10$ cm²/s in the base and $W_B = 0.5$ µm. Compute τ_B, the diffusion capacitance $C_D = \tau_F I_C/V_T$ at 1 mA, and the frequency $1/(2\pi\tau_B)$, the transit-time limit of the transistor (the analog course calls it the upper bound of f_T).

(b) What base width gives a transit-time limit of 5 GHz with the same D_p?

(c) The transistor of (b) has junction capacitances $C_{je} + C_{jc} = 2$ pF. Compute C_D at 1 mA and 10 mA. Above what current does the diffusion capacitance dominate?

(d) Compare the charge stored at 1 mA with that of a long base, where the carriers leave only by recombination with $\tau = 1$ µs. Why would an NPN transistor of the same base width be faster?

(Answer: (a) 0.125 ns, 4.8 pF, 1.27 GHz; (b) 0.25 µm; (c) 1.23 pF and 12.3 pF, above 1.6 mA; (d) 32 fC against 1 nC, about 30 000 times less.)

SOLUTION

(a) $\tau_B = (0.5\times10^{-4})^2/20 = 1.25\times10^{-10}$ s; $C_D = 1.25\times10^{-10}\times10^{-3}/0.0259 = 4.8$ pF; $1/(2\pi\tau_B) = 1.27$ GHz.
(b) $W_B = \sqrt{2D/(2\pi f)} = \sqrt{20/(2\pi\times5\times10^9)} = 2.5\times10^{-5}$ cm = 0.25 µm.
(c) $\tau_F = 31.8$ ps: $C_D = 31.8$ ps $\times\, I_C/25.9$ mV = 1.23 pF at 1 mA and 12.3 pF at 10 mA. $C_D = 2$ pF at $I_C = 2\times10^{-12}\times0.0259/31.8\times10^{-12} = 1.6$ mA: below, the junction capacitances dominate; above, the stored charge of the base.
(d) $Q = \tau_B I_C = 32$ fC against $\tau I_C = 1$ nC for a long base: the two exits of the course; the collector drains the carriers 3×10^4 times faster than recombination would. Electrons diffuse faster than holes ($D_n \approx 3D_p$ in silicon): for the same width τ_B is three times shorter. This is one reason why fast bipolar circuits use NPN transistors.

Practice exercises from the course notes

These exercises closed Chapter 8 of the course notes; they now live here, with their answers. Tables, equations and sections cited by number are those of the course notes. Silicon at 300 K, the reference transistor of Table 8.1, unless stated otherwise.

Problem 8.12 Equilibrium

B · Guided calculation

From the course notes, Chapter 8

(i) Compute V_{dE}, V_{dC}, n_{B0}, p_{E0} and p_{C0}. (ii) Compute the widths of the two SCRs at equilibrium and the part of each in the base. (iii) Check (8.4), $n_{B0} = N_E e^{-V_{dE}/V_T} = N_C e^{-V_{dC}/V_T}$.

(Answer: (i) $V_{dE} = 0.835$ V, $V_{dC} = 0.656$ V, $n_{B0} = 10^4$, $p_{E0} = 10^2$, $p_{C0} = 10^5$ cm^{-3}. (ii) EB: $W = 0.330$ µm, 0.327 µm in the base; BC: $W = 0.966$ µm, 0.088 µm in the base. (iii) $10^{18}e^{-32.2} = 10^4$

and $10^{15}\mathrm{e}^{-25.3} = 10^{4}\,\mathrm{cm}^{-3}$.)

Problem 8.13 Injection

B · Guided calculation

From the course notes, Chapter 8

(i) At $V_{BE} = 0.65$ V, compute $n(0)$ and I_C. (ii) What V_{BE} gives $I_C = 10$ mA, and what is then $n(0)/N_B$? (iii) Up to what collector current is low injection ($n(0) < N_B/10$) satisfied?

(Answer: (i) $n(0) = 10^{4}\mathrm{e}^{0.65/0.0259} = 7.9\times10^{14}\,\mathrm{cm}^{-3}$; $I_C = 4.4$ mA. (ii) $V_{BE} = V_T \ln(I_C/I_S) = 0.671$ V; $n(0) = 1.8\times10^{15}\,\mathrm{cm}^{-3}$, 18 % of N_B: beyond low injection. (iii) $n(0) = 10^{15}\,\mathrm{cm}^{-3}$ at $I_C = I_S\times10^{11} = 5.6$ mA.)

Problem 8.14 Gain and design

B · Guided calculation

From the course notes, Chapter 8

(i) Compute γ, α_T, α_F and β_F. (ii) Recompute β_F with $N_E = 10^{19}\,\mathrm{cm}^{-3}$ (keep D_p). (iii) Recompute it with $W_B = 0.5$ µm. Which design constraint does each change use?

(Answer: (i) $\gamma = 0.9932$, $\alpha_T = 0.99857$, $\alpha_F = 0.9918$, $\beta_F = 121$. (ii) The emitter term drops to 6.9×10^{-4}: $\beta_F = 473$ (heavily doped emitter; real emitters at 10^{19} are degenerate and the gain is somewhat lower). (iii) Both terms decrease, to 3.4×10^{-3} and 3.6×10^{-4}: $\beta_F = 264$ (short base).)

Problem 8.15 Ebers–Moll parameters

B · Guided calculation

From the course notes, Chapter 8

Neglect recombination in the base. (i) Compute I_{ES}, I_{CS}, α_F, α_R and β_R. (ii) Check the reciprocity relation. (iii) Why is α_R so much smaller than α_F? (iv) Where does I_C cross zero at $V_{BE} = 0.611$ V?

(Answer: (i) $I_{pE0} = 3.8\times10^{-16}$ A, $I_{pC0} = 1.9\times10^{-14}$ A; $I_{ES} = I_S + I_{pE0} = 5.645\times10^{-14}$ A, $I_{CS} = I_S + I_{pC0} = 7.53\times10^{-14}$ A; $\alpha_F = 0.9932$, $\alpha_R = 0.745$, $\beta_R = 2.92$. (ii) $\alpha_F I_{ES} = \alpha_R I_{CS} = 5.607\times10^{-14}$ A $= I_S$. (iii) The collector is lightly doped: when the BC junction is forward biased it injects many holes into the collector ($p_{C0} = 10^{5}$), which do not reach the emitter. (iv) At $V_{CB} = -V_{BE} + V_T \ln(1/\alpha_R) = -0.611 + 0.0076 = -0.603$ V.)

Problem 8.16 Saturation

B · Guided calculation

From the course notes, Chapter 8

The transistor carries $I_B = 0.1$ mA and $I_C = 1$ mA. Using (8.18)–(8.19) with $\beta_F = 121$ and $\beta_R = 2.9$ (neglect the -1 terms), compute V_{BE}, V_{CB} and V_{CE}. Draw the electron line across the base.

(Answer: With $x = \mathrm{e}^{V_{BE}/V_T}$, $y = \mathrm{e}^{-V_{CB}/V_T}$: $I_S(x - 1.345\,y) = 1$ mA and $I_S(x/121 + y/2.9) = 0.1$ mA give $V_{BE} = 0.619$ V, $V_{CB} = -0.576$ V, $V_{CE} = 43$ mV. The line goes from $n(0) = 2.4\times10^{14}$ to $n(W_B) = 4.6\times10^{13}\,\mathrm{cm}^{-3}$: both ends raised, smaller slope.)

Problem 8.17 Early effect

B · Guided calculation

From the course notes, Chapter 8

(i) Compute W_B at $V_{CB} = 2$ V and 10 V, knowing that it is 1.0 µm at 5 V and that the base side of the BC SCR is $N_C/(N_B + N_C)$ of its width. (ii) By what factor does I_C change between these voltages, at fixed V_{BE}? (iii) Compute the advance δ of the SCR edge per volt at 2 V and the Early voltage $V_A = W_B/\delta$; check with qSN_BW_B/C_{jc}.

(Answer: (i) $W_{BC} = 1.94$ µm at 2 V and 3.89 µm at 10 V, base parts 0.177 and 0.354 µm, against 0.258 µm at 5 V: $W_B = 1.081$ µm and 0.904 µm. (ii) $I_C \propto 1/W_B$: a factor 1.196. (iii) $dW_{BC}/dV = W_{BC}/2(V_{dC}+V_{CB}) = 0.366$ µm/V, $\delta = 33.3$ nm/V, $V_A = 32.5$ V; $C_{jc} = 0.533$ pF, $qSN_BW_B = 17.3$ pC, the same 32.5 V.)

Problem 8.18 Capacitances

B · Guided calculation

From the course notes, Chapter 8

At $I_C = 2$ mA and $V_{CB} = 5$ V, compute the diffusion capacitance C_D of the EB junction (with $\tau_F \approx \tau_B$) and the transition capacitance C_{jc} of the BC junction. What would C_D be with a typical $\tau_F = 0.5$ ns? How much charge would a long base ($\tau_n = 0.1$ µs) store at the same current?

(Answer: $C_D = 0.143$ ns × 2 mA/25.9 mV = 11.0 pF; $C_{jc} = 0.365$ pF; with $\tau_F = 0.5$ ns, $C_D = 39$ pF. Short base: $Q = \tau_B I_C = 0.29$ pC; long base: $\tau_n I_C = 200$ pC, 700 times more.)

Problem 8.19 A faster transistor

B · Guided calculation

From the course notes, Chapter 8

The base width is halved, everything else unchanged. How do τ_B, the diffusion capacitance at a given current, β_F, the Early voltage and the sheet resistance of the base change? Comment on the trade-offs.

(Answer: τ_B and C_D are divided by 4 (36 ps); β_F rises to 264 (Problem 8.16); V_A is roughly halved, since qSN_BW_B is halved while C_{jc} hardly changes (a shorter base, a larger relative modulation); the sheet resistance $1/(q\mu_p N_B W_B)$ doubles to about 27 kΩ per square. Speed and gain are bought with output resistance and base resistance.)

Challenge: a new situation

These problems apply the method of the chapter to a device, a material or a situation that the chapter did not treat. The reasoning does not change: state the physical picture, choose the model, compute, check.

Problem 8.20 An implanted base with a built-in field

C · Synthesis

Adapted from Sze & Lee, Ch. 4, Problems 16 and 17, p. 158

In a real NPN transistor the base is made by implantation, and its doping decreases from the emitter to the collector: $N(x) = N_{A0}\mathrm{e}^{-x/\ell}$ with $N_{A0} = 2\times10^{18}$ cm^{-3} and $\ell = 0.3$ µm, over a neutral base of width $W_B = 0.8$ µm ($x = 0$ at the emitter side).

(a) Compute the number of acceptors per cm² in the neutral base, $Q_G = \int_0^{W_B} N\,\mathrm{d}x$ (the 'Gummel number'), and the average doping.

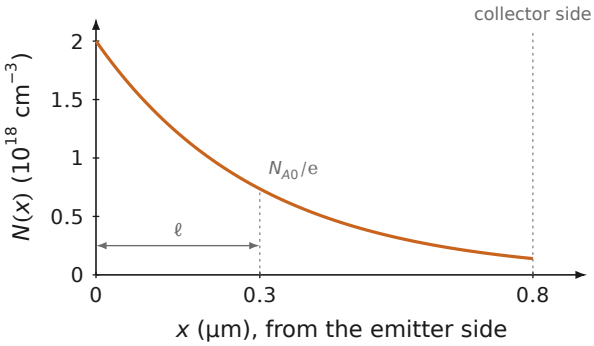
(b) In equilibrium, and approximately under low injection, the hole current in the base is zero. Use the drift–diffusion expression of the hole current (Chapters 1 and 4) to show that there is a built-in field $\mathcal{E} = \frac{kT}{q}\frac{1}{N}\frac{\mathrm{d}N}{\mathrm{d}x}$. Compute it, give its direction and show that it pushes electrons towards the collector.

(c) Write the electron current $J_n = q\mu_n n\mathcal{E} + qD_n \mathrm{d}n/\mathrm{d}x$ with this field and show that it can be written $J_n = \frac{qD_n}{N}\frac{\mathrm{d}(nN)}{\mathrm{d}x}$. Integrate across the base (with $nN = n_i^2 \mathrm{e}^{V_{BE}/V_T}$ at $x = 0$ and $n \approx 0$ at W_B) to obtain

$$|J_n| = \frac{qD_n n_i^2}{Q_G}\,\mathrm{e}^{V_{BE}/V_T}.$$

Check that it reduces to the course formula for a uniform base.

(d) The textbook gives, at 300 K, $\mu_n = 88 + 1252/(1 + 0.698\times10^{-17}N)$ cm²/(V·s). Compute D_n for the average doping. The emitter is long, with $L_E = 1$ µm, $N_E = 10^{19}\,\mathrm{cm}^{-3}$, $D_E = 1$ cm²/s, and the base lifetime is 10^{-6} s. Estimate γ, α_T (uniform-base formula, a slight underestimate) and β_F.



The implanted base doping.

HINT

(c) $\mu_n\mathcal{E} = D_n\frac{1}{N}\frac{\mathrm{d}N}{\mathrm{d}x}$ by Einstein's relation, so $J_n = qD_n\left(\frac{\mathrm{d}n}{\mathrm{d}x} + \frac{n}{N}\frac{\mathrm{d}N}{\mathrm{d}x}\right)$.

(Answer: (a) $Q_G = 5.6\times10^{13}$ cm^{-2}, average 7.0×10^{17} cm^{-3}; (b) 863 V/cm, pointing towards the emitter; (d) $\mu_n = 301$ cm²/(V·s), $D_n = 7.8$ cm²/s, $\gamma = 0.9929$, $\alpha_T = 0.99959$, $\beta_F \approx 132$.)

SOLUTION

(a) $Q_G = N_{A0}\ell\,(1 - \mathrm{e}^{-W_B/\ell}) = 2\times10^{18}\times0.3\times10^{-4}\times(1-\mathrm{e}^{-2.67}) = 5.6\times10^{13}$ cm^{-2}, and the average doping is $Q_G/W_B = 7.0\times10^{17}\,\mathrm{cm}^{-3}$.

(b) $J_p = qp\mu_p\mathcal{E} - qD_p\,\mathrm{d}p/\mathrm{d}x = 0$ with $p \simeq N$ gives $\mathcal{E} = \frac{D_p}{\mu_p}\frac{1}{N}\frac{\mathrm{d}N}{\mathrm{d}x} = \frac{kT}{q}\frac{1}{N}\frac{\mathrm{d}N}{\mathrm{d}x}$ (Einstein's relation). For the exponential profile $\mathrm{d}N/\mathrm{d}x = -N/\ell$, so $\mathcal{E} = -kT/(q\ell) = -0.0259/0.3\times10^{-4} = -863$ V/cm, uniform across the base. It points towards $-x$, the emitter: it holds the holes back against their diffusion towards the lightly doped side. Electrons feel the force $-q\mathcal{E}$, towards $+x$: the field pushes them towards the collector, an aiding field.

(c) With Einstein's relation $\mu_n = D_n/V_T$ and the field of (b), $q\mu_n n\mathcal{E} = qD_n\frac{n}{N}\frac{\mathrm{d}N}{\mathrm{d}x}$, so $J_n = qD_n\left(\frac{\mathrm{d}n}{\mathrm{d}x} + \frac{n}{N}\frac{\mathrm{d}N}{\mathrm{d}x}\right) = \frac{qD_n}{N}\frac{\mathrm{d}(nN)}{\mathrm{d}x}$. In steady state without recombination J_n is constant: $J_n\int_0^{W_B} N\,\mathrm{d}x = qD_n[nN]_0^{W_B} = -qD_n n(0)N(0)$, and $n(0)N(0) = n_i^2 \mathrm{e}^{V_{BE}/V_T}$ (the law of the junction). Hence $|J_n| = qD_n n_i^2 \mathrm{e}^{V_{BE}/V_T}/Q_G$. For a uniform base $Q_G = N_B W_B$ and $I_S = qSD_n n_i^2/(N_B W_B)$, the course formula: the collector current depends only on the *total* base charge, not on its distribution.
(d) $\mu_n = 88 + 1252/(1 + 0.698\times10^{-17}\times7.0\times10^{17}) = 301$ cm²/(V·s) and $D_n = 0.0259\times301 = 7.8$ cm²/s. The injection loss is the ratio of hole to electron currents: $\frac{D_E n_i^2/(N_E L_E)}{D_n n_i^2/Q_G} = \frac{1\times5.6\times10^{13}}{7.8\times10^{19}\times10^{-4}} = 7.2\times10^{-3}$, so $\gamma = 0.9929$. $L_n = \sqrt{7.8\times10^{-6}} = 27.9$ µm and $\alpha_T = 1 - W_B^2/2L_n^2 = 0.99959$. Then $1/\beta_F \simeq 7.2\times10^{-3} + 0.4\times10^{-3}$ and $\beta_F \approx 132$. The built-in field shortens the transit (electrons drift as well as diffuse), so the true α_T is even closer to 1; the emitter limits the gain.

Problem 8.21 A silicon–germanium base

C · Synthesis

Adapted from Sze & Lee, Ch. 4, Problem 25, p. 159

In a SiGe heterojunction transistor, the base contains 10 % germanium, which reduces its bandgap by 9.8 % compared with silicon; the emitter and the collector are pure silicon. Assume that the base current is due only to the holes injected into the emitter, and that the effective densities of states are unchanged.

(a) How does n_i^2 depend on E_g (Chapter 3)? By what factor is n_i^2 of the base larger than that of silicon at 300 K?

(b) Using the formula $I_S = qSD_n n_{iB}^2/(N_B W_B)$, with n_{iB} the intrinsic density of the base, show that β_F is multiplied by $\exp(\Delta E_g/kT)$ compared with an all-silicon transistor of the same doping and geometry.

(c) Compute the ratio of the current gains at 0 °C and at 100 °C. Does the gain increase or decrease with temperature?

(d) Why is this an advantage for the designer? What can be done with the base doping, and which other property of the transistor improves?

(Answer: (a) $n_i^2 \propto \exp(-E_g/kT)$; $\Delta E_g = 0.110$ eV, factor 69; (c) $\beta(0\,°\mathrm{C})/\beta(100\,°\mathrm{C}) = 3.5$, the gain decreases.)

SOLUTION

(a) $n_i^2 = N_c N_v \mathrm{e}^{-E_g/kT}$, so a smaller gap gives $n_{iB}^2/n_i^2 = \exp(\Delta E_g/kT)$. $\Delta E_g = 0.098\times1.12 = 0.110$ eV and $\exp(0.110/0.0259) = 69$.
(b) The electron current injected into the base is proportional to n_{iB}^2/N_B, the hole current injected into the emitter to n_i^2/N_E (silicon emitter). Their ratio, β_F, is multiplied by $n_{iB}^2/n_i^2 = \exp(\Delta E_g/kT)$.
(c) $\beta(0\,°\mathrm{C})/\beta(100\,°\mathrm{C}) = \exp\left[\frac{\Delta E_g}{k}\left(\frac{1}{273} - \frac{1}{373}\right)\right] = \exp(1274\times9.81\times10^{-4}) = 3.5$. The gain decreases with temperature, because the advantage $\exp(\Delta E_g/kT)$ shrinks as kT grows. (In an all-silicon transistor the same temperature dependence is much weaker; the temperature dependence of D and of the gap itself are ignored here.)

(d) A gain of 69 times the silicon value is more than needed: the designer spends it on a much more heavily doped base. That lowers the base resistance (the lateral resistance of the thin base layer), raises the Early voltage ($Q_{B,maj}$ larger) and allows a thinner base without punch-through, hence a shorter τ_B and a higher f_T. This is how SiGe transistors reach several hundred gigahertz.

SEMICONDUCTOR DEVICES PHYSICS · SOLUTIONS

ANNEX A

Energy: One Thread Through the Course

WHAT THIS SET USES

State and transfer; the zero of energy; hydrogen and negative energies; quantum states; atomic levels to bands; the band edge and $E - E_C$; occupation; transitions and where the energy goes.

DATA

Use $q = 1.602 \times 10^{-19}$ C, $m_0 = 9.109 \times 10^{-31}$ kg, $\hbar = 1.055 \times 10^{-34}$ J·s, $\varepsilon_0 = 8.854 \times 10^{-12}$ F/m, $g = 9.8$ m/s^2, $kT = 25.9$ meV at 300 K and $\lambda(\text{nm}) = 1240/E(\text{eV})$.

Problem A.1 Two zeros

B · Guided calculation

From the course notes, Annex A

A 0.5 kg ball rolls off a table 0.8 m high. (i) Give its potential energy on the table and on the floor, first with the zero at the floor, then with the zero at the table top. (ii) Compute the work done by the weight and the speed on reaching the floor in both cases. What does not depend on the zero?

(Answer: (i) Zero at the floor: $U = 0.5 \times 9.8 \times 0.8 = 3.92$ J on the table, 0 on the floor. Zero at the table: 0 on the table, -3.92 J on the floor. (ii) In both cases the weight does $W = -\Delta U = 3.92$ J and $v = \sqrt{2 \times 3.92/0.5} = 3.96$ m/s (for a ball leaving the table at negligible speed). The work, the change of U and the speed do not depend on the zero.)

Problem A.2 Hydrogen from the confinement estimate

B · Guided calculation

From the course notes, Annex A

(i) Minimise (A.3) and compute a_0 and E with the constants above. (ii) Show that at the minimum $K = -U/2$. (iii) Using $E_n = -13.6\,\text{eV}/n^2$, give the energy and the wavelength of the photon emitted in the transitions $3 \to 2$ and $2 \to 1$. Which one is visible? (iv) How much energy is needed to ionise the atom from $n = 2$?

(Answer: (i) $a_0 = 4\pi\varepsilon_0\hbar^2/mq^2 = 5.29 \times 10^{-11}$ m; $E = -\hbar^2/2ma_0^2 = -2.18 \times 10^{-18}$ J $= -13.6$ eV. (ii) At the minimum $\hbar^2/ma_0^3 = q^2/4\pi\varepsilon_0 a_0^2$, hence $2K = -U$. (iii) $3 \to 2$: $13.6(1/4 - 1/9) = 1.89$ eV, 656 nm, red, visible; $2 \to 1$: 10.2 eV, 122 nm, ultraviolet. (iv) 3.4 eV.)

Problem A.3 A donor in gallium arsenide

B · Guided calculation

From the course notes, Annex A

In GaAs, $m^* = 0.067\,m_0$ and $\varepsilon_r = 13.1$. Estimate the ionisation energy and the radius of a hydrogen-like donor. Compare the energy with kT at 300 K and at 77 K. What fraction of the donors do you expect to be ionised at 300 K?

(Answer: $E_d = 13.6 \times 0.067/13.1^2 = 5.3$ meV and $a_d = 0.053 \times 13.1/0.067 = 10$ nm. At 300 K, $kT = 25.9$ meV is five times E_d: the donors are practically all ionised. At 77 K, $kT = 6.6$ meV is still above E_d: GaAs donors freeze out only at much lower temperatures than in silicon.)

Problem A.4 A state near the band edge

B · Guided calculation

From the course notes, Annex A

An electron of the silicon conduction band is 0.05 eV above E_C ($m^* = 0.26\,m_0$). (i) Compute $|k - k_0|$ and its speed. (ii) Compare $|k - k_0|$ with $2\pi/a$, where $a = 0.543$ nm is the lattice constant of silicon. Is the parabolic approximation reasonable? (iii) Compare 0.05 eV with the mean kinetic energy of the conduction electrons at 300 K.

(Answer: (i) $|k - k_0| = \sqrt{2m^*E}/\hbar = 5.8 \times 10^8$ m^{-1}; $v = \hbar|k - k_0|/m^* = 2.6 \times 10^5$ m/s. (ii) $2\pi/a = 1.16 \times 10^{10}$ m^{-1}: the state is 5 % of the width of the zone from the minimum, close enough for the parabola. (iii) $\frac{3}{2}kT = 39$ meV: 0.05 eV is 1.3 times the mean, an ordinary thermal electron.)

Problem A.5 Energy per flight

B · Guided calculation

From the course notes, Annex A

Using $l = 46$ nm for electrons in silicon, compute the energy gained per free flight in fields of 100 V/cm, 10 kV/cm and 100 kV/cm, and compare each with kT. In which case do you expect Ohm's law to fail?

(Answer: $q\mathcal{E}l$: 100 V/cm = 10^4 V/m gives 0.46 meV; 10 kV/cm gives 46 meV; 100 kV/cm gives 0.46 eV. Compared with $kT = 25.9$ meV: negligible, comparable, much larger. Ohm's law (constant mobility) starts to fail when the gain per flight approaches kT, around 10 kV/cm, and fails clearly at 100 kV/cm (velocity saturation).)

Problem A.6 Energy budget of illuminated silicon

B · Guided calculation

From the course notes, Annex A

A silicon sample absorbs 1 mW of green light at 500 nm. (i) How many photons per second are absorbed? (ii) If each photon creates one pair, what power is turned into heat by thermalisation? (iii) The pairs then recombine through defects. What happens to the remaining energy? (iv) What would change for light at 1300 nm?

(Answer: (i) $E = 1240/500 = 2.48$ eV $= 3.97 \times 10^{-19}$ J: $10^{-3}/3.97 \times 10^{-19} = 2.5 \times 10^{15}$ photons per second. (ii) A fraction $(2.48 - 1.12)/2.48 = 55$ % of the power, 0.55 mW, becomes heat within about a picosecond. (iii) The remaining 1.12 eV per pair (0.45 mW) is released at recombination; through defects it also goes to the lattice, so all 1 mW ends as heat unless the carriers are extracted by a junction (solar cell). (iv) $1240/1300 = 0.95$ eV $< E_g$: the light is not absorbed by band-to-band transitions; silicon is transparent to it.)

SEMICONDUCTOR DEVICES PHYSICS · SOLUTIONS

ANNEX B

A Uniformly Biased Intrinsic Silicon Rod

WHAT THIS SET USES

Drift and diffusion currents in a homogeneous rod; the general solution of the transport equations; the barometric length $\ell = kT/q\mathcal{E}$; the quasi-Fermi level and $J_n = \mu_n n\, dE_{Fn}/dx$.

DATA

Silicon at 300 K: $n_i = 10^{10}\,\text{cm}^{-3}$, $\mu_n = 1350$ and $\mu_p = 460\,\text{cm}^2/(\text{V·s})$, $kT/q = 25.9$ mV.

Problem B.1 A small rod

B · Guided calculation

From the course notes, Annex B

An intrinsic silicon rod has $L = 2$ mm and a square cross-section of 0.5×0.5 mm^2; 5 V are applied. Compute $\mathcal{E}$, J, I, R, and the share of the current carried by the holes.

(Answer: $\mathcal{E} = 5/0.2 = 25$ V/cm; $J = 1.602 \times 10^{-19} \times 10^{10} \times 1810 \times 25 = 7.2 \times 10^{-5}$ A/cm^2; $S = 2.5 \times 10^{-3}$ cm^2, $I = 0.18$ µA; $R = 28$ MΩ. Holes: 460/1810 = 25 %.)

Problem B.2 The barometric length

B · Guided calculation

From the course notes, Annex B

For the rod of Problem B.1, compute ℓ and L/ℓ. How large would the voltage have to be for ℓ to equal 1 µm?

(Answer: $\ell = 0.0259/25 = 1.04 \times 10^{-3}$ cm = 10 µm, $L/\ell = 190$. $\ell = 1$ µm needs $\mathcal{E} = 259$ V/cm, i.e. 52 V across 2 mm.)

Problem B.3 A layer with no current

B · Guided calculation

From the course notes, Annex B

(i) Show that $n(x) = C_2\, e^{-x/\ell}$ gives $J_n = 0$. (ii) Show that it is the Boltzmann distribution $n \propto e^{-E_C(x)/kT}$ of electrons in the tilted band. (iii) Do the same for holes with $C_2'\, e^{+x/\ell}$.

(Answer: (i) $dn/dx = -n/\ell$, so $qD_n\, dn/dx = -qD_n n/\ell = -q\mu_n \mathcal{E} n$, which cancels the drift term. (ii) $E_C(x) = E_C(0) + q\mathcal{E}x$, so $e^{-x/\ell} = e^{-[E_C(x) - E_C(0)]/kT}$. (iii) $dp/dx = p/\ell$; $-qD_p\, p/\ell = -q\mu_p \mathcal{E} p$ cancels the hole drift term; $e^{+x/\ell} = e^{-[E_V(L) - E_V(x)]/kT} \times$const: the holes crowd where E_V is highest.)

Problem B.4 An injecting contact

B · Guided calculation

From the course notes, Annex B

Suppose that the contact at $x = 0$ holds $n(0) = 2n_i$ while the contact at $x = L$ holds $n(L) = n_i$ (consider the electrons only). (i) Find C_1 and C_2 for $L/\ell \gg 1$. (ii) Compare J_n with the value for ideal contacts. (iii) Where are the extra electrons? Why does the charge they carry make the uniform-field assumption only approximate?

(Answer: (i) $C_1 + C_2 = 2n_i$ and $C_1 + C_2 e^{-L/\ell} = n_i$; for $e^{-L/\ell} \to 0$, $C_1 = n_i$ and $C_2 = n_i$. (ii) $J_n = q\mu_n \mathcal{E} C_1$, the same as for ideal contacts (exactly, $C_1 = n_i(1-2e^{-L/\ell})/(1-e^{-L/\ell})$). (iii) Within a few ℓ of $x = 0$. They are not matched by holes, so the layer is charged; Gauss's law then bends the field near the contact, and a full treatment must solve Poisson's equation with the transport equations.)

Problem B.5 Fermi-level slope

B · Guided calculation

From the course notes, Annex B

For the rod of Problem B.1, compute $\mathrm{d}E_{Fn}/\mathrm{d}x$ in meV/µm, and check that $J_n = \mu_n n \, \mathrm{d}E_{Fn}/\mathrm{d}x$ gives the electron current of Problem B.1.

(Answer: $\mathrm{d}E_{Fn}/\mathrm{d}x = q\mathcal{E} = 25$ meV/cm $= 2.5 \times 10^{-3}$ meV/µm. $J_n = \mu_n n_i \, q\mathcal{E} = 1350 \times 10^{10} \times 1.602 \times 10^{-19} \times 25 = 5.4 \times 10^{-5}$ A/cm^2, 75 % of J in Problem B.1.)

Problem B.6 A warmer rod

B · Guided calculation

From the course notes, Annex B

At 350 K, n_i is 27.8 times larger than at 300 K and both mobilities are 1.26 times smaller (Chapter 3, Problem 3.15). At the same voltage, by what factor do the current and the dissipated power change? What happens to ℓ? Why can this lead to a runaway if the rod is poorly cooled?

(Answer: The conductivity is multiplied by 27.8/1.26 = 22, and so are the current and the power V^2/R. $\ell = kT/q\mathcal{E}$ grows in proportion to T, by 17 %. More power heats the rod, which raises n_i, hence the current and the power again: without enough cooling the temperature runs away (thermal runaway).)